

33.1 INTRODUCTION

In earlier classes, we have learnt about two approaches to the theory of probability, namely, (i) Statistical approach and (ii) Classical approach. The statistical approach has been discussed in class IX. It is also known as repeated experiments and observed frequency approach. In this approach, we have defined the probability of an event as the ratio of observed frequency to the total frequency. The classical approach has been discussed in class X. In this approach, we define the probability of occurrence of an event as the ratio of favourable number of outcomes to the total number of equally likely outcomes. These equally likely outcomes are also known as elementary events associated to the experiment. Both the theories have some serious deficiencies and limitations. For instance, these approaches cannot be applied to the experiments which have large number of outcomes. The classical definition of probability cannot be applied whenever it is not possible to make a simple enumeration of cases which can be considered equally likely. For instance, how does it apply to probability of rain? What are the possible outcomes? We might think that there are two cases 'rain' and 'no rain'. But at any given locality it will not usually be agreed that they are equally like. The classical approach also fails to answer questions like "what is the probability that a male will die before the age of 60", "what is the probability that a bulb will burn in less than 2000 hours? etc. In fact, the classical definition is difficult to apply as soon as we deviate from the experiments pertaining to coins, dice, cards and other simple games of chance.

The statistical definition has difficulties from a mathematical point of view because an actual limiting number may not really exist. For this reason, modern probability theory has been developed axiomatically. This theory of probability was developed by A.N. Kolmogorov (1903-1987) a Russian Mathematician in 1933. He laid down certain axioms to interpret probability, in his book 'Foundation of Probability' published in 1933. The axiomatic definition of probability includes 'both' the classical and statistical approaches as particular cases and overcomes the deficiencies of each of them. In order to understand this approach we must know about some basic terms viz. random experiment, elementary events, sample space, compound events etc. So, let us begin with the term random experiment as discussed in the following section.

33.2 RANDOM EXPERIMENTS

The word experiment means an operation which can produce some well-defined outcome(s). There are two types of experiments viz. (i) Deterministic experiments and (ii) Random or Probability experiments.

DETERMINISTIC EXPERIMENTS In our day-to-day life, we perform many activities/experiments which have a fixed outcome or result no matter any number of times they are repeated. Such experiments are known as deterministic experiments. For example, from the set of all triangles in a plane if a triangle is chosen, then even without knowing the three angles, we can definitely say that the sum of the measures of the angles is 180° . In fact, when experiments in science and engineering are repeated under identical conditions, we get the same result every time.

RANDOM OR PROBABILISTIC EXPERIMENTS If an experiment, when repeated under identical conditions, do not produce the same outcome every time but the outcome in a trial is one of the several possible outcomes then such an experiment is known as a probabilistic experiment or a random experiment. In other words, an experiment whose outcomes cannot be predicted or determined in advance is called a random experiment.

For example, in tossing of a coin one is not sure if a head or a tail will be obtained so it is a random experiment. Similarly, rolling an unbiased die and drawing a card from a well shuffled pack of playing cards are examples of random experiments.

33.3 SAMPLE SPACES

In the previous section, we have learnt about random experiments. Throughout this chapter the term experiment will mean random experiment. Associated to every random experiment there are two basic terms viz. outcomes (or elementary events) and sample space. In this section, we will discuss about these two for different random experiments.

ELEMENTARY EVENT *If a random experiment is performed, then each of its outcomes is known as an elementary event.*

In other words, outcomes of a random experiment are known as elementary events associated to it. Elementary events are also known as simple events.

SAMPLE SPACE *The set of all possible outcomes of a random experiment is called the sample space associated with it and it is generally denoted by S .*

If $E_1, E_2, E_3, \dots, E_n$ are the possible outcomes (or elementary events) of a random experiment, then $S = \{E_1, E_2, \dots, E_n\}$ is the sample space associated to it.

ILLUSTRATION 1 *Consider the random experiment of tossing of a coin. The possible outcomes of this experiment are H and T . Thus, if we define*

E_1 = Getting head (H) on the upper face and, E_2 = Getting tail (T) on the upper face.

Then, E_1 and E_2 are elementary events associated to the random experiment of tossing of a coin. The sample space associated to this experiment is given by $S = \{E_1, E_2\}$.

E_1 and E_2 are generally denoted by H and T respectively. Thus, we have $S = \{H, T\}$.

ILLUSTRATION 2 *Consider the experiment of throwing a die. Let the six faces of a die be marked as 1, 2, 3, 4, 5 and 6. If the die is thrown, then any one of the six faces may come upward. So, there are six possible outcomes of this experiment, namely, 1, 2, 3, 4, 5, 6. Thus, if we define*

E_i = Getting a face marked with number i , where $i = 1, 2, 3, 4, 5, 6$

Then, $E_1, E_2, \dots, E_6$ are six elementary events associated to this experiment. The sample space associated to this experiment is $S = \{E_1, E_2, \dots, E_6\}$.

In this experiment, elementary even E_i is denoted by i , where $i = 1, 2, \dots, 6$. Thus, we have

$$S = \{1, 2, 3, 4, 5, 6\}.$$

ILLUSTRATION 3 *Consider the experiment of tossing two coins together or a coin twice. In this experiment the possible outcomes are:*

Head on first and Head on second,

Head on first and Tail on second,

Tail on first and Head on second,

Tail on first and Tail on second.

If we define

HH = Getting head on both coins,

HT = Getting head on first and tail on second,

TH = Getting tail on first and head on second,

TT = Getting tail on both coins.

Then,

HH, HT, TH and TT are elementary events associated to the random experiment of tossing of two coins. The sample space associated to this experiment is given by $S = \{HH, HT, TH, TT\}$.

Similarly, the sample space associated to the random experiment of tossing three coins simultaneously or tossing a coin three times is given by

$$S = \{HHH, HHT, HTH, THH, TTH, HTT, THT, TTT\}$$

ILLUSTRATION 4 Consider the random experiment in which two dice are tossed together or a die is tossed twice. If we define

E_{ij} = Getting number i on the upper face of first die and number j on the upper face of second die,

where $i = 1, 2, \dots, 6$ and $j = 1, 2, \dots, 6$.

Then, E_{ij} are elementary events associated to this experiment and are generally denoted by (i, j) .

Thus, $(1, 1), (1, 2), \dots, (1, 6), (2, 1), \dots, (2, 6), (3, 1), \dots, (3, 6), (4, 1), \dots, (4, 6), (5, 1), \dots, (5, 6)$ and $(6, 1), \dots, (6, 6)$ are 36 elementary events associated to the random experiment of tossing two dice and the sample space associated to it is given by

$$S = \{(1, 1), \dots, (1, 6), (2, 1), \dots, (2, 6), (3, 1), \dots, (3, 6), \dots, (6, 1), \dots, (6, 6)\}.$$

ILLUSTRATION 5 Let there be a bag containing 3 white and 2 black balls. Let the white balls be denoted by W_1, W_2, W_3 and black balls be denoted by B_1, B_2 . If we draw two balls from the bag, then there are 5C_2 elementary events associated to this experiment. These elementary events are:

$B_1 W_1, B_1 W_2, B_1 W_3, B_2 W_1, B_2 W_2, B_2 W_3, W_1 W_2, W_1 W_3, W_2 W_3$, and $B_1 B_2$. The set of all these elementary events is the sample space associated to the experiment.

ILLUSTRATION 6 A coin is tossed. If it shows head, we draw a ball from a bag consisting of 3 red and 4 black balls; if it shows a tail, we throw a die. If we denote three red balls as R_1, R_2 and R_3 and four black balls as B_1, B_2, B_3 and B_4 . Then the elementary events associated to this experiment are:

$HR_1, HR_2, HR_3, HB_1, HB_2, HB_3, HB_4, T1, T2, T3, T4, T5$ and $T6$.

The set of these elementary events is the sample space associated to the given random experiment.

REMARK 1 Elementary events associated to a random experiment are also known as indecomposable events.

ILLUSTRATIVE EXAMPLES

LEVEL-1

EXAMPLE 1 From a group of 2 boys and 3 girls, two children are selected. Find the sample space associated to this random experiment.

SOLUTION Let the two boys be taken as B_1 and B_2 and the three girls be taken as G_1, G_2 and G_3 . Clearly, there are 5 children, out of which two children can be chosen in 5C_2 ways. So, there are ${}^5C_2 = 10$ elementary events associated to this experiments and are given by

$B_1 B_2, B_1 G_1, B_1 G_2, B_1 G_3, B_2 G_1, B_2 G_2, B_2 G_3, G_1 G_2, G_1 G_3$ and $G_2 G_3$

Consequently, the sample space S associated to this random experiment is given by

$$S = \{B_1 B_2, B_1 G_1, B_1 G_2, B_1 G_3, B_2 G_1, B_2 G_2, B_2 G_3, G_1 G_2, G_1 G_3, G_2 G_3\}.$$

EXAMPLE 2 A coin is tossed. If it shows head, we draw a ball from a bag consisting of 3 red and 4 black balls; if it shows tail, we throw a die. What is the sample associated to this experiment? **[NCERT]**

SOLUTION Let the three red balls be taken as R_1, R_2, R_3 and four black balls be taken as B_1, B_2, B_3 and B_4 .

If the coin shows head, we draw a ball which can be any one of the 7 balls. So, possible outcomes are $(H, R_1), (H, R_2), (H, R_3), (H, B_1), (H, B_2), (H, B_3), (H, B_4)$.

If the coin shows tail, then we through a die which may produce any one of the six numbers on its upper face. In this case, possible outcomes are $(T, 1), (T, 2), (T, 3), (T, 4), (T, 5), (T, 6)$.

Thus, all elementary events associated to the experiment are:

$$(H, R_1), (H, R_2), (H, R_3), (H, B_1), (H, B_2), (H, B_3), (H, B_4), (T, 1), (T, 2), (T, 3), (T, 4), (T, 5), (T, 6).$$

Consequently, the sample space S is given by

$$S = \{(H, R_1), (H, R_2), (H, R_3), (H, B_1), (H, B_2), (H, B_3), (H, B_4), (T, 1), (T, 2), (T, 3), (T, 4), (T, 5), (T, 6)\}.$$

EXAMPLE 3 An experiment consists of rolling a die and then tossing a coin once if the number on the die is even. If the number on the die is odd, the coin is tossed twice. Write the sample space for this experiment.

[NCERT]

SOLUTION If the die is rolled and we get an even number (2 or 4 or 6) on its upper face, then we toss a coin which may result in head (H) or tail (T). So the possible outcomes in this case are :

$$(2, H), (4, H), (6, H), (2, T), (4, T), (6, T)$$

If the die is rolled and we get an odd number (1 or 3 or 5) on its upper face, then the coin is tossed twice which may result in one of the following ways: HH, HT, TH, TT. So, the possible outcomes, in this case, are

$$(1, HH), (3, HH), (5, HH), (1, HT), (3, HT), (5, HT), (1, TH), (3, TH), (5, TH), (1, TT), (3, TT), (5, TT).$$

Thus, all elementary events associated to the experiment are:

$$(2, H), (4, H), (6, H), (2, T), (4, T), (6, T), (1, HH), (1, HT), (1, TH), (1, TT), (3, HH), (3, HT), (3, TH), (3, TT), (5, HH), (5, HT), (5, TH), (5, TT).$$

So, the sample space associated to the random experiment is

$$S = \{(2, H), (4, H), (6, H), (2, T), (4, T), (6, T), (1, HH), (1, HT), (1, TH), (1, TT), (3, HH), (3, HT), (3, TH), (3, TT), (5, HH), (5, HT), (5, TH), (5, TT)\}.$$

EXAMPLE 4 The numbers 1, 2, 3 and 4 are written separately on four slips of paper. The slips are then put in a box and mixed thoroughly. A person draws two slips from the box, one after the other, without replacement. Describe the sample space for the experiment.

[NCERT]

SOLUTION It is given that two slips are drawn from the box one after the other without replacement.

If the slip drawn in first draw bears number 1, then the slip drawn in second draw may bear any one of the remaining 3 numbers viz. 2, 3 and 4. Possible outcomes in this case are $(1, 2), (1, 3)$ and $(1, 4)$.

If the slip drawn in first draw bears number 2, then the slip drawn in second draw may bear any one of the remaining three numbers viz. 1, 3 and 4.

Thus, possible outcomes, in this case, are $(2, 1), (2, 3)$ and $(2, 4)$.

Similarly, possible outcomes when the slip drawn in first draw bears number 3 and 4 are respectively $(3, 1), (3, 2), (3, 4)$ and $(4, 1), (4, 2), (4, 3)$.

Thus, all elementary events associated to the random experiment are $(1, 2), (1, 3), (1, 4), (2, 1), (2, 3), (2, 4), (3, 1), (3, 2), (3, 4), (4, 1), (4, 2)$ and $(4, 3)$.

The set of all these elementary events is the required sample space.

EXAMPLE 5 A coin is tossed. If the result is a head, a die is thrown. If the die shows up an even number, the die is thrown again. What is the sample space for this experiment.

[NCERT]

SOLUTION A coin is tossed, if the outcome is tail (T). The experiment is over. If the outcome is head (H), a die is thrown and if the die shows up an odd number the experiment is stopped. Possible outcomes in this case are:

$(H, 1), (H, 3), (H, 5).$

If the die shows up an even number it is thrown again. In this case, possible outcomes are:

$(H, 2, 1), (H, 2, 2), (H, 2, 3), (H, 2, 4),$

$(H, 2, 5), (H, 2, 6)$

$(H, 4, 1), (H, 4, 2), (H, 4, 3), (H, 4, 4),$

$(H, 4, 5), (H, 4, 6)$

$(H, 6, 1), (H, 6, 2), (H, 6, 3), (H, 6, 4),$

$(H, 6, 5), (H, 6, 6)$

So, all elementary events associated to the given experiment are

$T, (H, 1), (H, 3), (H, 5), (H, 2, 1), (H, 2, 2),$

$(H, 2, 3), (H, 2, 4), (H, 2, 5), (H, 2, 6)$

$(H, 4, 1), (H, 4, 2), (H, 4, 3), (H, 4, 4),$

$(H, 4, 5), (H, 4, 6)$

$(H, 6, 1), (H, 6, 2), (H, 6, 3), (H, 6, 4),$

$(H, 6, 5), (H, 6, 6)$

The set of all these elementary events is the required sample space.

REMARK There are three stages in the above experiment. Possible outcomes at various stages can be depicted as shown in Fig. 33.1.

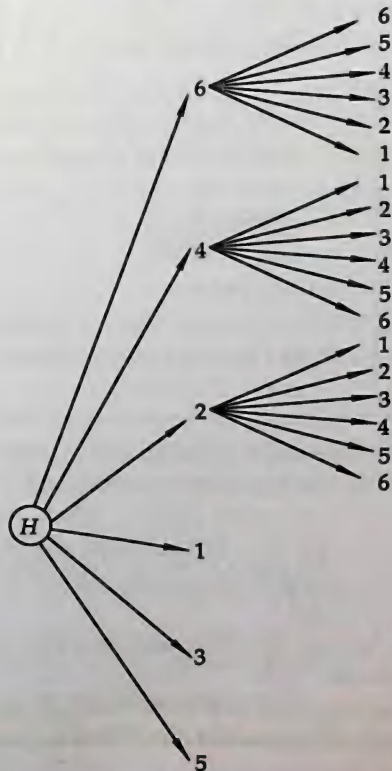


Fig. 33.1

EXAMPLE 6 A coin is tossed repeatedly until a head comes for the first time. Describe the sample space.

SOLUTION In this experiment, a coin is tossed. If the outcome is head the experiment is over. Otherwise, the coin is tossed again. In the second toss also if the outcome is head the experiment is over. Otherwise, the coin is tossed again. In the third toss, if the outcome is head the experiment is over, otherwise the coin is tossed again. This process continues indefinitely.

Possible outcomes in various tosses may be exhibited as follows:

Hence, the sample space S associated to this random experiment is

$$S = \{H, TH, TTH, TTTH, TTTTH, \dots\}$$

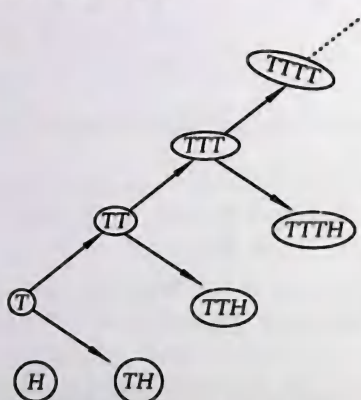


Fig. 33.2

REMARK 1 In the above example, the sample space is an infinite set.

REMARK 2 Let us consider the random experiment of drawing two cards from a well shuffled pack of 52 playing cards. There are ${}^{52}C_2 = 1326$ elementary events associated to this experiment. So, the sample space consists of 1326 elements. Clearly, it is not convenient to describe the sample space completely. In the remaining part of this chapter, we will describe the sample space associated to a given random experiment only if it is convenient and does not contain large number of elementary events.

EXERCISE 33.1

LEVEL-1

1. A coin is tossed once. Write its sample space
2. If a coin is tossed two times, describe the sample space associated to this experiment.
3. If a coin is tossed three times (or three coins are tossed together), then describe the sample space for this experiment. [NCERT]
4. Write the sample space for the experiment of tossing a coin four times. [NCERT]
5. Two dice are thrown. Describe the sample space of this experiment. [NCERT]
6. What is the total number of elementary events associated to the random experiment of throwing three dice together?
7. A coin is tossed and then a die is thrown. Describe the sample space for this experiment.
8. A coin is tossed and then a die is rolled only in case a head is shown on the coin. Describe the sample space for this experiment.
9. A coin is tossed twice. If the second throw results in a tail, a die is thrown. Describe the sample space for this experiment.
10. An experiment consists of tossing a coin and then tossing it second time if head occurs. If a tail occurs on the first toss, then a die is tossed once. Find the sample space. [NCERT]

11. A coin is tossed. If it shows tail, we draw a ball from a box which contains 2 red 3 black balls; if it shows head, we throw a die. Find the sample space of this experiment.
12. A coin is tossed repeatedly until a tail comes up for the first time. Write the sample space for this experiment.
13. A box contains 1 red and 3 black balls. Two balls are drawn at random in succession without replacement. Write the sample space for this experiment. [NCERT]
14. A pair of dice is rolled. If the outcome is a doublet, a coin is tossed. Determine the total number of elementary events associated to this experiment.
15. A coin is tossed twice. If the second draw results in a head, a die is rolled. Write the sample space for this experiment.
16. A bag contains 4 identical red balls and 3 identical black balls. The experiment consists of drawing one ball, then putting it into the bag and again drawing a ball. What are the possible outcomes of the experiment?
17. In a random sampling three items are selected from a lot. Each item is tested and classified as defective (D) or non-defective (N). Write the sample space of this experiment.
18. An experiment consists of boy-girl composition of families with 2 children.
 - (i) What is the sample space if we are interested in knowing whether it is a boy or girl in the order of their births?
 - (ii) What is the sample space if we are interested in the number of boys in a family? [NCERT]
19. There are three coloured dice of red, white and black colour. These dice are placed in a bag. One die is drawn at random from the bag and rolled, its colour and the number on its uppermost face is noted. Describe the sample space for this experiment. [NCERT]
20. 2 boys and 2 girls are in room P and 1 boy 3 girls are in room Q . Write the sample space for the experiment in which a room is selected and then a person. [NCERT]
21. A bag contains one white and one red ball. A ball is drawn from the bag. If the ball drawn is white it is replaced in the bag and again a ball is drawn. Otherwise, a die is tossed. Write the sample space for this experiment.
22. A box contains 1 white and 3 identical black balls. Two balls are drawn at random in succession without replacement. Write the sample space for this experiment.
23. An experiment consists of rolling a die and then tossing a coin once if the number on the die is even. If the number on the die is odd, the coin is tossed twice. Write the sample space for this experiment. [NCERT]
24. A die is thrown repeatedly until a six comes up. What is the sample space for this experiment. [NCERT]

ANSWERS

1. $S = \{H, T\}$
2. $S = \{HH, HT, TH, TT\}$
3. $S = \{HHH, HHT, HTH, THH, HTT, THT, TTH, TTT\}$
4. $S = \{HHHH, HHHT, HTHH, THHH, HHTH, HHTT, HTTH, TTHH, THHT, HTHT, THTH, TTTH, TTHT, THTT, HTTT, TTTT\}$
 $\{(1, 1), (1, 2), (1, 3), (1, 4), (1, 5), (1, 6), (2, 1), (2, 2), (2, 3), (2, 4), (2, 5), (2, 6)$
5. $S = \{(3, 1), (3, 2), (3, 3), (3, 4), (3, 5), (3, 6), (4, 1), (4, 2), (4, 3), (4, 4), (4, 5), (4, 6)$
 $(5, 1), (5, 2), (5, 3), (5, 4), (5, 5), (5, 6), (6, 1), (6, 2), (6, 3), (6, 4), (6, 5), (6, 6)\}$
6. 216
7. $S = \{(H, 1), (H, 2), (H, 3), (H, 4), (H, 5), (H, 6), (T, 1), (T, 2), (T, 3), (T, 4), (T, 5), (T, 6)\}$
8. $S = \{T, (H, 1), (H, 2), (H, 3), (H, 4), (H, 5), (H, 6)\}$
9. $S = \{HH, TH, (HT, 1), (HT, 2), (HT, 3), (HT, 4), (HT, 5), (HT, 6), (TT, 1), (TT, 2), (TT, 3),$
 $(TT, 4), (TT, 5), (TT, 6)\}$

10. $S = \{(T, 1), (T, 2), (T, 3), (T, 4), (T, 5), (T, 6), (H, H), (H, T)\}$
11. $S = \{(T, R_1), (T, R_2), (T, B_1), (T, B_2), (T, B_3), (H, 1), (H, 2), (H, 3), (H, 4), (H, 5), (H, 6)\}$
12. $S = \{T, HT, HHT, HHH, HHHH, \dots\}$
13. $S = \{(R, B_1), (R, B_2), (R, B_3), (B_1, R), (B_1, B_2), (B_1, B_3), (B_2, B_1), (B_2, B_3), (B_3, R), (B_3, B_1), (B_3, B_2)\}$
14. 42
15. $\{TT, HT, (TH, 1), (TH, 2), (TH, 3), (TH, 4), (TH, 5), (TH, 6), (HH, 1), (HH, 2), (HH, 3), (HH, 4), (HH, 5), (HH, 6)\}$
16. RR, RB, BR, BB
17. $S = \{DDD, DDN, DND, NDD, DNN, NDN, NND, NNN\}$
18. (i) $S = \{(B_1, B_2), (B_1, G_2), (G_1, B_2), (G_1, G_2)\}$ (ii) $S = \{0, 1, 2\}$
19. $S = \{(R, 1), (R, 2), (R, 3), (R, 4), (R, 5), (R, 6), (B, 1), (B, 2), (B, 3), (B, 4), (B, 5), (B, 6), (W, 1), (W, 2), (W, 3), (W, 4), (W, 5), (W, 6)\}$
20. $S = \{(P, B_1), (P, B_2), (P, G_1), (P, G_2), (Q, B_3), (Q, G_3), (Q, G_4), (Q, G_5)\}$
21. $S = \{(W, W), (W, R), (R, 1), (R, 2), (R, 3), (R, 4), (R, 5), (R, 6)\}$
22. $S = \{WB, BW, BB\}$
23. $S = \{(2, H), (2, T), (4, H), (4, T), (6, H), (6, T), (1, HH), (1, HT), (1, TH), (1, TT), (3, HH), (3, HT), (3, TH), (3, TT), (5, HH), (5, HT), (5, TH), (5, TT)\}$
24. $S = \{6, (1, 6), (2, 6), (3, 6), (4, 6), (5, 6), (1, 1, 6), (1, 2, 6), (1, 3, 6), (1, 4, 6), (1, 5, 6), (2, 1, 6), (2, 2, 6), (2, 3, 6), \dots\}$

33.4 EVENT

In the previous section, we have learnt about sample spaces associated with several random experiments. In this section, we will introduce an important term associated with a random experiment.

EVENT A subset of the sample space associated with a random experiment is called an event.

Consider the random experiment of throwing a die. The sample space associated with this experiment is $S = \{1, 2, 3, 4, 5, 6\}$. Clearly, S has $2^6 = 64$ subsets.

Each one of these 64 subsets is an event associated with the random experiment of throwing a die.

For Example, $A = \{2, 4, 6\}$, $B = \{1, 3, 5\}$, $C = \{3, 4, 5, 6\}$, $D = \{1, 2, 6\}$ etc. are events as they are subsets of S .

These events A , B and C can also be described in words as follows:

A = Getting an even number, B = Getting an odd number,
 C = Getting a number greater than 2

However, there is no general description in words for the event D . Thus, we find that some events associated with a random experiment may be described in words. However, it is not possible for every event.

Consider the experiment of tossing three coins at a time. The sample space S associated with this experiment is $S = \{HHH, HHT, THH, HTH, TTH, THT, HTT, TTT\}$. Let

$A = \{HHT, HTH, THH\}$, $B = \{HHH, HHT, HTH, THH\}$

$C = \{HHH, HHT, HTH, THH, TTH, HTT, THT\}$ and $D = \{HHH, TTT, HTH\}$

Clearly, A , B , C and D , being subsets of S , are events associated with the random experiment of tossing three coins (or tossing a coin three times). These events can also be described in words as follows:

A = Getting two heads, B = Number of heads exceeds the number of tails,
 C = Getting at least one head.

But, event D cannot be described in words.

REMARK Single element subsets of sample space associated with a random experiment define elementary events associated with the random experiment.

OCCURRENCE OF AN EVENT An event A associated to a random experiment is said to occur if any one of the elementary events associated to it is an outcome.

Thus, if an elementary event E is an outcome of a random experiment and A is an event such that $E \in A$, then we say that the event A has occurred.

Consider the random experiment of throwing an unbiased die. Let A be an event of getting an even number. Then, $A = \{2, 4, 6\}$. Suppose in a trial the outcome is 4. Since $4 \in A$, so we say that the event A has occurred. In another trial, let the outcome be 3, since $3 \notin A$, so we say that in this trial the event A has not occurred.

Suppose a die is thrown and the outcome of the trial is 4. Then, we can say that each of the following events have occurred:

- (i) Getting a number greater than or equal to 2, represented by the set $\{2, 3, 4, 5, 6\}$
- (ii) Getting a number less than or equal to 5, represented by the set $\{1, 2, 3, 4, 5\}$.

On the basis of the same outcome, we can also say that the following events have not occurred:

- (i) Getting an odd number represented by the set $\{1, 3, 5\}$
- (ii) Getting a multiple of 3, represented by the set $\{3, 6\}$.

Let us now consider the random experiment of throwing a pair of dice. If $(2, 6)$ is an outcome of a trial, then we can say that each of the following events has occurred:

- (i) Getting an even number on first die. (ii) Getting even numbers on both dice.
- (iii) Getting 8 as the sum of the numbers on two dice.

However, on the basis of the same outcome, one can also say that following events have not occurred:

- (i) Getting a multiple of 3 on first die. (ii) Getting an odd number on first die.
- (iii) Getting a doublet.

33.5 ALGEBRA OF EVENTS

In this section, we shall see how new events can be constructed by combining two or more events associated to a random experiment.

Let A and B be two events associated to a random experiment with sample space S . We define the event " A or B " which is said to occur if an elementary event favourable to either A or B or both is an outcome. In other words, the event " A or B " occurs if either A or B or both occur i.e. at least one of A and B occurs. Thus, " A or B " is represented by the subset $A \cup B$ of the sample space S .

For example, in a single throw of a die consider the following events:

A = Getting an even number, B = Getting a multiple of 3.

These two events are described by the sets $\{2, 4, 6\}$ and $\{3, 6\}$ respectively.

Clearly,

$A \cup B$ = Getting a number which is either even or a multiple of 3 or both = $\{2, 3, 4, 6\}$.

Similarly, if A , B and C are three events associated to a random experiment, then $A \cup B \cup C$ denotes the occurrence of at least one of the three events.

The event " A and B " is said to occur if an elementary event favourable to both A and B is an outcome. In other words, the event " A and B " occurs if A and B both occur. The event A and B is denoted by $A \cap B$.

For example, in a single throw of a pair of dice if we define

A = Getting an even number on first-die

and, $B = \text{Getting 8 as the sum of the numbers on two dice,}$

Then,

$$A \cap B = \text{Getting an even number on first die such that the sum of the numbers is 8} \\ = \{(2, 6), (6, 2), (4, 4)\}.$$

NEGATION OF AN EVENT Corresponding to every event A associated to a random experiment, we define an event "not A " which is said to occur when and only when A does not occur.

For example, in a single throw of a die if A denotes the event that the outcome is an odd number. Then $A = \{1, 3, 5\}$ and A does not occur if the outcome is any one of the outcomes 2, 4, 6. Thus, the event "not A " is represented by the set $\bar{A}$ and is called the *complementary event of A or negation of A* .

Sometimes the occurrence of one event implies the occurrence of other. For example, in a single throw of a die if A denotes the event that the outcome is 2 or 4 and B denotes the event that the outcome is even. Then, $A = \{2, 4\}$ and $B = \{2, 4, 6\}$. Clearly, the occurrence of A implies the occurrence of B . For if 2 or 4 occurs, we say that the outcome is an even number.

Thus, if the occurrence of an event A implies the occurrence of event B , then we say that " A implies B ". Clearly, if A implies B , then we have $A \subset B$.

Verbal description of the event	Equivalent set theoretic notation
Not A	$\bar{A}$
A or B (at least one of A or B)	$A \cup B$
A and B	$A \cap B$
A but not B	$A \cap \bar{B}$
Neither A nor B	$\bar{A} \cap \bar{B}$
At least one of A , B or C	$A \cup B \cup C$
Exactly one of A and B	$(A \cap \bar{B}) \cup (\bar{A} \cap B)$
All three of A , B and C	$A \cap B \cap C$
Exactly two of A , B and C	$(A \cap B \cap \bar{C}) \cup (A \cap \bar{B} \cap C) \cup (\bar{A} \cap B \cap C)$

In the above discussion and in the previous sections, we have seen that the events associated to a random experiment are generally described verbally, and it is very important to have the ability of conversion of verbal description to equivalent set theoretical notations. In the following table, we give verbal descriptions of some events and their equivalent set theoretic notations for ready reference.

ILLUSTRATION If A , B and C are three arbitrary events. Find the expression for the events noted below, in the context of A , B and C .

- | | |
|----------------------------------|---|
| (i) Only A occurs | (ii) Both A and B , but not C occur |
| (iii) All the three events occur | (iv) At least one occurs |
| (v) At least two occur | (vi) One and no more occurs |
| (vii) Two and no more occur | (viii) None occurs |
| (ix) Not more than two occur. | |

SOLUTION (i) $A \cap \bar{B} \cap \bar{C}$ (ii) $A \cap B \cap \bar{C}$ (iii) $A \cap B \cap C$ (iv) $A \cup B \cup C$
 (v) $(A \cap B) \cup (B \cap C) \cup (A \cap C) \cup (A \cap B \cap C)$
 (vi) $(A \cap \bar{B} \cap \bar{C}) \cup (\bar{A} \cap B \cap \bar{C}) \cup (\bar{A} \cap \bar{B} \cap C)$
 (vii) $(A \cap B \cap \bar{C}) \cup (\bar{A} \cap B \cap C) \cup (A \cap \bar{B} \cap C)$

$$(viii) \overline{A \cap B \cap C} = \overline{A} \cup \overline{B} \cup \overline{C}$$

$$(ix) (A \cap \overline{B} \cap \overline{C}) \cup (\overline{A} \cap B \cap \overline{C}) \cup (\overline{A} \cap \overline{B} \cap C) \cup (A \cap B \cap \overline{C}) \cup (\overline{A} \cap B \cap C) \cup (A \cap \overline{B} \cap C).$$

33.6 TYPES OF EVENTS

Let there be n elementary events associated with a random experiment. Then the corresponding sample space has n elements and hence 2^n subsets. Each subset of S is an event associated to the random experiment and the sample space is the universal set of these events. These 2^n events are divided into different types on the basis of their nature of occurrence. In this section, we shall learn about such types.

CERTAIN (OR SURE) EVENT *An event associated with a random experiment is called a certain event if it always occurs whenever the experiment is performed.*

For example, associated with the random experiment of rolling a die, the event A "Getting an even number or an odd number" is a certain event. Clearly, this event is represented by the set $\{1, 2, 3, 4, 5, 6\}$ which is the sample space of the experiment.

If S is the sample space associated with a random experiment. Then, S , being subset of itself, defines an event. Also, every outcome of the experiment is an element of S , so the event represented by S always occurs whenever we perform the experiment. Consequently, the event represented by S is a certain event.

Thus, the sample space associated with a random experiment defines a certain event.

IMPOSSIBLE EVENT *An event associated with a random experiment is called an impossible event if it never occurs whenever the experiment is performed.*

Consider the experiment of rolling a die. Let A be the event "The number turns up is divisible by 7". Clearly, none of the possible outcomes 1, 2, 3, 4, 5, 6 is divisible by 7. So, the event A cannot occur at all. In other words, there is no outcome belonging to set representing event A . So, the set A is the null set.

If S is the sample space associated with a random experiment, then the null (empty) set ϕ is a subset of S and no outcome of the experiment is a member of ϕ . So, the event represented by ϕ is an impossible event.

COMPOUND EVENT *An event associated with a random experiment is a compound event, if it is the disjoint union of two or more elementary events.*

In other words, an event having more than one sample point is called a compound event.

In fact, other than elementary events and impossible events associated with a random experiment, all events are compound events as they are obtained by combining two or more elementary events.

For example, in a single throw of an ordinary die there are, 6 elementary events and the total number of events is $2^6 = 64$. So, $2^6 - (6 + 1) = 57$ is the total number of compound events.

REMARK *If there are n elementary events associated to a random experiment, then the sample space associated to it has n elements and so there are 2^n subsets of it. Out of these 2^n subsets there are n single element subsets. These single element subsets define n elementary events and the remaining $2^n - (n + 1)$ subsets (excluding null set) define compound events. Some of these compound events can be described in words whereas for others there may not be any general description.*

If a pair of dice is thrown together, then there are 36 elementary events associated to this experiment. The sample space associated to this experiment is:

$$S = \{(1, 1), (1, 2), \dots, (1, 6), (2, 1), (2, 2), \dots, (2, 6), \dots, (6, 1), (6, 2), \dots, (6, 6)\}$$

If we define the event A as "Getting a doublet". i.e. $A = \{(1, 1), (2, 2), \dots, (6, 6)\}$

Clearly, it is a compound event obtained by combining 6 elementary events.

Similarly, the event B given by "Getting 8 as the sum" can be written as

$$B = \{(2, 6), (6, 2), (4, 4), (3, 5), (5, 3)\}$$

It is also a compound event obtained by combining 5 elementary events.

MUTUALLY EXCLUSIVE EVENTS Two or more events associated with a random experiment are said to be mutually exclusive or incompatible events if the occurrence of any one of them prevents the occurrence of all others i.e., if no two or more of them can occur simultaneously in the same trial.

Clearly, elementary events associated with a random experiment are always mutually exclusive, because elementary events are outcomes (results) of an experiment when it is performed and at a time only one outcome is possible.

Consider the random experiment of rolling a die. Let A, B, C be three events associated with the experiment as given below:

A = Getting an even number, B = Getting an odd number, C = Getting a multiple of 3.

These events in set theoretical notations are: $A = \{2, 4, 6\}$, $B = \{1, 3, 5\}$ and $C = \{3, 6\}$.

Clearly, $A \cap B = \phi$, $A \cap C \neq \phi$, $B \cap C \neq \phi$ and $A \cap B \cap C = \phi$.

So, A and B are mutually exclusive events but A and C as well as B and C are not mutually exclusive. However, A, B and C taken all the three together are mutually exclusive events.

In the experiment of throwing a pair of dice events A = Getting 8 as the sum and B = Getting an even number on first die are not mutually exclusive, because $A \cap B = \{(2, 6), (6, 2), (4, 4)\} \neq \phi$

Let two cards be drawn from a well-shuffled pack of 52 cards. Consider the following events:

A = Getting both red cards, B = Getting both black cards.

Clearly, A and B are mutually exclusive events because two cards drawn cannot be both red and black at the same time.

EXHAUSTIVE EVENTS Two or more events associated with a random experiment are exhaustive if their union is the sample space i.e. events $A_1, A_2, \dots, A_n$ associated with a random experiment with sample space S are exhaustive if $A_1 \cup A_2 \cup \dots \cup A_n = S$.

Thus, a set of events associated with a random experiment is an exhaustive set of events if one of them necessarily occurs whenever the experiment is performed.

It is evident from the above definition that all elementary events associated with a random experiment form a set of exhaustive events.

Consider the experiment of drawing a card from a well shuffled deck of playing cards. Let A be the event "card is red", B be the event "card is black." Clearly, A and B are exhaustive events because $A \cup B = S$.

In a single throw of an ordinary die, let us consider the following events:

A_1 = Getting an even number = $\{2, 4, 6\}$, A_2 = Getting an odd number = $\{1, 3, 5\}$,

A_3 = Getting a multiple of 3 = $\{3, 6\}$, A_4 = Getting a number greater than 3 = $\{4, 5, 6\}$

We observe that $A_1 \cup A_2 = S$. Also, $A_1 \cup A_2 \cup A_3 \cup A_4 = S$. But, $A_1 \cup A_3 \neq S$. So, A_1 and A_2 are exhaustive events. Also, A_1, A_2, A_3, A_4 are exhaustive events but A_1 and A_3 are not exhaustive events.

MUTUALLY EXCLUSIVE AND EXHAUSTIVE SYSTEM OF EVENTS Let S be the sample space associated with a random experiment. A set of events $A_1, A_2, \dots, A_n$ is said to form a set of mutually exclusive and exhaustive system of events if

- (i) $A_1 \cup A_2 \dots \cup A_n = S$ i.e. events $A_1, A_2, \dots, A_n$ form an exhaustive set of events.
- (ii) $A_i \cap A_j = \phi$ for $i \neq j$ i.e. events $A_1, A_2, \dots, A_n$ are mutually exclusive.

Clearly, elementary events associated with a random experiment always form a system of mutually exclusive and exhaustive events.

In a single throw of a die, the events A = Getting an even number and, B = Getting an odd number are mutually exclusive and exhaustive events.

Consider the experiment of drawing a card from a well-shuffled deck of 52 playing cards. Let A_1, A_2, A_3, A_4 be four events defined as follows:

A_1 = Card drawn is spades, A_2 = Card drawn is clubs,

A_3 = Card drawn is hearts, A_4 = Card drawn is diamonds

Since the card drawn is one of the four types of cards, so one of these events surely occurs whenever the experiment is performed. Also, if one of these events occurs, the others cannot occur. So, A_1, A_2, A_3 and A_4 form a mutually exclusive and exhaustive system of events.

Suppose a die is thrown once. Let A be the event "Getting a number greater than 3", B be the event "Getting a number less than 5". Then, $A = \{4, 5, 6\}$ and $B = \{1, 2, 3, 4\}$. Clearly, $A \cup B = S$ and $A \cap B = \{4\} \neq \phi$. So, events A and B are exhaustive but not mutually exclusive.

FAVOURABLE ELEMENTARY EVENTS Let S be the sample space associated with a random experiment and A be an event associated with the experiment. Then, elementary events belonging to A are known as favourable elementary events to the event A .

Thus, an elementary event E is favourable to an event A if the occurrence of E ensures the happening or occurrence of event A .

In a single throw of an ordinary die, let A be the event "Getting a multiple of 3". Clearly, $A = \{3, 6\}$. So, there are two elementary events favourable to A .

Consider the random experiment of throwing a pair of dice. Let A be the event "Getting 8 as the sum". Then, $A = \{(2, 6), (6, 2), (4, 4), (5, 3), (3, 5)\}$. Clearly, A occurs if any one of the elementary events $(2, 6), (6, 2), (4, 4), (5, 3)$, and $(3, 5)$ is an outcome of the experiment. So, all these elementary events are favourable to event A .

Consider a random experiment of drawing 4 cards from a well-shuffled deck of 52 playing cards. There are ${}^{52}C_4$ elementary events associated with this experiment as 4 cards can be drawn out of 52 cards in ${}^{52}C_4$ ways. Let A be the event "Getting all red cards". There are ${}^{26}C_4$ elementary events favourable to A , because 4 red cards can be chosen out of 26 red cards in ${}^{26}C_4$ ways. In this case, it is not convenient to list all favourable elementary events.

ILLUSTRATIVE EXAMPLES

LEVEL-1

EXAMPLE 1 An experiment consists of rolling die until a 2 appears.

(i) How many elements of the sample space correspond to the event that the 2 appears on the k^{th} roll of the die?

(ii) How many elements of the sample space correspond to the event that 2 appears not later than the k^{th} roll of the die?

[NCERT EXEMPLAR]

SOLUTION (i) 2 appears on k^{th} roll of the die means that each one of the first $(k-1)$ rolls have 5 outcomes $(1, 3, 4, 5, 6)$ and k^{th} roll results in 1 outcome i.e. 2.

$\therefore$ Number of elements of the sample space in the event = $5 \times 5 \times \dots \times 5 \times 1 = 5^{k-1}$
($k-1$) times

(ii) 2 appears not later than k^{th} roll means that 2 may appear in the first roll or in second roll or in third roll, ..., or in k^{th} roll.

From (i), the number of elements of the sample space corresponding to the event that 2 appears on the k^{th} roll of the die is 5^{k-1} .

Hence, The number of elements of the sample space corresponding to the event that 2 appears not later than the k^{th} roll of the die

$$= 5^{1-1} + 5^{2-1} + 5^{3-1} + \dots + 5^{k-1} = 1 + 5 + 5^2 + \dots + 5^{k-1} = \left(\frac{5^k - 1}{5 - 1} \right) = \frac{5^k - 1}{4}.$$

EXAMPLE 2 An experiment involves rolling a pair of dice and recording the numbers that come up. Describe the following events. [NCERT]

A = the sum is greater than 8, B = 2 occurs on either die., C = the sum is at least 7 and a multiple of 3.

Also, find $A \cap B$, $B \cap C$ and $A \cap C$.

Are: (i) A and B mutually exclusive? (ii) B and C mutually exclusive?

(iii) A and C mutually exclusive?

SOLUTION The sample space associated with the given random experiment is given by

$$S = \{(1, 1), (1, 2), (1, 3), (1, 4), (1, 5), (1, 6), (2, 1), (2, 2), (2, 3), (2, 4), (2, 5), (2, 6), (3, 1), (3, 2), (3, 3), (3, 4), (3, 5), (3, 6), (4, 1), (4, 2), (4, 3), (4, 4), (4, 5), (4, 6), (5, 1), (5, 2), (5, 3), (5, 4), (5, 5), (5, 6), (6, 1), (6, 2), (6, 3), (6, 4), (6, 5), (6, 6)\}$$

We have,

A = The sum is greater than 8

$$\Rightarrow A = \{(3, 6), (4, 5), (5, 4), (6, 3), (4, 6), (6, 4), (5, 5), (6, 5), (5, 6), (6, 6)\}$$

B = 2 occurs on either die

$$\Rightarrow B = \{(2, 1), (2, 2), (2, 3), (2, 4), (2, 5), (2, 6), (1, 2), (3, 2), (4, 2), (5, 2), (6, 2)\}.$$

and, C = The sum is atleast 7 and a multiple of 3

$$\Rightarrow C = \text{The sum is 9 or, 12.} = \{(3, 6), (6, 3), (4, 5), (5, 4), (6, 6)\}.$$

(i) Clearly, $A \cap B = \phi$. So, A and B are mutually exclusive events.

(ii) Clearly, $B \cap C = \phi$. So, B and C are mutually exclusive events.

(iii) Clearly, $A \cap C = \{(3, 6), (6, 3), (4, 5), (5, 4), (6, 6)\} \neq \phi$. So, A and C are not mutually exclusive events.

EXAMPLE 3 From a group of 2 boys and 3 girls, two children are selected at random. Describe the events.

(i) A = both selected children are girls. (ii) B = the selected group consists of one boy and one girl.

(iii) C = at least one boy is selected.

Which pair (s) of events is (are) mutually exclusive?

SOLUTION Let B_1, B_2 be two boys and G_1, G_2, G_3 be three girls. Then, the sample space associated with the random experiment is

$$S = \{B_1 B_2, B_1 G_1, B_1 G_2, B_1 G_3, B_2 G_1, B_2 G_2, B_2 G_3, G_1 G_2, G_1 G_3, G_2 G_3\}$$

(i) We have,

$$A = \text{Both selected children are girls} = \{G_1 G_2, G_1 G_3, G_2 G_3\}$$

(ii) We have,

B = The selected group consists of one boy and one girl.

$$\Rightarrow B = \{B_1 G_1, B_1 G_2, B_1 G_3, B_2 G_1, B_2 G_2, B_2 G_3\}$$

(iii) We have,

$$C = \text{At least one boy is selected} = \{B_1 B_2, B_1 G_1, B_1 G_2, B_1 G_3, B_2 G_1, B_2 G_2, B_2 G_3\}$$

Clearly, $A \cap B = \phi$ and $A \cap C = \phi$ So, A and B , A and C are two pairs of mutually exclusive events.

EXAMPLE 4 Two dice are thrown and the sum of the numbers which come up on the dice is noted. Let us consider the following events: [NCERT]

A = The sum is even, B = The sum is multiple of 3, C = The sum is less than 4,

D = The sum is greater than 11

Which pairs of these events are mutually exclusive?

SOLUTION The sample space associated with the random experiment is given in example 1.

We have,

A = The sum is even

= The sum is either 2 or 4 or 6, or 8 or 10 or 12

= $\{(1, 1), (2, 2), (1, 3), (1, 5), (2, 4), (2, 6), (3, 1), (3, 3), (3, 5), (4, 2), (4, 4), (4, 6), (5, 1), (5, 3), (5, 5), (6, 2), (6, 4), (6, 6)\}$

B = The sum is a multiple of 3

= The sum is either 3 or 6 or 9 or 12

= $\{(1, 2), (2, 1), (1, 5), (5, 1), (2, 4), (4, 2), (3, 3), (3, 6), (6, 3), (4, 5), (5, 4), (6, 6)\}$

C = The sum is less than 4 = (The sum is 2 or 3 = $\{(1, 1), (1, 2), (2, 1)\}$)

D = The sum is greater than 11 = The sum is 12 = $\{(6, 6)\}$

We observe that $A \cap B = \{(1, 5), (2, 4), (3, 3), (4, 2), (5, 1), (6, 6)\} \neq \phi$. So, A and B are not mutually exclusive events.

Similarly, we observe that $A \cap C \neq \phi$, $A \cap D \neq \phi$, $B \cap C \neq \phi$, $B \cap D \neq \phi$ and, $C \cap D = \phi$.

Hence, C and D are mutually exclusive events.

EXERCISE 33.2

LEVEL-1

1. A coin is tossed. Find the total number of elementary events and also the total number events associated with the random experiment.
2. List all events associated with the random experiment of tossing of two coins. How many of them are elementary events?
3. Three coins are tossed once. Describe the following events associated with this random experiment: [NCERT]

A = Getting three heads, B = Getting two heads and one tail,

C = Getting three tails, D = Getting a head on the first coin.

- (i) Which pairs of events are mutually exclusive?
 - (ii) Which events are elementary events?
 - (iii) Which events are compound events?
4. In a single throw of a die describe the following events:
 - (i) A = Getting a number less than 7
 - (ii) B = Getting a number greater than 7
 - (iii) C = Getting a multiple of 3
 - (iv) D = Getting a number less than 4
 - (v) E = Getting an even number greater than 4
 - (vi) F = Getting a number not less than 3.
 Also, find $A \cup B$, $A \cap B$, $B \cap C$, $E \cap F$, $D \cap F$ and $\bar{F}$.

5. Three coins are tossed. Describe

- (i) two events A and B which are mutually exclusive.
- (ii) three events A , B and C which are mutually exclusive and exhaustive.
- (iii) two events A and B which are not mutually exclusive.
- (iv) two events A and B which are mutually exclusive but not exhaustive.

[NCERT]

6. A die is thrown twice. Each time the number appearing on it is recorded. Describe the following events:
- (i) A = Both numbers are odd. (ii) B = Both numbers are even.
- (iii) C = sum of the numbers is less than 6
- Also, find $A \cup B$, $A \cap B$, $A \cup C$, $A \cap C$. Which pairs of events are mutually exclusive?
7. Two dice are thrown. The events A, B, C, D, E and F are described as follows:
- A = Getting an even number on the first die.
 B = Getting an odd number on the first die.
 C = Getting at most 5 as sum of the numbers on the two dice.
 D = Getting the sum of the numbers on the dice greater than 5 but less than 10.
 E = Getting at least 10 as the sum of the numbers on the dice.
 F = Getting an odd number on one of the dice.
- (i) Describe the following events: A and B , B or C , B and C , A and E , A or F , A and F
- (ii) State true or false:
- (a) A and B are mutually exclusive.
 (b) A and B are mutually exclusive and exhaustive events.
 (c) A and C are mutually exclusive events.
 (d) C and D are mutually exclusive and exhaustive events.
 (e) C, D and E are mutually exclusive and exhaustive events.
 (f) A' and B' are mutually exclusive events.
 (g) A, B, F are mutually exclusive and exhaustive events.
8. The numbers 1, 2, 3 and 4 are written separately on four slips of paper. The slips are then put in a box and mixed thoroughly. A person draws two slips from the box, one after the other, without replacement. Describe the following events:
- A = The number on the first slip is larger than the one on the second slip.
 B = The number on the second slip is greater than 2
 C = The sum of the numbers on the two slips is 6 or 7
 D = The number on the second slips is twice that on the first slip.
- Which pair(s) of events is (are) mutually exclusive?
9. A card is picked up from a deck of 52 playing cards.
- (i) What is the sample space of the experiment?
 (ii) What is the event that the chosen card is black faced card?

ANSWERS

1. 2, 4
2. $\{HH\}, \{HT\}, \{TH\}, \{TT\}, \{HH, HT\}, \{HH, TH\}, \{HH, TT\}, \{HT, TH\}, \{HT, TT\}, \{TH, TT\}, \{HH, HT, TH\}, \{HH, HT, TT\}, \{HH, TH, TT\}, \{HT, TH, TT\}, \{HH, HT, TH, TT\}$. 4
3. $A = \{HHH\}, B = \{HHT, THH, HTH\}, C = \{TTT\}, D = \{HHH, HHT, HTH, HTT\}$
- (i) $A, B; A, C; B, C; C, D$ (ii) A and C (iii) B and D
4. (i) $A = \{1, 2, 3, 4, 5, 6\}$ (ii) ϕ (iii) $C = \{3, 6\}$
- (iv) $D = \{1, 2, 3\}$ (v) $E = \{6\}$ (vi) $F = \{3, 4, 5, 6\}$
- $A \cup B = \{1, 2, 3, 4, 5, 6\}, A \cap B = \phi, B \cap C = \phi, E \cap F = \{6\}, D \cap F = \{3\}, \bar{F} = \{1, 2\}$
5. (i) A = Getting at least two heads, B = Getting at least two tails.
 (ii) A = Getting at most one head, B = Getting exactly two heads,
 C = Getting exactly three heads.
 (iii) A = Getting at most two tails, B = Getting exactly two heads
 (iv) A = Getting exactly one head, B = Getting exactly two heads.

6. (i) $A = \{(1, 1), (1, 3), (1, 5), (3, 1), (3, 3), (3, 5), (5, 1), (5, 3), (5, 5)\}$
 (ii) $B = \{(2, 2), (2, 4), (2, 6), (4, 2), (4, 4), (4, 6), (6, 2), (6, 4), (6, 6)\}$
 (iii) $C = \{(1, 1), (1, 2), (2, 1), (1, 3), (3, 1), (2, 2), (1, 4), (4, 1), (2, 3), (3, 2)\}$
 $A \cup B = \{(1, 1), (1, 3), (1, 5), (3, 1), (3, 3), (3, 5), (5, 1), (5, 3), (5, 5), (2, 2), (2, 4), (2, 6), (4, 2), (4, 4), (4, 6), (6, 2), (6, 4), (6, 6)\}$
 $A \cap B = \phi$
 $A \cup C = \{(1, 1), (1, 3), (1, 5), (3, 1), (3, 3), (3, 5), (5, 1), (5, 3), (5, 5), (1, 2), (1, 4), (2, 1), (2, 2), (2, 3), (3, 2), (4, 1)\}$
 $A \cap C = \{(1, 1), (1, 3), (3, 1)\}$ and $B \cap C = \phi$.
 A and B, B and C are pairs of mutually exclusive events.
7. (i) $A \cap B = \phi$
 $B \cup C = \{(1, 1), (1, 2), (1, 3), (1, 4), (1, 5), (1, 6), (2, 1), (2, 2), (2, 3), (3, 1), (3, 2), (3, 3), (3, 4), (3, 5), (3, 6), (4, 1), (5, 1), (5, 2), (5, 3), (5, 4), (5, 5), (5, 6)\}$
 $B \cap C = \{(1, 1), (1, 2), (1, 3), (1, 4), (3, 1), (3, 2)\}$
 $A \cap E = \{(4, 6), (6, 4), (6, 5), (6, 6)\}$
 $A \cap F = \{(1, 2), (1, 4), (1, 6), (2, 1), (2, 2), (2, 3), (2, 4), (2, 5), (2, 6), (3, 2), (3, 4), (3, 6), (4, 1), (4, 2), (4, 3), (4, 4), (4, 5), (4, 6), (5, 2), (5, 4), (5, 6), (6, 1), (6, 2), (6, 3), (6, 4), (6, 5), (6, 6)\}$
 $A \cap F = \{(2, 1), (2, 3), (2, 5), (4, 1), (4, 3), (4, 5), (6, 1), (6, 3), (6, 5)\}$
 (ii) (a) True, (b) True (c) False (d) False (e) True (f) True (g) False
8. $A = \{(2, 1), (3, 1), (3, 2), (4, 1), (4, 2), (4, 3)\}$, $B = \{(1, 3), (2, 3), (1, 4), (2, 4), (3, 4), (4, 3)\}$
 $C = \{(2, 4), (3, 4), (4, 2), (4, 3)\}$, $D = \{(1, 2), (2, 4)\}$
 A and D form a pair of mutually exclusive events.
9. (i) The sample space is the set of 52 cards.
 (ii) Required event is the set of jack, king and queen of spades and clubs.

HINTS TO NCERT & SELECTED PROBLEMS

3. We have,
 $A = \{HHH\}$, $B = \{HHT, HTH, THH\}$, $C = \{TTT\}$, $D = \{HHH, HHT, HTH, HTT\}$
 (i) We observe that $A \cap B = \phi$, $A \cap C = \phi$ but $A \cap D \neq \phi$
 So, A, B and A, C are pair of mutually exclusive events.
 Also, $B \cap C = \phi$ but $B \cap D \neq \phi$ So, B, C is a pair of mutually exclusive events.
 Finally, $C \cap D = \phi$. So, C and D form a pair of mutually exclusive events.
- (ii) Clearly, HHH and TTT may be outcomes of the random experiment of tossing three coins. So, A and C are elementary events.
- (iii) We observe that events B and D are obtained by combining more than one elementary events. So, B and D are compound events.
5. The sample space associated to the random experiment of tossing three coins is
 $S = \{HHH, HHT, THH, HTH, HTT, THT, TTH, TTT\}$
 (i) Clearly, $A = \{HHT, THH, HHT\}$ and $B = \{TTH, THT, HTT\}$ are mutually exclusive events.
- (ii) We observe that $A = \{HHH, TTT\}$, $B = \{HHT, HTH, THH\}$ and $C = \{HTT, THT, TTH\}$ are exhaustive and mutually exclusive events. Because, $A \cap B = \phi = B \cap C = C \cap A$ and $A \cup B \cup C = S$.
- (iii) We observe that the events $A = \{HHH, HHT, HTH, THH\}$ and $B = \{HHT, THH, HTH, HTT, THT, TTH, TTT\}$ are not mutually exclusive, because $A \cap B \neq \phi$.

- (iv) Events $A = \{HHT, HTH, THH\}$ $B = \{TTT, TTH, HTT, THT\}$ are mutually exclusive but not exhaustive as $A \cap B = \phi$ but $A \cup B \neq S$.

33.7 AXIOMATIC APPROACH TO PROBABILITY

The axiomatic approach to probability is deduced from the mathematical concepts laid down in the previous sections. It is based upon certain axioms. In this approach, for a given sample space associated to a random experiment, the probability is considered as a function which assigns a non-negative real number $P(A)$ to every event A . This non-negative real number is called the probability of the event A .

PROBABILITY FUNCTION Let $S = \{w_1, w_2, w_3, \dots, w_n\}$ be the sample space associated to a random experiment. Then a function P which assigns every event $A \subset S$ to a unique non-negative real number $P(A)$ is called the probability function, if the following axioms hold:

Axiom 1: $0 \leq P(w_i) \leq 1$ for all $w_i \in S$

Axiom 2: $P(S) = 1$ i.e., $P(w_1) + P(w_2) + \dots + P(w_n) = 1$.

Axiom 3: For any event $A \subset S$, $P(A) = \sum_{w_k \in A} P(w_k)$, the number $P(w_k)$ is called the probability of elementary event w_k .

Consider the experiment 'tossing a coin'. The sample space associated to this random experiment is $S = \{H, T\}$. If we assign the number $\frac{1}{2}$ to each of the outcomes (elementary events)

H and T i.e. $P(H) = \frac{1}{2}$ and $P(T) = \frac{1}{2}$. Then this assignment satisfies first two axioms i.e.

$0 \leq P(H) \leq 1$, $0 \leq P(T) \leq 1$, and $P(H) + P(T) = \frac{1}{2} + \frac{1}{2} = 1$. So, P is the probability function on S

and we can say that the probability of getting head is $\frac{1}{2}$ and the probability of getting tail is also

$\frac{1}{2}$. If we assign the number $\frac{1}{4}$ to H and $\frac{3}{4}$ to T i.e. $P(H) = \frac{1}{4}$ and $P(T) = \frac{3}{4}$. This assignment also

defines a probability function on $S = \{H, T\}$. However, if we take $P(H) = \frac{1}{8}$ and $P(T) = \frac{3}{8}$.

Then this assignment is not a probability function on $S = \{H, T\}$.

ILLUSTRATION Let $S = \{w_1, w_2, w_3, w_4, w_5, w_6\}$ be a sample space. Which of the following assignments of probability to each outcome are valid? [NCERT]

Outcomes or

Elementary events:	w_1	w_2	w_3	w_4	w_5	w_6
Probabilities: (i)	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$
(ii)	1	0	0	0	0	0
(iii)	$\frac{1}{8}$	$\frac{2}{3}$	$\frac{1}{3}$	$\frac{1}{3}$	$\frac{1}{4}$	$\frac{1}{4}$
(iv)	$\frac{1}{12}$	$\frac{1}{12}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{3}{2}$
(v)	0.1	0.2	0.3	0.4	0.5	0.6

SOLUTION (i) We have, $P(w_i) = \frac{1}{6}$, $i = 1, 2, 3, 4, 5, 6$.

$\therefore 0 \leq P(w_i) \leq 1$ for all $w_i \in S$

and, $P(w_1) + P(w_2) + \dots + P(w_6) = \frac{1}{6} + \frac{1}{6} + \frac{1}{6} + \frac{1}{6} + \frac{1}{6} + \frac{1}{6} = 1$

Hence, the assignment of the probabilities is valid.

(ii) We have,

$$P(w_1) = 1, P(w_2) = P(w_3) = P(w_4) = P(w_5) = P(w_6) = 0$$

$\therefore 0 \leq P(w_i) \leq 1$ for all $w_i \in S$

Also, $P(w_1) + P(w_2) + P(w_3) + P(w_4) + P(w_5) + P(w_6) = 1$

Hence, the assignment of the probabilities is valid.

(iii) We have,

$$P(w_5) = -\frac{1}{4} < 0 \text{ and } P(w_6) = -\frac{1}{3} < 0.$$

So, axiom 1 is not satisfied. Hence, the assignment of the probabilities is not valid.

(iv) We have, $P(w_6) = \frac{3}{2} > 1$.

So, axiom 1 is not satisfied. Hence, the assignment of the probabilities is not valid.

(v) We observe that axiom 1 is satisfied. But,

$$P(w_1) + P(w_2) + P(w_3) + P(w_4) + P(w_5) + P(w_6) = 0.1 + 0.2 + 0.3 + 0.4 + 0.5 + 0.6 = 2.1 \neq 1.$$

So, axiom 2 is not satisfied. Hence, the assignment is not valid.

Let $S = \{w_1, w_2, \dots, w_n\}$ be the sample space associated to a random experiment such that all the outcomes (elementary events) $w_1, w_2, w_3, \dots, w_n$ are equally likely to occur i.e. the chance of occurrence of each elementary event is same. i.e. $P(w_i) = p$ for all $w_i \in S$ where $0 \leq p \leq 1$.

Using axiom 2, we have

$$\sum_{i=1}^n P(w_i) = 1 \Rightarrow p + p + \dots + p = 1 \Rightarrow np = 1 \Rightarrow p = \frac{1}{n}$$

(n-times)

If A is an event such that m elementary events are favourable to A . Then,

$$P(A) = \sum_{w_k \in A} P(w_k)$$

$$\Rightarrow P(A) = \frac{1}{n} + \frac{1}{n} + \frac{1}{n} + \dots + \frac{1}{n}$$

m-times

$$\Rightarrow P(A) = \frac{m}{n} = \frac{\text{Favourable number of elementary events}}{\text{Total number of elementary events}}$$

Thus, we have the following definition of probability of an event when all the elementary events are equally likely to occur.

PROBABILITY OF AN EVENT If there are n elementary events associated with a random experiment and m of them are favourable to an event A , then the probability of happening or occurrence of A is denoted by $P(A)$ and is defined as the ratio $\frac{m}{n}$.

$$\text{Thus, } P(A) = \frac{m}{n}$$

Clearly, $0 \leq m \leq n$. Therefore,

$$0 \leq \frac{m}{n} \leq 1 \Rightarrow 0 \leq P(A) \leq 1$$

If $P(A) = 1$, then A is called certain event and A is called an impossible event, if $P(A) = 0$.

The number of elementary events which will ensure the non-occurrence of A i.e. which ensure the occurrence of $\bar{A}$ is $(n - m)$. Therefore,

$$P(\bar{A}) = \frac{n - m}{n}$$

$$\Rightarrow P(\bar{A}) = 1 - \frac{m}{n}$$

$$\Rightarrow P(\bar{A}) = 1 - P(A)$$

$$\Rightarrow P(A) + P(\bar{A}) = 1$$

The odds in favour of occurrence of the event A are defined by $m : (n - m)$ i.e. $P(A) : P(\bar{A})$ and the odds against the occurrence of A are defined by $n - m : m$ i.e. $P(\bar{A}) : P(A)$.

ILLUSTRATIVE EXAMPLES

LEVEL-1

Type I PROBLEMS BASED ON CONSTRUCTION OF SAMPLE SPACE

EXAMPLE 1 Find the probability of getting a head in a toss of an unbiased coin.

SOLUTION The sample space associated with the random experiment is $S = \{H, T\}$.

$\therefore$ Total number of elementary events = 2.

We observe that there are two elementary events viz. H, T associated to the given random experiment. Out of these two elementary events only one is favourable i.e. H .

$\therefore$ Favourable number of elementary events = 1

Hence, required probability = $\frac{1}{2}$.

EXAMPLE 2 In a simultaneous toss of two coins, find the probability of getting:

- (i) 2 heads (ii) exactly one head (iii) exactly 2 tails
(iv) exactly one tail (v) no tails.

SOLUTION The sample space associated to the given random experiment is given by

$$S = \{HH, HT, TH, TT\}$$

Clearly, there are 4 elements in S .

$\therefore$ Total number of elementary events = 4.

(i) There is only one elementary event i.e. HH favourable to the given event

So, required probability = $\frac{1}{4}$.

(ii) We observe that exactly one head can be obtained in two ways: HT or, TH .

So, favourable number of elementary events = 2.

Hence, required probability = $\frac{2}{4} = \frac{1}{2}$.

(iii) Exactly 2 tails can be obtained in one way i.e. TT . So, favourable number of elementary events = 1.

Hence, required probability = $\frac{1}{4}$.

(iv) Exactly one tail can be obtained in one of the following two ways: HT, TH

$\therefore$ Favourable number of elementary events = 2.

Hence, required probability = $\frac{2}{4} = \frac{1}{2}$.

(v) There is only one elementary event viz. HH favourable to the event "getting no tails".

So, required probability = $\frac{1}{4}$.

EXAMPLE 3 Three coins are tossed once. Find the probability of getting:

- (i) all heads (ii) at least two heads (iii) at most two heads
(iv) no heads (v) exactly one tail (vi) exactly 2 tails
(vii) a head on first coin.

SOLUTION Let S be the sample space associated with the random experiment of tossing three coins. Then, $S = \{HHH, HHT, HTH, THH, HTT, THT, TTH, TTT\}$.

Clearly, there are 8 elements in S .

$\therefore$ Total number of elementary events = 8.

(i) There is only one elementary event, namely HHH , favourable to the given event.

$\therefore$ Required probability = $\frac{1}{8}$.

(ii) At least two heads can be obtained if we obtain one of the following elementary events as an outcome: HHH, HHT, HTH, THH

$\therefore$ Favourable number of elementary events = 4.

Hence, required probability = $\frac{4}{8} = \frac{1}{2}$.

(iii) At most two heads can be obtained in any one of the following ways:

$HHT, THH, HTH, HTT, THT, TTH, TTT$

$\therefore$ Favourable number of elementary events = $\frac{7}{8}$.

(iv) "Getting no heads" means "Getting all tails". So, there is only one elementary event viz. TTT favourable to the given event.

Hence, required probability = $\frac{1}{8}$.

(v) Elementary events favourable to "Getting exactly one tail" are: HHT, THH, HTH .

$\therefore$ Favourable number of elementary events = 3

Hence, required probability = $\frac{3}{8}$.

(vi) Elementary events favourable to "Exactly 2 tails" are: HTT, THT, TTH .

$\therefore$ Favourable number of elementary events = 3

Hence, required probability = $\frac{3}{8}$.

(vii) A head on first coin can be obtained in one of the following ways: HTT, HHH, HTH, HHT .

$\therefore$ Favourable number of elementary events = 4.

Hence, required probability = $\frac{4}{8} = \frac{1}{2}$.

EXAMPLE 4 A die is thrown. Find the probability of getting:

- | | |
|---|--|
| (i) an even number | (ii) a prime number |
| (iii) a number greater than or equal to 3 | (iv) a number less than or equal to 4 |
| (v) a number more than 6 | (vi) a number less than or equal to 6. |

SOLUTION The sample space associated with the random experiment of rolling a die is given by $S = \{1, 2, 3, 4, 5, 6\}$. Clearly, there are 6 elements in S .

$\therefore$ Total number of elementary events = 6.

(i) An even number is obtained, if we obtain any one of 2, 4, 6 as an outcome.

So, favourable number of elementary events = 3.

Hence, required probability = $\frac{3}{6} = \frac{1}{2}$.

(ii) A prime number is obtained, if we get any one of 2, 3, 5 as an outcome.

So, favourable number of elementary events = 3.

Hence, required probability = $\frac{3}{6} = \frac{1}{2}$.

(iii) A number greater than or equal to 3 is obtained, if we get any one of 3, 4, 5, 6 as an outcome.

So, favourable number of elementary events = 4

$$\text{Hence, required probability} = \frac{4}{6} = \frac{2}{3}.$$

(iv) A number less than or equal to 4 is obtained, if we get any one of 1, 2, 3, 4 as an outcome.

So, favourable number of elementary events = 4

$$\text{Hence, required probability} = \frac{4}{6} = \frac{2}{3}.$$

(v) Since no face of the die is marked with a number greater than 6.

So, favourable number of elementary events = 0

$$\text{Hence, required probability} = \frac{0}{6} = 0.$$

In fact, the given event is an impossible event. So, probability of its occurrence is zero.

(vi) Since every face of a die is marked with a number less than or equal to 6.

So, favourable number of elementary events = 6.

$$\text{Hence, required probability} = \frac{6}{6} = 1.$$

In fact, the given event is a certain event. So, probability of its occurrence is 1.

EXAMPLE 5 Two dice are thrown simultaneously. Find the probability of getting:

- | | |
|---|----------------------------------|
| (i) an even number as the sum | (ii) the sum as a prime number |
| (iii) a total of at least 10 | (iv) a doublet of even number |
| (v) a multiple of 2 on one dice and a multiple of 3 on the other dice | |
| (vi) same number on both dice | (vii) a multiple of 3 as the sum |

SOLUTION When two dice are thrown together the sample space S associated with the random experiment is given by

$$S = \{(1, 1), (1, 2), \dots, (1, 6), (2, 1), (2, 2), \dots, (2, 6), (3, 1), (3, 2), \dots, (3, 6), \\ (4, 1), (4, 2), \dots, (4, 6), (5, 1), (5, 2), \dots, (5, 6), (6, 1), (6, 2), \dots, (6, 6)\}$$

Clearly, total number of elementary events = 36.

(i) Let A be the event "getting an even number as the sum" i.e., 2, 4, 6, 8, 10, 12 as the sum. Then,

$$A = \{(1, 1), (1, 3), (3, 1), (2, 2), (1, 5), (5, 1), (3, 3), (2, 4), (4, 2), (3, 5), (5, 3), (4, 4), \\ (6, 2), (2, 6), (5, 5), (6, 4), (4, 6), (6, 6)\}$$

$\therefore$ Favourable number of elementary events = 18

$$\text{So, required probability} = \frac{18}{36} = \frac{1}{2}.$$

(ii) Let A be the event "getting the sum as a prime number.. i.e., 2, 3, 5, 7, 11 as the sum. Then,

$$A = \{(1, 1), (1, 2), (2, 1), (1, 4), (4, 1), (2, 3), (3, 2), (1, 6), (6, 1), (2, 5), (5, 2), (3, 4), (4, 3), \\ (6, 5), (5, 6)\}$$

$\therefore$ Favourable number of elementary events = 15

$$\text{So, required probability} = \frac{15}{36} = \frac{5}{12}.$$

(iii) Let A be the event "getting a total of at least 10" i.e., 10, 11, 12 as the sum. Then,

$$A = \{(6, 4), (4, 6), (5, 5), (6, 5), (5, 6), (6, 6)\}$$

$\therefore$ Favourable number of elementary events = 6

So, required probability = $\frac{6}{36} = \frac{1}{6}$.

(iv) Let A be the event "getting a doublet of even number". Then, $A = \{(2, 2), (4, 4), (6, 6)\}$

$\therefore$ Favourable number of elementary events = 3

So, required probability = $\frac{3}{36} = \frac{1}{12}$.

(v) Let A be the event "getting a multiple of 2 on one die and a multiple of 3 on the other". Then,

$$A = \{(2, 3), (2, 6), (4, 3), (4, 6), (6, 3), (6, 6), (3, 2), (3, 4), (3, 6), (6, 2), (6, 4)\}$$

$\therefore$ Favourable number of elementary events = 11

So, required probability = $\frac{11}{36}$.

(vi) Let A be the event "getting the same number on both the dice". Then,

$$A = \{(1, 1), (2, 2), (3, 3), (4, 4), (5, 5), (6, 6)\}$$

$\therefore$ Favourable number of elementary events = 6

So, required probability = $\frac{6}{36} = \frac{1}{6}$.

(vii) Let A be the event "getting a multiple of 3 as the sum" 3, 6, 9, 12 as the sum. Then,

$$A = \{(1, 2), (2, 1), (1, 5), (5, 1), (2, 4), (4, 2), (3, 3), (3, 6), (6, 3), (5, 4), (4, 5), (6, 6)\}$$

$\therefore$ Favourable number of elementary events = 12

So, required probability = $\frac{12}{36} = \frac{1}{3}$.

EXAMPLE 6 A fair coin with 1 marked on one face and 6 on the other and a fair die are both tossed, find the probability that the sum of numbers that turn up is (i) 3 (ii) 12. [NCERT]

SOLUTION The sample space S associated to the given random experiment is given by

$$S = \{(1, 1), (1, 2), (1, 3), (1, 4), (1, 5), (1, 6), (6, 1), (6, 2), (6, 3), (6, 4), (6, 5), (6, 6)\}$$

$\therefore$ Total number of elementary events = 12.

(i) Let A be the event that the sum of the number is 3. Then, $A = \{(1, 2)\}$

$\therefore$ Favourable number of elementary events = 1

Hence, required probability = $P(A) = \frac{1}{12}$

(ii) Let B denote the event that the sum of the numbers is 12. Then, $B = \{(6, 6)\}$.

$\therefore$ Favourable number of elementary events = 1

Hence, required probability = $\frac{1}{12}$

EXAMPLE 7 Suppose each child born is equally likely to be a boy or a girl. Consider the family with exactly three children.

(a) List the eight elements in the sample space whose outcomes are all possible gender of three children.

(b) Write each of the following events as a set and find its probability:

(i) The event that exactly one child is a girl. (ii) The event that at least two children are girls.

(iii) The event that no child is a girl.

SOLUTION (a) All possible genders of three children are:

$$BBB, BBG, BGB, GBB, BGG, GBG, GGB, GGG$$

So the sample space S is given by $S = \{BBB, BBG, BGB, GBB, BGG, GBG, GGB, GGG\}$.

(b) (i) Let A denote the event "Exactly one child is a girl". Then, $A = \{BBG, BGB, GBB\}$

Clearly, favourable number of elementary events to A is 3.

$$\text{Hence, } P(A) = \frac{3}{8}.$$

(ii) Let B denote the event that at least two children are girls. Then, $B = \{BGG, GBG, GGB, GGG\}$

Clearly, favourable number of elementary events to B is 4.

$$\text{Hence, } P(B) = \frac{4}{8} = \frac{1}{2}$$

(iii) Let C denote the event: "No child is a girl". Then,

$C = \{BBB\}$ and favourable number of elementary events to C is 1.

$$\text{Hence, } P(C) = \frac{1}{8}.$$

EXAMPLE 8 Find the probability that a leap year, selected at random, will contain 53 Sundays.

SOLUTION In a leap year there are 366 days.

366 days = 52 weeks and 2 days.

Thus, a leap year has always 52 Sundays. The remaining 2 days can be:

- (i) Sunday and Monday, (ii) Monday and Tuesday, (iii) Tuesday and Wednesday,
- (iv) Wednesday and Thursday, (v) Thursday and Friday, (vi) Friday and Saturday,
- (vii) Saturday and Sunday.

If S is the sample space associated with this experiment, then S consists of the above seven points.

$\therefore$ Total number of elementary events = 7.

Let A be the event that a leap year has 53 Sundays. In order that a leap year, selected at random, should contain 53 Sundays, one of the 'over' days must be a Sunday. This can be in any one of the following two ways:

- (i) Sunday and Monday
- (ii) Saturday and Sunday

$\therefore$ Favourable number of elementary events = 2.

$$\text{Hence, required probability} = \frac{2}{7}.$$

EXAMPLE 9 Three dice are thrown together. Find the probability of getting a total of at least 6.

SOLUTION Since one die can be thrown in six ways to obtain any one of the six numbers marked on its six faces. Therefore, if three dice are thrown, the total number of elementary events = $6 \times 6 \times 6 = 216$.

Let A be the event of getting a total of at least 6. Then, $\bar{A}$ denotes the event of getting a total of less than 6 i.e., 3, 4, 5.

$$\therefore \bar{A} = \{(1, 1, 1), (1, 1, 2), (1, 2, 1), (2, 1, 1), (1, 1, 3), (1, 3, 1), (3, 1, 1), (1, 2, 2), (2, 1, 2), (2, 2, 1)\}$$

So, favourable number of elementary events = 10

$$\therefore P(\bar{A}) = \frac{10}{216}$$

$$\Rightarrow 1 - P(A) = \frac{10}{216} \Rightarrow P(A) = 1 - \frac{10}{216} = \frac{103}{108}.$$

EXAMPLE 10 What is the probability that a number selected from the numbers 1, 2, 3, ..., 25, is prime number, when each of the given numbers is equally likely to be selected?

SOLUTION Let S be the sample space associated with the given experiment and A be the event "selecting a prime number". Then, $S = \{1, 2, 3, \dots, 25\}$ and $A = \{2, 3, 5, 7, 11, 13, 17, 19, 23\}$

$\therefore$ Total number of elementary events = 25 and, Favourable number of elementary events = 9

Hence, required probability = $\frac{9}{25}$.

EXAMPLE 11 Tickets numbered from 1 to 20 are mixed up together and then a ticket is drawn at random. What is the probability that the ticket has a number which is a multiple of 3 or 7?

SOLUTION Let S be the sample space associated with the given random experiment and A denote the event "getting a ticket bearing a number which is a multiple of 3 or 7". Then,

$S = \{1, 2, \dots, 20\}$ and $A = \{3, 6, 9, 12, 15, 18, 7, 14\}$

$\therefore$ Total number of elementary events = 20

Favourable number of elementary events = 8

Hence, required probability = $8 / 20 = 2 / 5$

EXAMPLE 12 A coin is tossed. If head comes up, a die is thrown but if tail comes up, the coin is tossed again. Find the probability of obtaining:

(i) two tails

(ii) head and number 6

(iii) head and an even number,

SOLUTION The sample space S associated with the given random experiment is

$S = \{(H, 1), (H, 2), (H, 3), (H, 4), (H, 5), (H, 6), (T, H), (T, T)\}$

Clearly, it has 8 elements.

$\therefore$ Total number of elementary events = 8

(i) If the outcome is (T, T) , then we say that two tails are obtained.

$\therefore$ Favourable number of elementary events = 1

Hence, required probability = $\frac{1}{8}$

(ii) Head and the number 6 is obtain in only one way i.e. when the outcome is $(H, 6)$

$\therefore$ Favourable number elementary events = 1

Hence, required probability = $\frac{1}{8}$

(iii) Head and an even number can be obtained in any one of the following ways:

$(H, 2), (H, 4), (H, 6)$.

$\therefore$ Favourable number of elementary events = 3

Hence, required probability = $\frac{3}{8}$.

EXAMPLE 13 One urn contains two black balls (labelled B_1 and B_2) and one white ball. A second urn contains one black ball and two white balls (labelled W_1 and W_2). Suppose the following experiment is performed. One of the two urns is chosen at random. Next a ball is randomly chosen from the urn. Then a second ball is chosen at random from the same urn without replacing the first ball.

(i) Write the sample space showing all possible outcomes.

(ii) What is the probability that two black balls are chosen?

(iii) What is the probability that two balls of opposite colour are chosen? [NCERT EXEMPLAR]

SOLUTION (i) Let the contents of first urn be W, B_1, B_2 , and that of second urn be B, W_1, W_2 .

When two balls are drawn in succession from first urn, we may get any one of the following outcomes as an outcome:

WB1, WB2, B1, B2, B1W, B2W, B2B1

Similarly, when we draw two balls in succession from the second urn, we may obtain any one of the following as an outcome:

BW1, BW2, W1B, W1W2, W2W1, W2B

Thus, the sample space S is

$S = \{WB1, WB2, B1B2, B1W, B2W, B2B1, BW1, BW2, W1BW1W2, W2W1, W2B\}$

(ii) We obtain two black balls if the outcome is one of the following outcomes: B1B2, B2B1.

$\therefore$ Favourable number of elementary events = 2

Hence, required probability = $\frac{2}{12} = \frac{1}{6}$.

(iii) Two balls of opposite colour can be drawn in any one of the following outcomes:

WB1, WB2, B1W, B2W, BW1, BW2, W1B, W2B

$\therefore$ Favourable number of elementary events = 8

Hence, required probability = $\frac{8}{12} = \frac{2}{3}$.

NOTE Consider an experiment of drawing 2 cards from a pack of 52 cards. The sample space associated with this experiment consists of ${}^{52}C_2 = 1426$ points and therefore it is not easy to list all the elements of the sample space. So, in future we will not be writing the sample space associated with the given random experiment.

EXAMPLE 14 On her vacations Veena visits four cities A, B, C and D in a random order. What is the probability that she visits:

(i) A before B?

(ii) A before B and B before C?

(iii) A first and B last?

(iv) A either first or second?

(v) A just before B?

[NCERT]

SOLUTION Veena can visit four cities A, B, C and D in any one of the following orders:

ABCD, ABDC, ACDB, ACBD, ADBC, ADCB, BACD, BADC, BCAD, BCDA, BDAC, BDCA, CABD, CADB, CBDA, CBAD, CDAB, CDBA, DABC, DACB, DBCA, DBAC, DCAB, DCBA

$\therefore$ Total number of arrangements (orders) in which Veena can visit four cities A, B, C and D is $4! = 24$.

(i) Out of these 24 ordered arrangements Veena can visit city A before city B in the following arrangements:

ABCD, ABDC, ACDB, ACBD, ADBC, ADCB, CABD, CADB, CDAB, DABC, DACB, DCAB

So, there are 12 ways in which Veena can visit city A before city B.

$\therefore$ Required probability = $\frac{12}{24} = \frac{1}{2}$

(ii) Veena can visit A before B and B before C in any one of the following four ways:

ABCD, ABDC, DABC, ADBC

$\therefore$ Required probability = $\frac{4}{24} = \frac{1}{6}$

(iii) Veena can visit city A first and city B last in any one of the following two ways:

ACDB, ADCB

$\therefore$ Required probability = $\frac{2}{24} = \frac{1}{12}$

(iv) Veena can visit city A first in $3!$ ways or city A second in $3!$ ways.

$\therefore$ Number of ways in which Veena can visit city A either first or second $= 3! + 3! = 12$

$\therefore$ Required probability $= \frac{12}{24} = \frac{1}{2}$

(v) Taking AB together A, B, C, D can be arranged in $3!$ ways.

So, required probability $= \frac{3!}{24} = \frac{1}{4}$

EXAMPLE 15 A die has two faces each with number '1', three faces each with number '2' and one face with number '3'. If die is rolled once determine: [NCERT]

(i) $P(2)$ (ii) $P(1 \text{ or } 3)$ (iii) $P(\text{not } 3)$

SOLUTION Total number of elementary events $= 6$.

(i) Out of six faces, 3 faces are marked with number 2.

$\therefore P(2) = \frac{3}{6} = \frac{1}{2}$

(ii) Two faces are marked with number 1 and one face with number 3. Therefore, a face marked with 1 or 3 can be chosen in 3 ways.

$\therefore P(1 \text{ or } 3) = \frac{3}{6} = \frac{1}{2}$

(iii) There is only one face marked with number 3. Therefore, $P(3) = \frac{1}{6}$.

Hence, $P(\text{not } 3) = 1 - P(3) = 1 - \frac{1}{6} = \frac{5}{6}$

EXAMPLE 16 What is the probability that a randomly chosen two-digit integer is a multiple of 3?

SOLUTION 2-digit positive integers are 10, 11, 12, ..., 98, 99. These are 90 numbers out of which one number can be chosen in 90 ways.

$\therefore$ Total number of elementary events $= 90$.

Out of these 90 numbers, 30 numbers (12, 15, 18, ..., 96, 99) are multiples of 3. One number out of these 30 numbers can be chosen in 30 ways.

$\therefore$ Favourable number of elementary events $= 30$.

Hence, required probability $= \frac{30}{90} = \frac{1}{3}$.

Type II PROBLEMS BASED UPON COMBINATIONS OR SELECTIONS

EXAMPLE 17 One card is drawn from a pack of 52 cards, each of the 52 cards being equally likely to be drawn. Find the probability that the card drawn is:

(i) an ace (ii) red (iii) either red or king (iv) red and a king.

SOLUTION Out of 52 cards, one card can be drawn in ${}^{52}C_1$ ways.

$\therefore$ Total number of elementary events $= {}^{52}C_1 = 52$.

(i) There are four aces in a pack of 52 cards, out of which one ace can be drawn in 4C_1 ways.

$\therefore$ Favourable number of elementary events $= {}^4C_1 = 4$.

So, required probability $= \frac{4}{52} = \frac{1}{13}$.

(ii) There are 26 red cards, out of which one red card can be drawn in ${}^{26}C_1$ ways.

$\therefore$ Favourable number of elementary events $= {}^{26}C_1 = 26$.

So, required probability $= \frac{26}{52} = \frac{1}{2}$.

(iii) There are 26 red cards including 2 red kings and there are 2 more kings. Therefore, there are 28 cards which are either red or king, out of 28 cards, one can be drawn in ${}^{28}C_1$ ways.

$\therefore$ Favourable number of elementary events = ${}^{28}C_1 = 28$.

So, required probability = $\frac{28}{52} = \frac{7}{13}$.

(iv) There are 2 cards which are red and king i.e., red kings.

$\therefore$ Favourable number of elementary events = ${}^2C_1 = 2$

So, required probability = $\frac{2}{52} = \frac{1}{26}$.

EXAMPLE 18 An urn contains 9 red, 7 white and 4 black balls. If two balls are drawn at random, find the probability that:

(i) both the balls are red,

(ii) one ball is white

(iii) the balls are of the same colour

(iv) one is white and other red.

SOLUTION There are 20 balls in the bag out of which 2 balls can be drawn in ${}^{20}C_2$ ways.

So, total number of elementary events = ${}^{20}C_2 = 190$.

(i) There are 9 red balls out of which 2 balls can be drawn in 9C_2 ways.

$\therefore$ Favourable number of elementary events = ${}^9C_2 = 36$.

So, required probability = $\frac{36}{190} = \frac{18}{95}$.

(ii) There are 7 white balls out of which one white can be drawn in 7C_1 ways. One ball from the remaining 13 balls can be drawn in ${}^{13}C_1$ ways. Therefore, one white and one other colour ball can be drawn in ${}^7C_1 \times {}^{13}C_1$ ways.

So, favourable number of elementary events = ${}^7C_1 \times {}^{13}C_1$.

Hence, required probability = $\frac{{}^7C_1 \times {}^{13}C_1}{{}^{20}C_2} = \frac{91}{190}$.

(iii) Two balls drawn are of the same colour means that either both are red or both are white or both are black. Out of 9 red balls two red balls can be drawn in 9C_2 ways. Similarly, two white balls can be drawn from 7 white balls in 7C_2 ways and two black balls from 4 black balls in 4C_2 ways. Therefore,

The number of ways of drawing 2 balls of the same colour = ${}^9C_2 + {}^7C_2 + {}^4C_2 = 36 + 21 + 6 = 63$

$\therefore$ Favourable number of elementary events = 63.

So, required probability = $\frac{63}{190}$.

(iv) Out of 7 white balls one white ball can be drawn in 7C_1 ways and out of 9 red balls one red ball can be drawn in 9C_1 ways. Therefore,

One white and one red ball can be drawn in ${}^7C_1 \times {}^9C_1$ ways.

So, favourable number of elementary events = ${}^7C_1 \times {}^9C_1 = 63$.

So, required probability = $63/190$.

EXAMPLE 19 A box contains 10 red marbles, 20 blue marbles and 30 green marbles. Five marbles are drawn from the box, what is the probability that (i) all will be blue? (ii) at least one will be green?

[NCERT]

SOLUTION Out of 60 marbles, 5 marbles can be drawn in ${}^{60}C_5$ ways.

$\therefore$ Total number of elementary events = ${}^{60}C_5$

(i) Out of 20 blue marbles, 5 blue marbles can be chosen in ${}^{20}C_5$ ways.

∴ Favourable number of ways = ${}^{20}C_5$

Hence, required probability = $\frac{{}^{20}C_5}{{}^{60}C_5}$

(ii) Clearly,

Required probability = 1 – Probability that no ball is green

$$= 1 - \text{Probability that 5 balls drawn are red or blue.} = 1 - \frac{{}^{30}C_5}{{}^{60}C_5}$$

EXAMPLE 20 In a lottery 10,000 tickets are sold and ten equal prizes are awarded. What is the probability of not getting a prize if you buy (i) 1 ticket (ii) two tickets (iii) 10 tickets. [NCERT]

SOLUTION (i) Out of 10,000 tickets, one ticket can be chosen in ${}^{10000}C_1 = 10000$ ways.

There are 9990 tickets not containing a prize. Out of these 9990 tickets one can be chosen in ${}^{9990}C_1$ ways.

∴ Probability of not getting a prize = $\frac{9990}{10000} = \frac{999}{1000}$

(ii) Out of 10,000 tickets, two tickets can be chosen in ${}^{10000}C_2$ ways. As there are 9990 tickets without any prize. Therefore, two drawn tickets will not contain any prize, if they are chosen from the remaining 9990 tickets. This can be done in ${}^{9990}C_2$ ways.

So, required probability = $\frac{{}^{9990}C_2}{{}^{10000}C_2}$

(iii) 10 tickets can be drawn out of 10,000 tickets in ${}^{10000}C_{10}$ ways. There are 9990 tickets without any prize. Out of these tickets 10 tickets can be chosen in ${}^{9990}C_{10}$ ways. So, 10 drawn tickets will not contain any prize, if they are chosen from the remaining 9990 tickets.

Hence, required probability = $\frac{{}^{9990}C_{10}}{{}^{10000}C_{10}}$

EXAMPLE 21 The number lock of a suitcase has 4 wheels, each labelled with ten digits i.e. from 0 to 9. The lock opens with a sequence of four digits with no repeats. What is the probability of a person getting the right sequence to open the suitcase. [NCERT]

SOLUTION There are ${}^{10}C_4 \times 4! = 5040$ sequences of 4 distinct digits out of which there is only one sequence in which the lock opens.

∴ Required probability = $\frac{1}{5040}$

EXAMPLE 22 Out of 100 students, two sections of 40 and 60 students are formed. If you and your friends are among the 100 students, what is the probability that [NCERT]

(i) you both enter the same section?

(ii) you both enter the different sections?

SOLUTION Out of 100 students, two sections of 40 and 60 students can be formed in ${}^{100}C_{40} \times {}^{60}C_{60} = \frac{100!}{40!60!}$ ways.

(i) You and your friend can be in the same section in ${}^{98}C_{38} \times {}^{60}C_{60} + {}^{98}C_{58} \times {}^{40}C_{40}$

$$= \left(\frac{98!}{60!38!} + \frac{98!}{58!40!} \right) \text{ways}$$

$$\therefore \text{Probability that you and your friend enter the same section} = \frac{\frac{98!}{60! 38!} + \frac{98!}{58! 40!}}{\frac{100!}{60! 40!}}$$

$$= \frac{40 \times 39}{100 \times 99} + \frac{60 \times 59}{100 \times 99} = \frac{17}{33}$$

(ii) Required probability = 1 - Probability that you and your friend enter the same section

$$= 1 - \frac{17}{33} = \frac{16}{33}$$

EXAMPLE 23 Four cards are drawn at random from a pack of 52 playing cards. Find the probability of getting

- | | |
|---|--|
| (i) all the four cards of the same suit | (ii) all the four cards of the same number |
| (iii) one card from each suit | (iv) two red cards and two black cards |
| (v) all cards of the same colour | (vi) all face cards. |

SOLUTION Four cards can be drawn from a pack of 52 cards in ${}^{52}C_4$ ways.

So, total number of elementary events = ${}^{52}C_4$

(i) There are four suits viz. club, spade, heart and diamond, each of 13 cards. All the four cards are of the same suit means that either four cards drawn are club cards or spade cards or heart cards or diamond cards. So, the total number of ways of getting all the four cards of the same suit is ${}^{13}C_4 + {}^{13}C_4 + {}^{13}C_4 + {}^{13}C_4 = 4 ({}^{13}C_4)$

So, required probability = $\frac{4 ({}^{13}C_4)}{{}^{52}C_4} = \frac{198}{20825}$

(ii) Four cards drawn can be of the same number in any one of the following ways:

(1, 1, 1, 1), (2, 2, 2, 2), (3, 3, 3, 3), ..., (13, 13, 13, 13)

$\therefore$ Favourable number of elementary events = 13.

So, required probability = $\frac{13}{{}^{52}C_4} = \frac{13}{270725}$

(iii) There are four suits each of 13 cards. One card from each suit means that there is one diamond card, one club card, one spade card and one heart card. There are 13 diamond cards, out of which one can be selected in ${}^{13}C_1$ ways. Similarly, one club, one spade and one heart, each can be selected in ${}^{13}C_1$ ways.

$\therefore$ The number of ways of selecting 4 cards, one from each suit = ${}^{13}C_1 \times {}^{13}C_1 \times {}^{13}C_1 \times {}^{13}C_1$
 $= ({}^{13}C_1)^4$

So, required probability = $\frac{({}^{13}C_1)^4}{{}^{52}C_4} = \frac{2197}{20825}$

(iv) There are 26 red cards and 26 black cards. Out of 26 red cards, 2 cards can be drawn in ${}^{26}C_2$ ways. Similarly, 2 black cards can be drawn in ${}^{26}C_2$ ways. Therefore, 2 red and 2 black cards can be drawn in ${}^{26}C_2 \times {}^{26}C_2$ ways.

So, required probability = $\frac{{}^{26}C_2 \times {}^{26}C_2}{{}^{52}C_4}$

(v) There are two colours viz. red and black. Out of 26 red colour cards, 4 cards can be drawn in ${}^{26}C_4$ ways. 4 black cards can be drawn in ${}^{26}C_4$ ways. Therefore, 4 red or 4 black cards can be drawn in ${}^{26}C_4 + {}^{26}C_4 = 2 ({}^{26}C_4)$ ways.

$$\text{So, required probability} = \frac{2({}^{26}C_4)}{{}^{52}C_4}$$

(vi) There are 12 face cards (4 kings, 4 queens and 4 jacks). Out of these 12 face cards, 4 cards can be selected in ${}^{12}C_4$ ways.

$$\therefore \text{Favourable number of elementary events} = {}^{12}C_4$$

$$\text{So, required probability} = \frac{{}^{12}C_4}{{}^{52}C_4}$$

EXAMPLE 24 In a lottery of 50 tickets numbered 1 to 50, two tickets are drawn simultaneously. Find the probability that:

(i) both the tickets drawn have prime numbers, (ii) none of the tickets drawn has prime number,

(iii) one ticket has prime number.

SOLUTION Out of 50 tickets 2 tickets can be drawn in ${}^{50}C_2$ ways.

$$\text{So, total number of elementary events} = {}^{50}C_2 = 1225$$

(i) There are 15 prime numbers between 1 and 50 viz. 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47. Out of these 15 prime numbers 2 numbers can be selected in ${}^{15}C_2$ ways.

$$\therefore \text{Favourable number of elementary events} = {}^{15}C_2 = 105$$

$$\text{So, required probability} = \frac{105}{1225} = \frac{21}{245}$$

(ii) Number of non-primes from 1 to 50 = $50 - 15 = 35$. Out of these 35 numbers 2 can be selected in ${}^{35}C_2$ ways.

$$\therefore \text{Favourable number of elementary events} = {}^{35}C_2 = 595$$

$$\text{So, required probability} = \frac{595}{1225} = \frac{17}{35}$$

(iii) Out of 15 primes from 1 to 50, one prime number can be selected in ${}^{15}C_1$ ways. Therefore, one prime and one non-prime can be selected in ${}^{15}C_1 \times {}^{35}C_1$ ways.

$$\therefore \text{Favourable number of elementary events} = {}^{15}C_1 \times {}^{35}C_1 = 525$$

$$\text{So, required probability} = \frac{525}{1225} = \frac{3}{7}$$

EXAMPLE 25 Four persons are to be chosen at random from a group of 3 men, 2 women and 4 children. Find the probability of selecting:

(i) 1 man, 1 woman and 2 children (ii) exactly 2 children (iii) 2 women

SOLUTION There are 9 persons viz. 3 men, 2 women and 4 children. Out of these 9 persons 4 persons can be selected in ${}^9C_4 = 126$ ways.

$$\therefore \text{Total number of elementary events} = 126$$

(i) 1 man, 1 woman and 2 children can be selected in ${}^3C_1 \times {}^2C_1 \times {}^4C_2 = 36$ ways.

$$\therefore \text{Favourable number of elementary events} = 36$$

$$\text{So, required probability} = \frac{36}{126} = \frac{2}{7}$$

(ii) Exactly 2 children means: 2 children out of 4 children and 2 persons from 5 persons consisting of 3 men and 2 women. This can be done in ${}^4C_2 \times {}^5C_2$ ways.

$$\therefore \text{Favourable number of elementary events} = {}^4C_2 \times {}^5C_2 = 60$$

$$\text{So, required probability} = \frac{60}{126} = \frac{10}{21}$$

(iii) We have to select 4 persons of which 2 are women and the remaining 2 are chosen from 7 persons consisting of 3 men and 4 children. This can be done in ${}^2C_2 \times {}^7C_2$ ways.

$$\therefore \text{Favourable number of elementary events} = {}^2C_2 \times {}^7C_2 = 21$$

$$\text{So, required probability} = \frac{21}{126} = \frac{1}{6}$$

EXAMPLE 26 A box contains 10 bulbs, of which just three are defective. If a random sample of five bulbs is drawn, find the probabilities that the sample contains:

- (i) exactly one defective bulb, (ii) exactly two defective bulbs, (iii) no defective bulbs.

SOLUTION Out of 10 bulbs 5 can be chosen in ${}^{10}C_5$ ways.

$$\text{So, total number of elementary events} = {}^{10}C_5$$

(i) There are 3 defective and 7 non-defective bulbs. The number of ways of selecting one defective bulb out of 3 and 4 non-defective out of 7 is ${}^3C_1 \times {}^7C_4$.

$$\therefore \text{Favourable number of elementary events} = {}^3C_1 \times {}^7C_4$$

$$\text{So, required probability} = \frac{{}^3C_1 \times {}^7C_4}{{}^{10}C_5} = \frac{5}{12}$$

(ii) The number of ways of selecting 2 defective bulbs out of 3 defective bulbs and 3 non-defective bulbs out of 7 non defective bulbs is ${}^3C_2 \times {}^7C_3$.

$$\therefore \text{Favourable number of elementary events} = {}^3C_2 \times {}^7C_3$$

$$\text{So, required probability} = \frac{{}^3C_2 \times {}^7C_3}{{}^{10}C_5} = \frac{5}{12}$$

(iii) No defective bulbs means all non-defective bulbs. The number of ways of selecting all 5 non-defective bulbs out of 7 is 7C_5 .

$$\therefore \text{Favourable number of elementary events} = {}^7C_5$$

$$\text{So, required probability} = \frac{{}^7C_5}{{}^{10}C_5} = \frac{1}{12}$$

EXAMPLE 27 Five marbles are drawn from a bag which contains 7 blue marbles and 4 black marbles. What is the probability that: (i) all will be blue? (ii) 3 will be blue and 2 black?

SOLUTION There are $7 + 4 = 11$ marbles in the bag out of which 5 marbles can be drawn in ${}^{11}C_5$ ways.

$$\therefore \text{Total number of elementary events} = {}^{11}C_5.$$

(i) There are 7 blue marbles out of which 5 blue marbles can be drawn in 7C_5 ways.

$$\therefore \text{Favourable number of elementary events} = {}^7C_5$$

$$\text{Hence, required probability} = \frac{{}^7C_5}{{}^{11}C_5} = \frac{7!}{2!5!} \times \frac{5!6!}{11!} = \frac{1}{22}$$

(ii) Three blue out of 7 blue balls and 2 black out of 4 black balls can be drawn in ${}^7C_3 \times {}^4C_2$ ways.

$$\therefore \text{Favourable number of elementary events} = {}^7C_3 \times {}^4C_2$$

$$\text{Hence, required probability} = \frac{{}^7C_3 \times {}^4C_2}{{}^{11}C_5} = \frac{7!}{3!4!} \times \frac{4!}{2!2!} \times \frac{5! \times 6!}{11!} = \frac{5}{11}$$

EXAMPLE 28 Find the probability that when a hand of 7 cards is dealt from a well-shuffled deck of 52 cards, it contains: (i) all 4 kings (ii) exactly 3 kings (iii) at least 3 kings.

SOLUTION Out of 52 cards from a deck of 52 playing cards, 7 cards can be drawn in ${}^{52}C_7$ ways.

$\therefore$ Total number of elementary events = ${}^{52}C_7$

(i) There are 4 kings. Therefore, 4 kings out of 4 kings and 3 other cards from the remaining 48 cards can be chosen in ${}^4C_4 \times {}^{48}C_3$ ways.

$\therefore$ Favourable number of elementary events = ${}^4C_4 \times {}^{48}C_3$

$$\text{Hence, required probability} = \frac{{}^4C_4 \times {}^{48}C_3}{{}^{52}C_7} = \frac{1}{7735}$$

(ii) Three kings out of 4 kings and 4 other cards out of remaining 48 cards can be chosen in ${}^4C_3 \times {}^{48}C_4$ ways.

$\therefore$ Favourable number of elementary events = ${}^4C_3 \times {}^{48}C_4$

$$\text{Hence, required probability} = \frac{{}^4C_3 \times {}^{48}C_4}{{}^{52}C_7} = \frac{9}{1547}$$

(iii) When 7 cards are drawn from a deck of 52 playing cards, then getting at least 3 kings means: getting 3 kings and 4 other cards or getting 4 kings and 3 other cards. This can be done in ${}^4C_3 \times {}^{48}C_4 + {}^4C_4 \times {}^{48}C_3$ ways.

$\therefore$ Favourable number of elementary events = ${}^4C_3 \times {}^{48}C_4 + {}^4C_4 \times {}^{48}C_3$

$$\text{Hence, required probability} = \frac{{}^4C_3 \times {}^{48}C_4 + {}^4C_4 \times {}^{48}C_3}{{}^{52}C_7} = \frac{46}{7735}$$

EXAMPLE 29 In a single throw of three dice, determine the probability of getting

(i) a total of 5 (ii) a total of at most 5 (iii) a total of at least 5.

SOLUTION Total number of elementary events associated with the random experiment of throwing three dice simultaneously is $6 \times 6 \times 6 = 216$.

(i) A total of 5 can be obtained in one of the following ways:

(1, 1, 3), (3, 1, 1), (1, 3, 1), (2, 2, 1), (1, 2, 2), (2, 1, 2)

$\therefore$ Favourable number of elementary events = 6

$$\text{Hence, required probability} = \frac{6}{216} = \frac{1}{36}$$

(ii) A total of at most 5 can be obtained in any one of the following ways:

(1, 1, 1), (1, 1, 2), (2, 1, 1), (1, 2, 1), (1, 1, 3), (1, 3, 1), (3, 1, 1), (1, 2, 2), (2, 1, 2), (2, 2, 1)

$\therefore$ Favourable number of elementary events = 10

$$\text{Hence, required probability} = \frac{10}{216} = \frac{5}{108}$$

(iii) Let A be the event "getting a total of at least 5". Then,

$$P(A) = 1 - P(\bar{A}) = 1 - P(\text{Getting a total of at most 4})$$

A total of at most 4 can be obtained in any one of the following ways:

(1, 1, 1), (1, 1, 2), (2, 1, 1), (1, 2, 1).

So, Favourable number of elementary events to $\bar{A}$ is 4.

$$\therefore P(\bar{A}) = \frac{4}{216} \Rightarrow 1 - P(A) = \frac{4}{216} \Rightarrow P(A) = 1 - \frac{4}{216} = \frac{212}{216} = \frac{53}{54}$$

Type II PROBLEMS BASED UPON PERMUTATIONS OR ARRANGEMENTS

EXAMPLE 30 If the letters of the word ALGORITHM are arranged at random in a row what is the probability that the letters GOR must remain together as a unit? [NCERT EXEMPLAR]

SOLUTION There are 9 letters in the word ALGORITHM. These 9 letters can be arranged in a row in $9!$ ways.

$\therefore$ Total number of elementary events = $9!$

Considering GOR as one letter there are 7 letters which can be arranged in a row in $7!$ ways.

$\therefore$ Favourable number of elementary events = $7!$

Hence, required probability = $\frac{7!}{9!} = \frac{1}{72}$

EXAMPLE 31 If the letters of the word 'ATTRACTION' are written down at random, find the probability that (i) all the T's occur together (ii) no two T's occur together.

SOLUTION The total number of arrangements of the letters of the word 'ATTRACTION' is $\frac{10!}{3!2!}$

(i) Considering three T's as one letter there are 8 letters consisting of two identical A's. These 8 letters can be arranged in $\frac{8!}{2!}$ ways.

Hence, required probability = $\frac{\frac{8!}{2!}}{\frac{10!}{3!2!}} = \frac{3!8!}{10!} = \frac{1}{15}$

(ii) Other than 3 T's there are 7 letters which can be arranged in $\frac{7!}{2!}$ ways. There are 8 places, 6 between the 7 letters and one on extreme left and the other on extreme right. To separate three T's, we arrange them in these 8 places. This can be done in 8C_3 ways. Therefore,

Number of ways in which no two T's are together = $\frac{7!}{2!} \times {}^8C_3$

Hence, required probability = $\frac{\frac{7!}{2!} \times {}^8C_3}{\frac{10!}{3!2!}} = \frac{7}{15}$

EXAMPLE 32 A five digit number is formed by the digits 1, 2, 3, 4, 5 without repetition. Find the probability that the number is divisible by 4.

SOLUTION Total number of five digit numbers formed by the digits 1, 2, 3, 4, 5 is $5!$.

$\therefore$ Total number of elementary events = $5! = 120$.

We know that a number is divisible by 4 if the number formed by last two digits is divisible by 4. Therefore last two digits can be 12, 24, 32, 52 that is, last two digits can be filled in 4 ways. But corresponding to each of these ways there are $3! = 6$ ways of filling the remaining three places. Therefore the total number of five digit numbers formed by the digits 1, 2, 3, 4, 5 and divisible by 4 is $4 \times 6 = 24$

$\therefore$ Favourable number of elementary events = 24

So, required probability = $\frac{24}{120} = \frac{1}{5}$

Type III PROBLEMS BASED UPON COMBINATIONS OR SELECTIONS

EXAMPLE 33 Out of 9 outstanding students in a college, there are 4 boys and 5 girls. A team of four students is to be selected for a quiz programme. Find the probability that two are boys and two are girls.

SOLUTION Out of 9 students 4 students can be selected in 9C_4 ways.

So, total number of elementary events = 9C_4 .

There are 4 boys and 5 girls out of which 2 boys and 2 girls can be selected in ${}^4C_2 \times {}^5C_2$ ways.

So, favourable number of elementary events = ${}^4C_2 \times {}^5C_2$

Hence, required probability = $\frac{{}^4C_2 \times {}^5C_2}{{}^9C_4} = \frac{10}{21}$

EXAMPLE 34 In a lot of 12 Microwave ovens, there are 3 defective units. A person has ordered 4 of these units and since each is identically packed, the selection will be random. What is the probability that (i) all 4 units are good. (ii) exactly 3 units are good (iii) at least 2 units are good. [NCERT]

SOLUTION Out of 12 Microwave ovens, 4 can be chosen in ${}^{12}C_4$ ways.

$\therefore$ Total number of elementary events = ${}^{12}C_4$

(i) There are 9 good units out of which 4 can be chosen in 9C_4 ways.

$\therefore$ Favourable number of elementary events = 9C_4

Hence, required probability = $\frac{{}^9C_4}{{}^{12}C_4} = \frac{14}{55}$

(ii) Exactly 3 good units can be chosen in ${}^9C_3 \times {}^3C_1$ ways.

$\therefore$ Required probability = $\frac{{}^9C_3 \times {}^3C_1}{{}^{12}C_4} = \frac{28}{55}$

(iii) Required probability = $1 - P(\text{At most one unit is good}) = 1 - \frac{{}^9C_1 \times {}^3C_3}{{}^{12}C_4} = 1 - \frac{1}{55} = \frac{54}{55}$

EXAMPLE 35 In a relay race there are five teams A, B, C, D and E.

(i) What is the probability that A, B and C finish first, second and third respectively.

(ii) What is the probability that A, B and C are first three to finish (in any order). [NCERT]

SOLUTION Out of 5 teams first three positions can be occupied by 3 teams in any order in ${}^5C_3 \times 3!$ ways.

So, total number of elementary events = ${}^5C_3 \times 3! = 60$

(i) Teams A, B and C can finish first, second and third in only one way, because there is only one finishing order.

$\therefore$ Favourable number of elementary events = 1

So, required probability = $\frac{1}{60}$

(ii) Teams A, B and C finish at first three places in any order in $3!$ ways.

$\therefore$ Favourable number of elementary events = $3! = 6$

So, required probability = $\frac{6}{60} = \frac{1}{10}$

Type IV MISCELLANEOUS PROBLEMS

EXAMPLE 36 A card is drawn from an ordinary pack of 52 cards and a gambler bets that, it is a spade or an ace. What are the odds against his winning this bet?

SOLUTION Let A be the event of getting a spade or an ace from a pack of 52 cards. Then,

Total number of elementary events = ${}^{52}C_1 = 52$

Since there are 13 spade cards including an ace of spade and three aces other than an ace of spade.

$\therefore$ Favourable number of elementary events $= {}^{16}C_1 = 16$

$$\text{So, } P(A) = \frac{16}{52} = \frac{4}{13}.$$

Hence, odds against A are $P(\bar{A}) : P(A) = \frac{9}{13} : \frac{4}{13} = 9 : 4$

EXAMPLE 37 The odds in favour of an event are 3 : 5. Find the probability of occurrence of this event.

SOLUTION It is given that the odds in favour of an event are 3:5. Therefore,

Favourable number of elementary events $= 3x$

Unfavourable number of elementary events $= 5x$.

So, total number of elementary events $= 3x + 5x = 8x$.

Hence, probability of the occurrence of the event $= \frac{3x}{8x} = \frac{3}{8}$

LEVEL-2

Type V MIXED PROBLEMS ON PROBABILITY

EXAMPLE 38 A letter is chosen at random from the word 'ASSASSINATION'. Find the probability that letter is (i) a vowel (ii) a consonant. [NCERT]

SOLUTION There are 13 letters in the word 'ASSASSINATION' out of which there are 6 vowels viz., A, A, I, A, I, O and 7 consonants.

Total number of ways of selecting a word from 13 letters is $= 13$.

Number of ways of selecting a vowel out of 6 vowels $= 6$

Number of ways of selecting a consonant out of 7 consonants $= 7$.

$\therefore P(\text{Selecting a vowel}) = \frac{6}{13}$ and $P(\text{Selecting a consonant}) = \frac{7}{13}$

EXAMPLE 39 If the letters of the word ASSASSINATION are arranged at random. Find the probability that

- (i) Four S's come consecutively in the word. (ii) Two I's and two N's come together.
(iii) All A's are not coming together. (iv) No two A's are coming together.

[NCERT EXEMPLAR]

SOLUTION There are 13 letters in the word ASSASSINATION out of which there are 3A's 4S's 2I's 2N's, one O and one T. These 13 letters can be arranged in a row in $\frac{13!}{3!4!2!2!1!1!}$ ways.

(i) Considering 4S's as one letter there are 10 letters (3A's, 2I's, 2N's, one O; one T and one letter formed by 4S's). These 10 letters can be arranged in $\frac{10!}{3!2!2!1!1!1!}$ ways.

$$\therefore P(4S's \text{ come consecutively}) = \frac{\frac{10!}{3!2!2!1!1!1!}}{\frac{13!}{3!4!2!2!1!1!}} = \frac{4! \times 10!}{13!} = \frac{2}{143}$$

(ii) Two I's and two N's can be put together in $\frac{4!}{2!2!}$ ways. Considering these 4 letters as one, there are 10 letters which can be arranged in a row in $\frac{10!}{3!4!}$ ways.

$$\therefore \text{Number of arrangements in which two } I\text{'s and two } N\text{'s come together} = \frac{10!}{3!4!} \times \frac{4!}{2!2!} \\ = \frac{10!}{3!2!2!}$$

$$\text{Hence, } P(\text{Two } I\text{'s and two } N\text{'s come together}) = \frac{\frac{10!}{3!2!2!}}{\frac{13!}{3!4!2!2!}} = \frac{2}{143}$$

(iii) Considering all A's as one letter, there are 11 letters which can be arranged in a row in $\frac{11!}{4!2!2!}$ ways.

$$\therefore P(\text{All } A\text{'s come together}) = \frac{\frac{11!}{4!2!2!}}{\frac{13!}{3!4!2!2!}} = \frac{1}{26}$$

$$\text{Hence, } P(\text{All } A\text{'s are not coming together}) = 1 - \frac{1}{26} = \frac{25}{26}$$

(iv) Other than 3 A's there are 10 letters (4S's, 2I's 2N's, one O and one T). These 10 letters can be arranged in a row in $\frac{10!}{4!2!2!}$ ways. In each arrangement of these 10 letters there are 11 places which can be filled by 3 A's in ${}^{11}C_3$ ways.

$$\therefore \text{Number of arrangements in which no two } A\text{'s come together} = \frac{10!}{4!2!2!} \times {}^{11}C_3 \\ = \frac{10!}{4!2!2!} \times \frac{11!}{8!3!}$$

$$\text{Hence, } P(\text{No two } A\text{'s are coming together}) = \frac{\frac{10!}{4!2!2!} \times \frac{11!}{8!3!}}{\frac{13!}{3!4!2!2!}} = \frac{15}{26}$$

EXAMPLE 40 In a lottery, a person chooses six different natural numbers at random from 1 to 20, and if these six numbers match with the six numbers already fixed by the lottery committee, he wins the prize. What is the probability of winning the prize in the game?

SOLUTION Out of 20 numbers six numbers can be chosen in ${}^{20}C_6$ ways.

$$\therefore \text{Total number of elementary events} = {}^{20}C_6 = 38760$$

It is given that a person wins the prize if six selected numbers match with the six numbers already fixed by the committee.

$$\therefore \text{Favourable number of ways} = 1$$

$$\text{Hence, required probability} = \frac{1}{38760}$$

EXAMPLE 41 A typical PIN (personal identification number) is a sequence of any four symbols chosen from the 26 letters in the alphabet and the ten digits. If all PINs are equally likely, what is the probability that a randomly chosen PIN contains a repeated symbol?

SOLUTION It is given that a PIN is a sequence of four symbols selected from 36 (26 letters and 10 digits) symbols. Therefore,

$$\text{Total numbers of PINs} = 36 \times 36 \times 36 \times 36 = 36^4 = 1,679,616$$

Total number of PINs with distinct symbols = $36 \times 35 \times 34 \times 33 = 1,413,720$.

∴ The number of PINs that contain at least one repeated symbol = $1,679,616 - 1,413,720$
 $= 2,65,896$

Hence,

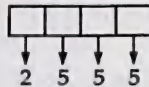
The probability that a randomly chosen PIN contains a repeated symbol = $\frac{2,65,896}{1,679,616}$
 $= 0.1583$

EXAMPLE 42 If 4-digit numbers greater than or equal to 5000 are randomly formed from the digits 0, 1, 3, 5 and 7, what is the probability of forming number divisible by 5 when

(i) the digits may be repeated (ii) the repetition of digits is not allowed.

[NCERT]

SOLUTION (i) Total number of 4-digit numbers formed from the digits 0, 1, 3, 5 and 7 and greater than or equal to 5000 is $2 \times 5 \times 5 \times 5 = 250$.



A number is divisible by 5, if units digit is 0 to 5. Therefore, number of 4 digit numbers formed from the digits 0, 1, 3, 5 and 7, divisible by 5 and greater than or equal to 5000 is $2 \times 5 \times 5 \times 2 = 100$

∴ Probability of forming a number divisible by 5 = $\frac{100}{250} = \frac{2}{5}$

(ii) If repetition of digits is not allowed, then the total number of 4 digit numbers formed from the digits 0, 1, 3, 5 and 7 is $2 \times 4 \times 3 \times 2 = 48$.

Now,

Number of 4 digit numbers divisible by 5 having 0 at one's place = $2 \times 3 \times 2 \times 1 = 12$

Number of 4 digit numbers divisible by 5 having 5 at one's place = $1 \times 3 \times 2 \times 1 = 6$

∴ Number of 4 digit numbers with distinct digits and divisible by 5 = $12 + 6 = 18$

Hence, probability of forming a number divisible by 5 = $\frac{18}{48} = \frac{3}{8}$

EXAMPLE 43 A fair coin is tossed four times, and a person wins Re 1 for each head and loses Rs. 1.50 for each tail that turns up. From the sample space calculate how many different amounts of money you can have after four tosses and the probability of having each of these amounts.

[NCERT]

SOLUTION The sample space associated with the random experiment of tossing four coins is given by

$S = \{HHHH, HHHT, HHTH, THHH, HTHH, HHTT, TTHH, HTHT, HTTH, THTH, THHT, TTHT, HTTT, THTT, TTTH, TTTT\}$

Let Rs X be the amount won by the person in a single throw of four coins.

If the person gets 4 heads in a throw of four coins, then amount won = ₹ (4×1) = ₹ 4 i.e. $X = 4$.

If the person gets 3 heads and one tail in a throw of four coins, then amount won
 $= ₹ (3 \times 1 - 1.50 \times 1) = ₹ 1.50$ i.e., $X = 1.50$

If the person gets 2 heads and 2 tails in a throw of four coins, then amount won
 $= ₹ (2 \times 1 - 1.50 \times 2) = - ₹ 1$ i.e. amount lost = -1

If the person gets 1 head and 3 tails in a throw of four coins, then amount won/lost
 $= ₹ (1 \times 1 - 3 \times 1.50) = - ₹ 3.50$. So, $X = -3.50$

If the person gets all tails in a throw of 4 coins, then amount lost
 $= ₹ (0 - 4 \times 1.50) = - ₹ 6$ i.e., $X = -6$

Now, $P(X = 4) = \text{Probability of getting all heads} = \frac{1}{16}$

$$P(X = 1.50) = \text{Probability of getting 3 heads and 1 tail} = \frac{4}{16} = \frac{1}{4}$$

$$P(X = -1) = \text{Probability of getting 2 heads and 2 tails} = \frac{6}{16} = \frac{3}{8}$$

$$P(X = -3.50) = \text{Probability of getting 1 head and 3 tails} = \frac{4}{16} = \frac{1}{4}$$

$$\text{and, } P(X = -6) = \text{Probability of getting all tails} = \frac{1}{16}$$

EXAMPLE 44 Three letters are dictated to three persons and an envelope is addressed to each of them, the letters are inserted into the envelopes at random so that each envelope contains exactly one letter. Find the probability that at least one letter is in its proper envelope. [NCERT]

SOLUTION Three letters can be inserted in three envelopes in $3! = 6$ ways.

Let us now find the number of ways of inserting 3 letters in three envelopes so that no letter is put in proper envelope.

Number of ways of inserting 2 letters in 2 envelopes so that no letter is in proper envelope = 1

$\therefore$ Number of ways of inserting 3 letters in 3 envelopes so that no letter is in proper envelope

= Total number of ways of inserting 3 letters in 3 envelopes

– Number of ways in which one letter is in proper envelope and remaining two are in wrong envelopes

– Number of ways in which all are in proper envelopes

$$= 3! - {}^3C_1 \times 1 - 1 = 6 - 3 - 1 = 2$$

$\therefore$ Probability that at least one letter is in its proper envelope

$$= 1 - \text{Probability that no letter is in its proper envelope} = 1 - \frac{2}{6} = \frac{2}{3}$$

EXAMPLE 45 A word consists of 9 letters; 5 consonants and 4 vowels. Three letters are chosen at random. What is the probability that more than one vowel will be selected?

SOLUTION Three letters can be chosen out of 9 letters in 9C_3 ways.

$\therefore$ Total number of elementary events = 9C_3

More than one vowels can be chosen in one of the following ways:

(i) 2 vowels and one consonant or, (ii) 3 vowels.

So, favourable number of elementary events = ${}^4C_2 \times {}^5C_1 + {}^4C_3$

$$\text{Hence, required probability} = \frac{{}^4C_2 \times {}^5C_1 + {}^4C_3}{{}^9C_3} = \frac{17}{42}$$

EXAMPLE 46 A bag contains 50 tickets numbered 1, 2, 3, ..., 50 of which five are drawn at random and arranged in ascending order of magnitude ($x_1 < x_2 < x_3 < x_4 < x_5$). Find the probability that $x_3 = 30$.

SOLUTION Five tickets out of 50 can be drawn in ${}^{50}C_5$ ways.

$\therefore$ Total number of elementary events = ${}^{50}C_5$

Since $x_1 < x_2 < x_3 < x_4 < x_5$ and $x_3 = 30$. Therefore, $x_1, x_2 < 30$ i.e., x_1 and x_2 should come from tickets numbered 1 to 29 and this may happen in ${}^{29}C_2$ ways. Remaining two i.e., $x_4, x_5 > 30$, should come from 20 tickets numbered from 31 to 50 in ${}^{20}C_2$ ways.

$\therefore$ Favourable number of elementary events = ${}^{29}C_2 \times {}^{20}C_2$

$$\text{Hence, required probability} = \frac{{}^{29}C_2 \times {}^{20}C_2}{{}^{50}C_5} = \frac{551}{15134}$$

EXAMPLE 47 A bag contains tickets numbered 1 to 30. Three tickets are drawn at random from the bag. What is the probability that the maximum number on the selected tickets exceeds 25?

SOLUTION It is given that the maximum number on the selected tickets exceeds 25. This means that at least one of the selected tickets should bear a number that exceeds 25. Note that the negation of 'at least one' is none and in this case it will be easier for us to find the probability that none of selected tickets bear number exceeding 25.

Let A be the event that none of the selected tickets bear number exceeding 25. Then, $\bar{A}$ denotes the event that at least one of the selected tickets bears a number that exceeds 25.

$$\therefore \text{Required probability} = P(\bar{A}) = 1 - P(A)$$

Now, we calculate $P(A)$.

The total number of ways of drawing three tickets out of 30 is ${}^{30}C_3$.

$$\therefore \text{Total number of elementary events} = {}^{30}C_3$$

Since none of the selected tickets bear number exceeding 25. Therefore, three tickets are drawn from tickets bearing number 1 to 25. This can be done in ${}^{25}C_3$ ways.

$$\therefore \text{Favourable number of elementary events} = {}^{25}C_3$$

$$\text{So, } P(A) = \frac{{}^{25}C_3}{{}^{30}C_3} = \frac{115}{203}$$

$$\text{Hence, required probability} = P(\bar{A}) = 1 - \frac{115}{203} = \frac{88}{203}$$

EXAMPLE 48 Twelve balls are distributed among three boxes, find the probability that the first box will contain three balls.

SOLUTION Since each ball can be put into any one of the three boxes. So, the total number of ways in which 12 balls can be put into three boxes is 3^{12} . Out of 12 balls, 3 balls can be chosen in ${}^{12}C_3$ ways. Put these three balls in the first box and put remaining 9 balls in the remaining two boxes, which can be done in 2^9 ways.

So, the total number of ways in 3 balls can be put in the first box and the remaining 9 in other two boxes is ${}^{12}C_3 \times 2^9$.

$$\text{Hence, required probability} = \frac{{}^{12}C_3 \times 2^9}{3^{12}}.$$

EXAMPLE 49 Find the probability that the birth days of six different persons will fall in exactly two calendar months.

SOLUTION Since each person can have his birth day in any one of the 12 calendar months. So, there are 12 options for each person.

$$\therefore \text{Total number of ways in which 6 persons can have their birth days} \\ = 12 \times 12 \times 12 \times 12 \times 12 \times 12 = 12^6$$

Out of 12 months, 2 months can be chosen in ${}^{12}C_2$ ways.

Now, birth days of six persons can fall in these two months in 2^6 ways. Out of these 2^6 ways, there are two ways when all six birth days fall in one month. So, there are $(2^6 - 2)$ ways in which six birth days fall in the chosen 2 months.

$$\therefore \text{Number of ways in which six birth days fall in exactly two calendar month} = {}^{12}C_2 \times (2^6 - 2)$$

$$\text{Hence, required probability} = \frac{{}^{12}C_2 \times (2^6 - 2)}{12^6} = \frac{341}{12^5}$$

EXAMPLE 50 Three dice are thrown simultaneously. Find the probability that:

- (i) all of them show the same face. (ii) all show distinct faces. (iii) two of them show the same face.

SOLUTION The total number of elementary events associated to the random experiment of throwing three dice simultaneously is $6 \times 6 \times 6 = 6^3$

- (i) All dice show the same face in one of the following mutually exclusive ways :

$(1, 1, 1), (2, 2, 2), (3, 3, 3), (4, 4, 4), (5, 5, 5), (6, 6, 6)$

So, favourable number of elementary events = 6.

$$\text{Hence, required probability} = \frac{6}{6^3} = \frac{1}{36}$$

- (ii) The total number of ways in which all dice show different faces is equal to the number of ways of arranging 6 distinct objects by taking three at a time i.e. ${}^6C_3 \times 3!$.

So, favourable number of elementary events = ${}^6C_3 \times 3!$

$$\text{Hence, required probability} = \frac{{}^6C_3 \times 3!}{6^3} = \frac{5}{9}$$

- (iii) Select a number which occurs on two dice out of the six numbers 1, 2, 3, 4, 5, 6 marked on the six faces of a die. This can be done in 6C_1 ways. Now, select a number from the remaining 5 numbers which occurs on the remaining one die. This can be done in 5C_1 ways. Now, we have three numbers like 1, 1, 2 ; 2, 2, 5 etc. These three digits can be arranged in $\frac{3!}{2!}$ ways.

So, the favourable number of elementary events = ${}^6C_1 \times {}^5C_1 \times \frac{3!}{2!}$

$$\text{Hence, required probability} = \frac{90}{6^3} = \frac{5}{12}$$

EXAMPLE 51 What is the probability that in a group of

- (i) 2 people, both will have the same birth-day? (ii) 3 people, at least two will have the same birth-day? assuming that there are 365 days in a year and no one has his/her birth day on 29th February.

SOLUTION (i) First person may have any one of the 365 days of the year as a birth day. Similarly, second person may have any one of 365 days of the year as a birth day.

So, the total number of ways in which two persons may have their birth days = $365 \times 365 = 365^2$

The number of ways in which two persons have the same birth-day = 365.

$$\text{Hence, required probability} = \frac{365}{365^2} = \frac{1}{365}$$

- (ii) Let A be the event "At least two people have the same birth day". Then,

$\bar{A}$ = No two or more people have the same birth day = All the three persons have distinct birth-days.

$$\therefore P(\bar{A}) = \frac{365 \times 364 \times 363}{365^3} = \frac{364 \times 363}{365^2}$$

$$\text{Hence, required probability} = 1 - P(\bar{A}) = 1 - \frac{364 \times 363}{365^2}$$

EXAMPLE 52 If n biscuits are distributed among N beggars, find the chance that a particular beggar will get r ($< n$) biscuits.

SOLUTION Since a biscuit can be given to any one of N beggars. Therefore, each biscuit can be distributed in N ways.

So, total number of ways of distributing n biscuits among N beggars = $N \times N \times \dots \times N = N^n$.
 n -times

Now, r biscuits can be given to a particular beggar in nC_r ways and the remaining $(n-r)$ biscuits can be distributed to $(N-1)$ beggars in $(N-1)^{n-r}$ ways. Thus, the number of ways in which a particular beggar receives r biscuits is ${}^nC_r \times (N-1)^{n-r}$.

$$\text{Hence, required probability} = \frac{{}^nC_r \times (N-1)^{n-r}}{N^n}$$

EXAMPLE 53 The letters of word 'SOCIETY' are placed at random in a row. What is the probability that three vowels come together?

SOLUTION There are 7 letters in the word 'Society'. These 7 letters can be arranged in a row in $7!$ ways.

$\therefore$ Total number of elementary events = $7!$

There are 3 vowels viz. O, I, E in word 'SOCIETY'. Considering these three vowels as one letter we have 5 letters which can be arranged in a row in $5!$ ways. But, three vowels O, I, E can be put together in $3!$ ways. Therefore, the total number of arrangements in which three vowels come together is $5! \times 3!$.

So, favourable number of elementary events = $5! \times 3!$

$$\text{Hence, required probability} = \frac{5! \times 3!}{7!} = \frac{1}{7}$$

EXAMPLE 54 Find the probability that in a random arrangement of the letters of the word 'UNIVERSITY' the two I's come together.

SOLUTION The total number of words which can be formed by permuting the letters of the word 'UNIVERSITY' is $\frac{10!}{2!}$ as there are two I's.

$$\therefore \text{Total number of elementary events} = \frac{10!}{2!}$$

Regarding 2I's as one letter, number of ways of arrangement in which both I's are together = $9!$

$\therefore$ Favourable number of elementary events = $9!$

$$\text{Hence, required probability} = \frac{9!}{10!/2!} = \frac{1}{5}$$

EXAMPLE 55 Six new employees, two of whom are married to each other, are to be assigned six desks that are lined up in a row. If the assignment of employees to desks is made randomly, what is the probability that the married couple will have non-adjacent desks? [NCERT EXEMPLAR]

SOLUTION Six employees can be seated in row in six desks in $6!$ ways. Married couple can occupy adjacent seats in the following 5 ways.

1-2, 2-3, 3-4, 4-5, 5-6,

Also, they can interchange their seats and the remaining 4 seats can be occupied by remaining 4 employees in $4!$ ways.

$\therefore$ Number of ways in which married couple will have adjacent seats = $5 \times 2! \times 4!$

So, Number of ways in which married couple will have non-adjacent seats = $6! - 5 \times 2! \times 4!$
 $= 480$.

$$\text{Hence, required probability} = \frac{480}{720} = \frac{2}{3}$$

EXAMPLE 56 In how many ways, can three girls and nine boys be seated in two vans, each having numbered seats, 3 in the front and 4 at the back? How many seating arrangements are possible if 3 girls should sit together in a back row on adjacent seats? Now, if all the seating arrangements are equally likely, what is the probability of 3 girls sitting together in a back row on adjacent seats?

SOLUTION Each van has 7 seats. So, there are 14 numbered seats in two vans.

The total number of ways in which 3 girls and 9 boys can sit on these seats is ${}^{14}C_{12} \times 12!$

So, total number of seating arrangements = ${}^{14}C_{12} \times 12!$

In a van 3 girls can choose adjacent seats in the back row in two ways (1, 2, 3, or 2, 3, 4). So, the number of ways in which 3 girls can sit in the back row on adjacent seats is 2 (3!) ways. The number of ways in which 9 boys can sit on the remaining 11 seats is ${}^{11}C_9 \times 9!$ ways.

So, the number of ways in which 3 girls and 9 boys can sit in two vans

$$= 2(3!) \times {}^{11}C_9 \times 9! + 2(3!) \times {}^{11}C_9 \times 9!$$

$$\text{Hence, required probability} = \frac{4 \times 3! \times {}^{11}C_9 \times 9!}{{}^{14}C_{12} \times 12!} = \frac{1}{91}$$

EXAMPLE 57 Five persons entered the lift cabin on the ground floor of an 8-floor house. Suppose that each of them independently and with equal probability can leave the cabin at any floor beginning with the first. Find out the probability of all five persons leaving at different floors.

SOLUTION Besides ground floor, there are 7 floors. Since a person can leave the cabin at any of the seven floors, therefore there are 7 ways for a person to leave the lift cabin. There are five persons in the cabin and each can leave the cabin in 7 ways. Therefore,

The total number of ways in which 5 persons can leave the cabin = $7 \times 7 \times 7 \times 7 \times 7 = 7^5$

$\therefore$ Total number of elementary events = 7^5

The total number of ways in which five persons can leave the lift cabin at different floors is same as the number of arrangements of 7 by taking 5 at a time i.e., ${}^7C_5 \times 5!$

$\therefore$ Favourable number of elementary events = ${}^7C_5 \times 5!$

$$\text{Hence, required probability} = \frac{{}^7C_5 \times 5!}{7^5}$$

EXAMPLE 58 If n persons are seated on a round table, what is the probability that two named individuals will be neighbours?

SOLUTION Total number of ways in which n persons can sit on a round table is $(n-1)!$.

$\therefore$ Total number of elementary events = $(n-1)!$

Considering two named individuals as one person there are $(n-1)$ persons who can sit on a round table in $(n-2)!$ ways. But, two named individual can be seated together in $2!$ ways.

$\therefore$ Favourable number of elementary events = $(n-2)! \times 2!$

$$\text{So, required probability} = \frac{(n-2)! \times 2!}{(n-1)!} = \frac{2}{n-1}$$

EXAMPLE 59 There are 4 letters and 4 addressed envelopes. Find the probability that all the letters are not dispatched in right envelopes.

SOLUTION Four letters can be put in four addressed envelopes in $4!$ ways.

$\therefore$ Total number of elementary events = $4!$

All four letters can be put in correct envelopes in exactly one way.

∴ Probability that all four letters are put in correct envelopes = $\frac{1}{4!}$

Hence, required probability = $1 - \frac{1}{4!} = \frac{23}{24}$

EXAMPLE 60 Each coefficient in the equation $ax^2 + bx + c = 0$ is determined by throwing an ordinary six faced die. Find the probability that the equation will have real roots.

SOLUTION Since each of the coefficients a , b and c can take the values from 1 to 6.

∴ Total number of equations = $6 \times 6 \times 6 = 216$.

The roots of the equation $ax^2 + bx + c = 0$ will be real if $b^2 - 4ac \geq 0$ i.e. $b^2 \geq 4ac$.

The favourable number of elementary events can be enumerated as follows:

ac	a	c	$4ac$	b (so that $b^2 \geq 4ac$)	No. of ways
1	1	1	4	2, 3, 4, 5, 6	$1 \times 5 = 5$
2	$\begin{Bmatrix} 1 \\ 2 \end{Bmatrix}$	$\begin{Bmatrix} 2 \\ 1 \end{Bmatrix}$	8	3, 4, 5, 6	$2 \times 4 = 8$
3	$\begin{Bmatrix} 1 \\ 3 \end{Bmatrix}$	$\begin{Bmatrix} 3 \\ 1 \end{Bmatrix}$	12	4, 5, 6	$2 \times 3 = 6$
4	$\begin{Bmatrix} 1 \\ 4 \\ 2 \end{Bmatrix}$	$\begin{Bmatrix} 4 \\ 1 \\ 2 \end{Bmatrix}$	16	4, 5, 6	$3 \times 3 = 9$
5	$\begin{Bmatrix} 1 \\ 5 \end{Bmatrix}$	$\begin{Bmatrix} 5 \\ 1 \end{Bmatrix}$	20	5, 6	$2 \times 2 = 4$
6	$\begin{Bmatrix} 1 \\ 6 \\ 2 \\ 3 \end{Bmatrix}$	$\begin{Bmatrix} 6 \\ 1 \\ 3 \\ 2 \end{Bmatrix}$	24	5, 6	$4 \times 2 = 8$
7	ac is not possible				0
8	$\begin{Bmatrix} 2 \\ 4 \end{Bmatrix}$	$\begin{Bmatrix} 4 \\ 2 \end{Bmatrix}$	32	6	$2 \times 1 = 2$
9	3	3	36	6	1
					Total = 43

Since $b^2 \geq 4ac$ and since the maximum value of b^2 is 36, therefore $ac = 10, 11, 12 \dots$ etc. is not possible.

∴ Total number of favourable elementary events = 43.

Hence, required probability = $\frac{43}{216}$.

EXAMPLE 61 Two numbers b and c are chosen at random with replacement from the numbers 1, 2, 3, 4, 5, 6, 7, 8 and 9. Find the probability that $x^2 + bx + c > 0$ for all $x \in R$.

SOLUTION Since b and c both can assume value from 1 to 9. So, total numbers of ways of choosing b and c is $9 \times 9 = 81$.

Now, $x^2 + bx + c > 0$ for all $x \in R$

$\Rightarrow$ Disc < 0 i.e. $b^2 - 4c < 0$

The following table shows the possible values of b and c for which $b^2 - 4c < 0$

c	b	Total
1	1	1
2	1, 2	2
3	1, 2, 3	3
4	1, 2, 3	3
5	1, 2, 3, 4	4
6	1, 2, 3, 4	4
7	1, 2, 3, 4, 5	5
8	1, 2, 3, 4, 5	5
9	1, 2, 3, 4, 5	5
		32

So, favourable number of favourable elementary events = 32

Hence, required probability = $32/81$.

EXAMPLE 62 Three squares of Chess board are selected at random. Find the probability of getting 2 squares of one colour and other of a different colour. [NCERT EXEMPLAR]

SOLUTION In a Chess board, there are 64 squares of which 32 are white and 32 are black. Out of 64 squares 3 square can be chosen in ${}^{64}C_3$. Since 2 of one colour and 1 ways of other colour can be 2W, 1B or 1W, 2B. Therefore, the number of ways of selecting 2 squares of one colour and one other colour is ${}^{32}C_2 \times {}^{32}C_1 + {}^{32}C_1 \times {}^{32}C_2 = 2({}^{32}C_2 \times {}^{32}C_1)$

$\therefore$ Favourable number of ways = $2({}^{32}C_2 \times {}^{32}C_1)$

Hence, required probability = $\frac{2({}^{32}C_2 \times {}^{32}C_1)}{{}^{64}C_3} = \frac{16}{21}$.

EXERCISE 33.3

LEVEL-1

1. Which of the following cannot be valid assignment of probability for elementary events or outcomes of sample space $S = \{w_1, w_2, w_3, w_4, w_5, w_6, w_7\}$:

Elementary events:	w_1	w_2	w_3	w_4	w_5	w_6	w_7
(i)	0.1	0.01	0.05	0.03	0.01	0.2	0.6
(ii)	$\frac{1}{7}$	$\frac{1}{7}$	$\frac{1}{7}$	$\frac{1}{7}$	$\frac{1}{7}$	$\frac{1}{7}$	$\frac{1}{7}$
(iii)	0.7	0.6	0.5	0.4	0.3	0.2	0.1
(iv)	$\frac{1}{14}$	$\frac{2}{14}$	$\frac{3}{14}$	$\frac{4}{14}$	$\frac{5}{14}$	$\frac{6}{14}$	$\frac{15}{14}$

2. A die is thrown. Find the probability of getting:
- (i) a prime number
 - (ii) 2 or 4
 - (iii) a multiple of 2 or 3.
3. In a simultaneous throw of a pair of dice, find the probability of getting:
- (i) 8 as the sum
 - (ii) a doublet
 - (iii) a doublet of prime numbers
 - (iv) a doublet of odd numbers
 - (v) a sum greater than 9
 - (vi) an even number on first
 - (vii) an even number on one and a multiple of 3 on the other
 - (viii) neither 9 nor 11 as the sum of the numbers on the faces
 - (ix) a sum less than 6
 - (x) a sum less than 7
 - (xi) a sum more than 7
 - (xii) neither a doublet nor a total of 10

- (xiii) odd number on the first and 6 on the second
(xiv) a number greater than 4 on each die
(xv) a total of 9 or 11 (xvi) a total greater than 8.
4. In a single throw of three dice, find the probability of getting a total of 17 or 18.
5. Three coins are tossed together. Find the probability of getting:
(i) exactly two heads (ii) at least two heads
(iii) at least one head and one tail.
6. What is the probability that an ordinary year has 53 Sundays?
7. What is the probability that a leap year has 53 Sundays and 53 Mondays?
8. A bag contains 8 red and 5 white balls. Three balls are drawn at random. Find the probability that:
(i) All the three balls are white. (ii) All the three balls are red.
(iii) One ball is red and two balls are white. [NCERT EXEMPLAR]
9. In a single throw of three dice, find the probability of getting the same number on all the three dice.
10. Two unbiased dice are thrown. Find the probability that the total of the numbers on the dice is greater than 10.
11. A card is drawn at random from a pack of 52 cards. Find the probability that the card drawn is:
(i) a black king (ii) either a black card or a king
(iii) black and a king (iv) a jack, queen or a king
(v) neither a heart nor a king (vi) spade or an ace
(vii) neither an ace nor a king (viii) a diamond card
(ix) not a diamond card (x) a black card
(xi) not an ace (xii) not a black card.
12. In shuffling a pack of 52 playing cards, four are accidentally dropped; find the chance that the missing cards should be one from each suit.
13. From a deck of 52 cards, four cards are drawn simultaneously, find the chance that they will be the four honours of the same suit.
14. Tickets numbered from 1 to 20 are mixed up together and then a ticket is drawn at random. What is the probability that the ticket has a number which is a multiple of 3 or 7?
15. A bag contains 6 red, 4 white and 8 blue balls. If three balls are drawn at random, find the probability that one is red, one is white and one is blue.
16. A bag contains 7 white, 5 black and 4 red balls. If two balls are drawn at random, find the probability that: (i) both the balls are white (ii) one ball is black and the other red (iii) both the balls are of the same colour.
17. A bag contains 6 red, 4 white and 8 blue balls. If three balls are drawn at random, find the probability that: (i) one is red and two are white (ii) two are blue and one is red (iii) one is red.
18. Five cards are drawn from a pack of 52 cards. What is the chance that these 5 will contain:
(i) just one ace (ii) at least one ace?
19. The face cards are removed from a full pack. Out of the remaining 40 cards, 4 are drawn at random. What is the probability that they belong to different suits?
20. There are four men and six women on the city councils. If one council member is selected for a committee at random, how likely is that it is a women? [NCERT]

21. A box contains 100 bulbs, 20 of which are defective. 10 bulbs are selected for inspection. Find the probability that: (i) all 10 are defective (ii) all 10 are good (iii) at least one is defective (iv) none is defective
22. Find the probability that in a random arrangement of the letters of the word 'SOCIAL' vowels come together.
23. The letters of the word 'CLIFTON' are placed at random in a row. What is the chance that two vowels come together?
24. The letters of the word 'FORTUNATES' are arranged at random in a row. What is the chance that the two 'T' come together.
25. A committee of two persons is selected from two men and two women. What is the probability that the committee will have (i) no man ? (ii) one man? (iii) two men?

[NCERT]

26. If odds in favour of an event be 2 : 3, find the probability of occurrence of this event.
27. If odds against an event be 7 : 9, find the probability of non-occurrence of this event.
28. Two balls are drawn at random from a bag containing 2 white, 3 red, 5 green and 4 black balls, one by one without, replacement. Find the probability that both the balls are of different colours.
29. Two unbiased dice are thrown. Find the probability that:
 - (i) neither a doublet nor a total of 8 will appear
 - (ii) the sum of the numbers obtained on the two dice is neither a multiple of 2 nor a multiple of 3
30. A bag contains 8 red, 3 white and 9 blue balls. If three balls are drawn at random, determine the probability that (i) all the three balls are blue balls (ii) all the balls are of different colours.
31. A bag contains 5 red, 6 white and 7 black balls. Two balls are drawn at random. What is the probability that both balls are red or both are black?
32. If a letter is chosen at random from the English alphabet, find the probability that the letter is (i) a vowel (ii) a consonant
33. In a lottery, a person chooses six different numbers at random from 1 to 20, and if these six numbers match with six numbers already fixed by the lottery committee, he wins the prize. What is the probability of winning the prize in the game? [NCERT]
34. 20 cards are numbered from 1 to 20. One card is drawn at random. What is the probability that the number on the cards is: (i) a multiple of 4? (ii) not a multiple of 4? (iii) odd? (iv) greater than 12? (v) divisible by 5? (vi) not a multiple of 6?
35. Two dice are thrown. Find the odds in favour of getting the sum
 - (i) 4
 - (ii) 5
 - (iii) What are the odds against getting the sum 6?
36. What are the odds in favour of getting a spade if a card is drawn from a well-shuffled deck of cards? What are the odds in favour of getting a king?
37. A box contains 10 red marbles, 20 blue marbles and 30 green marbles. 5 marbles are drawn at random. From the box, what is the probability that: (i) all are blue? (ii) at least one is green?
38. A box contains 6 red marbles numbered 1 through 6 and 4 white marbles numbered from 12 through 15. Find the probability that a marble drawn is (i) white (ii) white and odd numbered (iii) even numbered (iv) red or even numbered.
39. A class consists of 10 boys and 8 girls. Three students are selected at random. What is the probability that the selected group has (i) all boys? (ii) all girls? (iii) 1 boy and 2 girls? (iv) at least one girl? (v) at most one girl?

40. Five cards are drawn from a well-shuffled pack of 52 cards. Find the probability that all the five cards are hearts.
41. A bag contains tickets numbered from 1 to 20. Two tickets are drawn. Find the probability that (i) both the tickets have prime numbers on them (ii) on one there is a prime number and on the other there is a multiple of 4.
42. An urn contains 7 white, 5 black and 3 red balls. Two balls are drawn at random. Find the probability that (i) both the balls are red (ii) one ball is red and the other is black (iii) one ball is white.

LEVEL-2

43. A and B throw a pair of dice. If A throws 9, find B's chance of throwing a higher number.
44. In a hand at Whist, what is the probability that four kings are held by a specified player?
45. Find the probability that in a random arrangement of the letters of the word 'UNIVERSITY', the two I's do not come together.

ANSWERS

1. (i), (ii) $\frac{1}{2}$ (iii) $\frac{2}{3}$ 3. (i) $\frac{5}{36}$ (ii) $\frac{1}{6}$
- (iii) $\frac{1}{12}$ (iv) $\frac{1}{12}$ (v) $\frac{1}{6}$ (vi) $\frac{1}{2}$ (vii) $\frac{11}{36}$ (viii) $\frac{5}{6}$
- (ix) $\frac{5}{18}$ (x) $\frac{5}{12}$ (xi) $\frac{5}{12}$ (xii) $\frac{7}{9}$ (xiii) $\frac{1}{12}$ (xiv) $\frac{1}{9}$
- (xv) $\frac{1}{6}$ (xvi) $\frac{5}{18}$ 4. $\frac{1}{54}$ 5. (i) $\frac{3}{8}$ (ii) $\frac{1}{2}$ (iii) $\frac{3}{4}$
6. $\frac{1}{7}$ 7. $\frac{1}{7}$ 8. (i) $\frac{5}{143}$ (ii) $\frac{28}{143}$ (iii) $\frac{40}{143}$ 9. $\frac{1}{36}$ 10. $\frac{1}{12}$
11. (i) $\frac{1}{26}$ (ii) $\frac{7}{13}$ (iii) $\frac{1}{26}$ (iv) $\frac{3}{13}$ (v) $\frac{9}{13}$ (vi) $\frac{4}{13}$
- (vii) $\frac{11}{13}$ (viii) $\frac{1}{4}$ (ix) $\frac{3}{4}$ (x) $\frac{1}{2}$ (xi) $\frac{12}{13}$ (xii) $\frac{1}{2}$
12. $\frac{2197}{20825}$ 13. $\frac{4}{270725}$ 14. $\frac{2}{5}$ 15. $\frac{4}{17}$
16. (i) $\frac{7}{40}$ (ii) $\frac{1}{6}$ (iii) $\frac{37}{120}$ 17. (i) $\frac{3}{68}$ (ii) $\frac{7}{34}$ (iii) $\frac{33}{68}$
18. (i) $\frac{3243}{10829}$ (ii) $\frac{18472}{54145}$ 19. $\frac{1000}{9139}$ 20. $\frac{3}{5}$ 21. (i) $\frac{{}^{20}C_{10}}{{}^{100}C_{10}}$
- (ii) $\frac{{}^{80}C_{10}}{{}^{100}C_{10}}$ (iii) $1 - \frac{{}^{80}C_{10}}{{}^{100}C_{10}}$ (iv) $\frac{{}^{80}C_{10}}{{}^{100}C_{10}}$ 22. $\frac{1}{5}$ 23. $\frac{2}{7}$
24. $\frac{1}{5}$ 25. (i) $\frac{1}{6}$ (ii) $\frac{2}{3}$ (iii) $\frac{1}{6}$ 26. $\frac{2}{5}$ 27. $\frac{7}{16}$ 28. 0.78 29. (i) $\frac{13}{18}$
- (ii) $\frac{1}{3}$ 30. (i) $\frac{7}{95}$ (ii) $\frac{18}{95}$ 31. $\frac{31}{153}$ 32. (i) $\frac{5}{26}$ (ii) $\frac{21}{26}$
33. $\frac{1}{38760}$ 34. (i) $\frac{1}{4}$ (ii) $\frac{3}{4}$ (iii) $\frac{1}{2}$ (iv) $\frac{2}{5}$ (v) $\frac{1}{5}$
- (vi) $\frac{17}{20}$ 35. (i) 1 : 11 (ii) 1 : 8 (iii) 31 : 5 36. (i) 1 : 3, 1 : 12
37. (i) $\frac{{}^{20}C_5 \times {}^{40}C_0}{{}^{60}C_5} = \frac{34}{11977}$ (ii) $\frac{4367}{4484}$ 38. (i) $\frac{2}{5}$ (ii) $\frac{1}{5}$ (iii) $\frac{1}{2}$

- (iv) $\frac{4}{5}$ 39. (i) $\frac{5}{34}$ (ii) $\frac{7}{102}$ (iii) $\frac{35}{102}$ (iv) $\frac{29}{34}$ (v) $\frac{10}{17}$
40. $\frac{{}^{13}C_5}{{}^{52}C_5} = \frac{33}{66640}$ 41. (i) $\frac{14}{95}$ (ii) $\frac{4}{19}$ 42. (i) $\frac{1}{35}$ (ii) $\frac{1}{7}$ (iii) $\frac{8}{15}$
43. $\frac{1}{6}$ 44. $\frac{11}{4165}$ 45. $\frac{4}{5}$

HINTS TO NCERT & SELECTED PROBLEMS

12. Required probability = $\frac{{}^{13}C_1 \times {}^{13}C_1 \times {}^{13}C_1 \times {}^{13}C_1}{{}^{52}C_4}$
13. Four honours means king, queen, jack and ace.
So required probability = $\frac{{}^4C_4 + {}^4C_4 + {}^4C_4 + {}^4C_4}{{}^{52}C_4}$
14. Total number of elementary events = ${}^{20}C_1 = 20$.
A multiple of 3 or 7 can be obtained as follows: 3, 6, 9, 12, 15, 18, 7, 14.
So, favourable number of elementary events = ${}^8C_1 = 8$.
15. Total number of elementary events = ${}^{18}C_3$.
Favourable number of elementary events = ${}^6C_1 \times {}^4C_1 \times {}^8C_1$
20. Out of 4 men and 6 women one person can be chosen in ${}^{10}C_1 = 10$ ways.
The number of ways of selecting 1 women out of 6 women = ${}^6C_1 = 6$.
 $\therefore$ Required probability = $\frac{6}{10} = \frac{3}{5}$
25. A committee of two persons can be formed from two men and two women in ${}^4C_2 = 6$ ways.
(i) Number of committees having no man = ${}^2C_2 = 1$
 $\therefore$ Probability that a committee has no man = $\frac{1}{6}$
(ii) Number of committees having one man = ${}^2C_1 \times {}^2C_1 = 4$
 $\therefore$ Probability that the committee will have one man = $\frac{4}{6} = \frac{2}{3}$
 $\therefore$ Probability that the committee has two men = $\frac{1}{6}$
33. Total number of ways of selecting six numbers from numbers 1 to 20 = ${}^{20}C_6$
 $\therefore$ Total number of elementary events = ${}^{20}C_6 = 38760$
Favourable number of elementary events = 1
 $\therefore$ Required probability = $\frac{1}{38760}$
44. Total number of elementary events = ${}^{52}C_{13}$
Favourable number of elementary events = ${}^{48}C_9 \times {}^4C_4$

33.8 ADDITION THEOREMS ON PROBABILITY

Uptill now we have been computing the probability of occurrence or non-occurrence of an event by using favourable and total number of elementary events. But it is not always convenient to compute favourable number of elementary events to a given event. In such cases, we express the given event as the union of two or more events and the probability of the given event is expressed in terms of the probabilities of these events. Theorems which express the probability

of an event in terms of the probabilities of those events whose union is the given event are known as addition theorems on probability. In this section, we shall discuss addition theorems for two or more events.

THEOREM 1 (Addition Theorem for two events) If A and B are two events associated with a random experiment, then $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.

PROOF Let S be the sample space associated with the given random experiment. Suppose the random experiment results in n mutually exclusive ways. Then, S contains n elementary events.

Let m_1 , m_2 and m be the number of elementary events favourable to A , B and $A \cap B$ respectively. Then,

$$P(A) = \frac{m_1}{n}, \quad P(B) = \frac{m_2}{n} \quad \text{and} \quad P(A \cap B) = \frac{m}{n}.$$

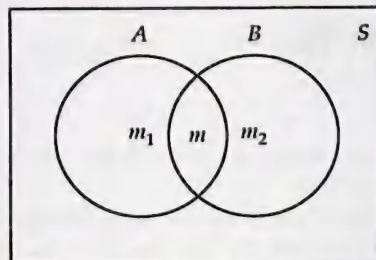


Fig. 33.3

The number of elementary events favourable to A only is $m_1 - m$. Similarly, the number of elementary events favourable to B only is $m_2 - m$. Since m elementary events are favourable to both A and B . Therefore, the number of elementary events favourable to A or B or both i.e. $A \cup B$ is $m_1 - m + m_2 - m + m = m_1 + m_2 - m$.

$$\text{So,} \quad P(A \cup B) = \frac{m_1 + m_2 - m}{n} = \frac{m_1}{n} + \frac{m_2}{n} - \frac{m}{n}$$

$$\Rightarrow \quad P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Q.E.D.

COROLLARY If A and B are mutually exclusive events, then $P(A \cap B) = 0$.

$$\therefore \quad P(A \cup B) = P(A) + P(B)$$

This is the addition theorem for mutually exclusive events.

THEOREM 2 (Addition Theorem for three events) If A , B , C are three events associated with a random experiment, then

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C) - P(A \cap C) + P(A \cap B \cap C)$$

PROOF Let $D = B \cup C$. Then,

$$P(A \cup B \cup C) = P(A \cup D) = P(A) + P(D) - P(A \cap D) \quad \dots(i)$$

$$\text{But,} \quad A \cap D = A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

$$\begin{aligned} \therefore \quad P(A \cap D) &= P[(A \cap B) \cup (A \cap C)] \\ &= P(A \cap B) + P(A \cap C) - P[(A \cap B) \cap (A \cap C)] \\ &= P(A \cap B) + P(A \cap C) - P(A \cap B \cap C) \end{aligned} \quad \dots(ii)$$

$$\text{and,} \quad P(D) = P(B \cup C) = P(B) + P(C) - P(B \cap C) \quad \dots(iii)$$

From (i), (ii) and (iii), we get

$$\begin{aligned} P(A \cup B \cup C) &= P(A) + P(B) + P(C) - P(B \cap C) - [P(A \cap B) + P(A \cap C) - P(A \cap B \cap C)] \\ \Rightarrow P(A \cup B \cup C) &= P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C) \end{aligned}$$

Q.E.D.

COROLLARY If A , B , C are mutually exclusive events, then

$$P(A \cap B) = P(B \cap C) = P(A \cap C) = P(A \cap B \cap C) = 0.$$

$$\therefore \quad P(A \cup B \cup C) = P(A) + P(B) + P(C)$$

This is the addition theorem for three mutually exclusive events.

THEOREM 3 Let A and B be two events associated to a random experiment. Then,

$$(i) \quad P(\bar{A} \cap B) = P(B) - P(A \cap B)$$

$$(ii) \quad P(A \cap \bar{B}) = P(A) - P(A \cap B)$$

$$(iii) \quad P((A \cap \bar{B}) \cup (\bar{A} \cap B)) = P(A) + P(B) - 2P(A \cap B)$$

PROOF (i) Since $A \cap B$ and $\bar{A} \cap B$ are mutually exclusive events such that

$$(A \cap B) \cup (\bar{A} \cap B) = B$$

$$\therefore P(A \cap B) + P(\bar{A} \cap B) = P(B)$$

$$\Rightarrow P(\bar{A} \cap B) = P(B) - P(A \cap B)$$

(ii) Since $A \cap B$ and $A \cap \bar{B}$ are mutually exclusive events such that

$$(A \cap B) \cup (A \cap \bar{B}) = A$$

$$\therefore P(A \cap B) + P(A \cap \bar{B}) = P(A)$$

$$\Rightarrow P(A \cap \bar{B}) = P(A) - P(A \cap B).$$

(iii) Since $A \cap \bar{B}$ and $\bar{A} \cap B$ are mutually exclusive events. Therefore,

$$\begin{aligned} P[(A \cap \bar{B}) \cup (\bar{A} \cap B)] &= P(A \cap \bar{B}) + P(\bar{A} \cap B) \\ &= P(A) - P(A \cap B) + P(B) - P(A \cap B) \quad [\text{Using (i) and (ii)}] \\ &= P(A) + P(B) - 2P(A \cap B) \end{aligned}$$

REMARK 1 $P(\bar{A} \cap B)$ is known as the probability of occurrence of B only.

Q.E.D.

REMARK 2 $P(A \cap \bar{B})$ is known as the probability of occurrence of A only.

REMARK 3 $P[(A \cap \bar{B}) \cup (\bar{A} \cap B)]$ is known as the probability of occurrence of exactly one of two events A and B.

REMARK 4 If A and B are two events associated to a random experiment such that $A \subset B$, then $\bar{A} \cap B \neq \phi$.

$$\therefore P(\bar{A} \cap B) \geq 0 \Rightarrow P(B) - P(A \cap B) \geq 0 \Rightarrow P(B) - P(A) \geq 0 \Rightarrow P(A) \leq P(B).$$

THEOREM 4 For any two events A and B, prove that

$$P(A \cap B) \leq P(A) \leq P(A \cup B) \leq P(A) + P(B).$$

PROOF Since $A \cap B \subset A$. Therefore, we have

$$P(A \cap B) \leq P(A). \quad \dots(i)$$

$$\text{Also, } A \subset A \cup B \Rightarrow P(A) \leq P(A \cup B) \quad \dots(ii)$$

$$\text{Now, } P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$\Rightarrow P(A \cup B) \leq P(A) + P(B) \quad \dots(iii)$$

From (i), (ii) and (iii), we get

$$P(A \cap B) \leq P(A) \leq P(A \cup B) \leq P(A) + P(B). \quad \text{Q.E.D.}$$

THEOREM 5 For any two events A and B, prove that the probability that exactly one of A, B occurs is given by $P(A) + P(B) - 2P(A \cap B) = P(A \cup B) - P(A \cap B)$.

PROOF We have,

$$\begin{aligned} P(\text{Exactly one of A, B occurs}) &= P[(A \cap \bar{B}) \cup (\bar{A} \cap B)] \\ &= P(A \cap \bar{B}) + P(\bar{A} \cap B) \\ &= P(A) - P(A \cap B) + P(B) - P(A \cap B) \\ &= P(A) + P(B) - 2P(A \cap B) \\ &= [P(A) + P(B) - P(A \cap B)] - P(A \cap B) \\ &= P(A \cup B) - P(A \cap B). \end{aligned}$$

Q.E.D.

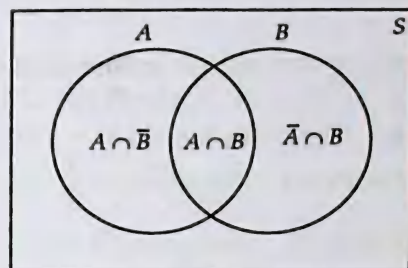


Fig. 33.4

ILLUSTRATIVE EXAMPLES**LEVEL-1****Type I PROBLEMS BASED UPON FORMULAE**

$$(i) P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$(ii) P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C) - P(A \cap C) + P(A \cap B \cap C)$$

EXAMPLE 1 Given $P(A) = \frac{3}{5}$ and $P(B) = \frac{1}{5}$. Find $P(A \text{ or } B)$, if A and B are mutually exclusive events.

SOLUTION Since A and B are mutually exclusive events.

$$\therefore P(A \text{ or } B) = P(A \cup B) = P(A) + P(B) = \frac{3}{5} + \frac{1}{5} = \frac{4}{5}$$

EXAMPLE 2 A and B are two mutually exclusive events of an experiment. If $P(\text{'not } A) = 0.65$, $P(A \cup B) = 0.65$ and $P(B) = p$, find the value of p .

SOLUTION By addition theorem for mutually exclusive events, we have

$$P(A \cup B) = P(A) + P(B)$$

$$\Rightarrow P(A \cup B) = 1 - P(\text{'not } A) + P(B)$$

$$[\because P(A) = 1 - P(\bar{A})]$$

$$\Rightarrow 0.65 = 1 - 0.65 + p \Rightarrow p = 0.30$$

EXAMPLE 3 A and B are two non-mutually exclusive events. If $P(A) = \frac{1}{4}$, $P(B) = \frac{2}{5}$ and $P(A \cup B) = \frac{1}{2}$, find the values of $P(A \cap B)$ and $P(A \cap \bar{B})$.

SOLUTION We have, $P(A) = \frac{1}{4}$, $P(B) = \frac{2}{5}$ and $P(A \cup B) = \frac{1}{2}$

By addition theorem, we have

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$\Rightarrow \frac{1}{2} = \frac{1}{4} + \frac{2}{5} - P(A \cap B)$$

$$\Rightarrow P(A \cap B) = \frac{1}{4} + \frac{2}{5} - \frac{1}{2} = \frac{3}{20}$$

$$\therefore P(A \cap \bar{B}) = P(A) - P(A \cap B) = \frac{1}{4} - \frac{3}{20} = \frac{1}{10}$$

EXAMPLE 4 If E and F are two events such that $P(E) = \frac{1}{4}$, $P(F) = \frac{1}{2}$ and $P(E \text{ and } F) = \frac{1}{8}$, find

(i) $P(E \text{ or } F)$ (ii) $P(\text{not } E \text{ and not } F)$.

SOLUTION We have,

$$P(E) = \frac{1}{4}, P(F) = \frac{1}{2} \text{ and } P(E \cap F) = \frac{1}{8}$$

$$(i) P(E \text{ or } F) = P(E \cup F) = P(E) + P(F) - P(E \cap F) = \frac{1}{4} + \frac{1}{2} - \frac{1}{8} = \frac{5}{8}$$

$$\begin{aligned} (ii) P(\text{not } E \text{ and not } F) &= P(\bar{E} \cap \bar{F}) \\ &= P(\overline{E \cup F}) \\ &= 1 - P(E \cup F) = 1 - [P(E) + P(F) - P(E \cap F)] \\ &= 1 - \left\{ \frac{1}{4} + \frac{1}{2} - \frac{1}{8} \right\} = 1 - \frac{5}{8} = \frac{3}{8} \end{aligned}$$

EXAMPLE 5 The probability that at least one of the events A and B occurs is 0.6. If A and B occur simultaneously with probability 0.2, then find $P(A) + P(B)$.

SOLUTION We have,

$$P(\text{At least one of the events } A \text{ and } B \text{ occurs}) = 0.6 \text{ i.e. } P(A \cup B) = 0.6$$

$$\text{and, } P(A \text{ and } B \text{ occur simultaneously}) = 0.2 \text{ i.e. } P(A \cap B) = 0.2$$

Now,

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$\Rightarrow 0.6 = P(A) + P(B) - 0.2$$

$$\Rightarrow 0.6 = 1 - P(\bar{A}) + 1 - P(\bar{B}) - 0.2$$

$$\Rightarrow 0.6 = 2 - 0.2 - [P(\bar{A}) + P(\bar{B})]$$

$$\Rightarrow 0.6 = 1.8 - [P(\bar{A}) + P(\bar{B})]$$

$$\Rightarrow P(\bar{A}) + P(\bar{B}) = 1.8 - 0.6 = 1.2$$

EXAMPLE 6 Check whether the following probabilities $P(A)$ and $P(B)$ are consistently defined:

$$(i) P(A) = 0.5, P(B) = 0.7, P(A \cap B) = 0.6 \quad (ii) P(A) = 0.5, P(B) = 0.4, P(A \cup B) = 0.8$$

SOLUTION (i) We have, $P(A) = 0.5, P(B) = 0.7, P(A \cap B) = 0.6$

We know that $P(A \cap B) \leq P(A)$ and $P(A \cap B) \leq P(B)$. But, for the given probabilities $P(A \cap B) \nless P(A)$. So, given probabilities are not consistently defined.

(ii) We have,

$$P(A) = 0.5, P(B) = 0.4 \text{ and } P(A \cup B) = 0.8$$

$$\therefore P(A \cap B) = P(A) + P(B) - P(A \cup B) = 0.5 + 0.4 - 0.8 = 0.1$$

Clearly, $P(A \cap B) \leq P(A)$ and $P(A \cap B) \leq P(B)$. Hence, the given probabilities are consistently defined.

EXAMPLE 7 Events E and F are such that $P(\text{not } E \text{ or not } F) = 0.25$. State whether E and F are mutually exclusive.

SOLUTION We have, $P(\text{not } E \text{ or not } F) = 0.25$

$$\text{i.e. } P(\bar{E} \cup \bar{F}) = 0.25 \Rightarrow P(\overline{E \cap F}) = 0.25 \Rightarrow 1 - P(E \cap F) = 0.25 \Rightarrow P(E \cap F) = 1 - 0.25 = 0.75 \neq 0$$

Hence, E and F are not mutually exclusive.

EXAMPLE 8 A, B, C are three mutually exclusive and exhaustive events associated with a random experiment. Find $P(A)$, it being given that $P(B) = \frac{3}{2} P(A)$ and $P(C) = \frac{1}{2} P(B)$.

SOLUTION Let $P(A) = p$. Then,

$$P(B) = \frac{3}{2} P(A) \Rightarrow P(B) = \frac{3}{2} p \text{ and } P(C) = \frac{1}{2} P(B) \Rightarrow P(C) = \frac{3}{4} p$$

Since A, B, C are mutually exclusive and exhaustive events associated with a random experiment.

$$\therefore A \cup B \cup C = S$$

$$\Rightarrow P(A \cup B \cup C) = P(S)$$

$$\Rightarrow P(A \cup B \cup C) = 1$$

$$\Rightarrow P(A) + P(B) + P(C) = 1$$

$$\Rightarrow p + \frac{3}{2} p + \frac{3}{4} p = 1 \Rightarrow p = \frac{4}{13} \Rightarrow P(A) = \frac{4}{13}$$

$$[\because P(S) = 1]$$

[By addition Theorem]

EXAMPLE 9 Four candidates A, B, C, D have applied for the assignment to coach a school cricket team. If A is twice as likely to be selected as B , and B and C are given about the same chance of being selected, while C is twice as likely to be selected as D , what are the probabilities that (i) C will be selected? (ii) A will not be selected?

[NCERT EXEMPLAR]

SOLUTION Let A_1, A_2, A_3 and A_4 be the events that candidates A, B, C and D respectively are selected as school cricket team coach. Then,

It is given that

$$P(A_3) = 2P(A_4), P(A_2) = P(A_3) \text{ and } P(A_1) = 2P(A_2)$$

$$\Rightarrow P(A_1) = 4P(A_4), P(A_2) = P(A_3) = 2P(A_4)$$

Clearly, A_1, A_2, A_3 and A_4 are mutually exclusive and exhaustive events. Therefore,

$$A_1 \cup A_2 \cup A_3 \cup A_4 = S$$

$$\Rightarrow P(A_1 \cup A_2 \cup A_3 \cup A_4) = P(S)$$

$$\Rightarrow P(A_1) + P(A_2) + P(A_3) + P(A_4) = 1$$

$$\Rightarrow 4P(A_4) + 2P(A_4) + 2P(A_4) + P(A_4) = 1$$

$$\Rightarrow 9P(A_4) = 1$$

$$\Rightarrow P(A_4) = \frac{1}{9}$$

$$(i) \text{ Required probability} = P(A_3) = 2P(A_4) = \frac{2}{9}$$

$$(ii) \text{ Required probability} = P(\bar{A}_1) = 1 - P(A_1) = 1 - 4P(A_4) = 1 - \frac{4}{9} = \frac{5}{9}$$

EXAMPLE 10 Probability that a truck stopped at a roadblock will have faulty brakes or badly worn tires are 0.23 and 0.24, respectively. Also, the probability is 0.38 that a truck stopped at the roadblock will have faulty brakes and or badly working tires. What is the probability that a truck stopped at this roadblock will have faulty brakes as well as badly worn tires? [NCERT EXEMPLAR]

SOLUTION Let B be the event that a truck stopped at the roadblock will have faulty brakes and T be the event that it will have badly worn tires.

It is given that $P(B) = 0.23$, $P(T) = 0.24$ and $P(B \cup T) = 0.38$. We have to find $P(B \cap T)$.

We know that

$$P(B \cup T) = P(B) + P(T) - P(B \cap T) \quad [\text{By addition theorem}]$$

$$\Rightarrow P(B \cap T) = P(B) + P(T) - P(B \cup T) = 0.23 + 0.24 - 0.38 = 0.09$$

EXAMPLE 11 The probability of two events A and B are 0.25 and 0.50 respectively. The probability of their simultaneous occurrence is 0.14. Find the probability that neither A nor B occurs.

SOLUTION We have, $P(A) = 0.25$, $P(B) = 0.50$ and $P(A \cap B) = 0.14$

$$\begin{aligned} \therefore \text{ Required probability} &= P(\bar{A} \cap \bar{B}) \\ &= P(\overline{A \cup B}) \quad [\because \overline{A \cup B} = \bar{A} \cap \bar{B}] \\ &= 1 - P(A \cup B) = 1 - [P(A) + P(B) - P(A \cap B)] \\ &= 1 - (0.25 + 0.50 - 0.14) = 0.39 \end{aligned}$$

Type II PROBLEMS BASED UPON ADDITION THEOREMS OF PROBABILITY BUT CAN BE SOLVED INDEPENDENTLY BY USING THE DEFINITION ONLY

NOTE Following problems will be solved by using addition theorems but these problems can be solved otherwise also. Students are advised to do these problems without using addition theorem.

EXAMPLE 12 Find the probability of getting an even number on the first die or a total of 8 in a single throw of two dice.

SOLUTION Let S be the sample space associated with the experiment of throwing a pair of dice. Then, $n(S) = 36$.

$$\therefore \text{ Total number of elementary events} = 36$$

Let A and B be two events given by

A = Getting an even number on first die, B = Getting a total of 8.

Then, $A \cap B$ = Getting an even number on first die and a total of 8.

Clearly, $A = \{(2, 1), \dots, (2, 6), (4, 1), (4, 2), \dots, (4, 6), (6, 1), (6, 2), \dots, (6, 6)\}$,

$B = \{(2, 6), (6, 2), (3, 5), (5, 3), (4, 4)\}$ and, $A \cap B = \{(2, 6), (6, 2), (4, 4)\}$.

$$\therefore P(A) = \frac{18}{36}, P(B) = \frac{5}{36} \text{ and } P(A \cap B) = \frac{3}{36}$$

$$\text{Required probability} = P(A \cup B) = P(A) + P(B) - P(A \cap B) = \frac{18}{36} + \frac{5}{36} - \frac{3}{36} = \frac{20}{36} = \frac{5}{9}$$

EXAMPLE 13 A die is thrown twice. What is the probability that at least one of the two throws comes up with the number 4?

SOLUTION Let S be the sample space associated with the random experiment of throwing a die twice. Then, $n(S) = 36$.

$\therefore$ Total number of elementary events = 36

Consider the events: A = First throw shows 4, B = Second throw shows 4

$\therefore A \cap B$ = First and Second throw show 4 i.e. getting 4 in each throw.

Clearly,

$A = \{(4, 1), (4, 2), (4, 3), (4, 4), (4, 5), (4, 6)\}$, $B = \{(1, 4), (2, 4), (3, 4), (4, 4), (5, 4), (6, 4)\}$

and, $A \cap B = \{(4, 4)\}$

$$\therefore P(A) = \frac{6}{36}, P(B) = \frac{6}{36} \text{ and } P(A \cap B) = \frac{1}{36}$$

$\therefore$ Required probability = Probability that at least one of the two throws shows 4.

$$= P(A \cup B) = P(A) + P(B) - P(A \cap B) = \frac{6}{36} + \frac{6}{36} - \frac{1}{36} = \frac{11}{36}$$

EXAMPLE 14 One number is chosen from numbers 1 to 200. Find the probability that it is divisible by 4 or 6?

SOLUTION Let S be the sample space. Then, $n(S) = 200$.

$\therefore$ Total number of elementary events = 200

Let A be the event that the number selected is divisible by 4 and B be the event that the number selected is divisible by 6. Then,

$A = \{4, 8, 12, \dots, 200\}$, $B = \{6, 12, \dots, 198\}$ and $A \cap B = \{12, 24, \dots, 192\}$

Clearly,

$$n(A) = \frac{200}{4} = 50, n(B) = \frac{198}{6} = 33 \text{ and } n(A \cap B) = \frac{192}{12} = 16$$

$$\therefore P(A) = \frac{50}{200}, P(B) = \frac{33}{200} \text{ and } P(A \cap B) = \frac{16}{200}$$

Required probability = P (a number is divisible by 4 or 6)

$$= P(A \cup B) = P(A) + P(B) - P(A \cap B) = \frac{50}{200} + \frac{33}{200} - \frac{16}{200} = \frac{67}{200}$$

EXAMPLE 15 A card is drawn from a deck of 52 cards. Find the probability of getting a king or a heart or a red card.

SOLUTION Consider the following events:

A = Getting a king, B = getting a heart card, C = Getting a red card.

Clearly,

$$P(A) = \frac{{}^4C_1}{{}^{52}C_1} = \frac{4}{52}, P(B) = \frac{{}^{13}C_1}{{}^{52}C_1} = \frac{13}{52}, P(C) = \frac{{}^{26}C_1}{{}^{52}C_1} = \frac{26}{52}$$

$$P(A \cap B) = P(\text{Getting a king of heart}) = \frac{1}{52}, P(B \cap C) = P(\text{Getting a heart card}) = \frac{13}{52}$$

$$P(C \cap A) = P(\text{Getting a red king}) = \frac{2}{52}, P(A \cap B \cap C) = P(\text{Getting a king of heart}) = \frac{1}{52}$$

$$\begin{aligned} \text{Required Probability} &= P(A \cup B \cup C) \\ &= P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C) - P(C \cap A) + P(A \cap B \cap C) \\ &= \frac{4}{52} + \frac{13}{52} + \frac{26}{52} - \frac{1}{52} - \frac{13}{52} - \frac{2}{52} + \frac{1}{52} = \frac{28}{52} = \frac{7}{13} \end{aligned}$$

EXAMPLE 16 A drawer contains 50 bolts and 150 nuts. Half of the bolts and half of the nuts are rusted. If one item is chosen at random, what is the probability that it is rusted or a bolt?

SOLUTION Let A be the event that the item chosen is rusted and B be the event that the item chosen is a bolt. Clearly, there are 200 items in all, out of which 100 are rusted.

$$\therefore P(A) = \frac{100}{200}, P(B) = \frac{50}{200} \text{ and } P(A \cap B) = \frac{25}{200}$$

$$\text{Required probability} = P(A \cup B)$$

$$= P(A) + P(B) - P(A \cap B) = \frac{100}{200} + \frac{50}{200} - \frac{25}{200} = \frac{125}{200} = \frac{5}{8}$$

EXAMPLE 17 Four cards are drawn at a time from a pack of 52 playing cards. Find the probability of getting all the four cards of the same suit.

SOLUTION Since 4 cards can be drawn at a time from a pack of 52 cards in ${}^{52}C_4$ ways. Therefore,

$$\text{Total number of elementary events} = {}^{52}C_4$$

Consider the following events:

A = Getting all spade cards; B = Getting all club cards;

C = Getting all diamond cards, and D = Getting all heart cards.

Clearly, A, B, C and D are mutually exclusive events such that

$$P(A) = \frac{{}^{13}C_4}{{}^{52}C_4}, P(B) = \frac{{}^{13}C_4}{{}^{52}C_4}, P(C) = \frac{{}^{13}C_4}{{}^{52}C_4} \text{ and } P(D) = \frac{{}^{13}C_4}{{}^{52}C_4}$$

$$\text{Required probability} = P(A \cup B \cup C \cup D)$$

$$= P(A) + P(B) + P(C) + P(D)$$

[By addition Theorem]

$$= 4 \left(\frac{{}^{13}C_4}{{}^{52}C_4} \right) = \frac{44}{4165}$$

EXAMPLE 18 An integer is chosen at random from the numbers ranging from 1 to 50. What is the probability that the integer chosen is a multiple of 2 or 3 or 10?

SOLUTION Out of 50 integers an integer can be chosen in ${}^{50}C_1$ ways.

$$\therefore \text{Total number of elementary events} = {}^{50}C_1 = 50.$$

Consider the following events:

A = Getting a multiple of 2, B = Getting a multiple of 3 and, C = Getting a multiple of 10.

Clearly,

$$A = \{2, 4, \dots, 50\}, B = \{3, 6, \dots, 48\}, C = \{10, 20, \dots, 50\}$$

$$A \cap B = \{6, 12, \dots, 48\}, B \cap C = \{30\}, A \cap C = \{10, 20, \dots, 50\} \text{ and } A \cap B \cap C = \{30\}$$

$$\therefore P(A) = \frac{25}{50}, P(B) = \frac{16}{50}, P(C) = \frac{5}{50}, P(A \cap B) = \frac{8}{50}, P(B \cap C) = \frac{1}{50},$$

$$P(A \cap C) = \frac{5}{50} \text{ and } P(A \cap B \cap C) = \frac{1}{50}$$

$$\text{Required probability} = P(A \cap B \cap C)$$

$$= P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap A) + P(A \cap B \cap C)$$

$$= \frac{25}{50} + \frac{16}{50} + \frac{5}{50} - \frac{8}{50} - \frac{1}{50} - \frac{5}{50} + \frac{1}{50} = \frac{33}{50}$$

EXAMPLE 19 In an essay competition, the odds in favour of competitors P, Q, R, S are $1:2, 1:3, 1:4$, and $1:5$ respectively. Find the probability that one of them wins the competition.

SOLUTION Let A, B, C, D be the events that the competitors P, Q, R and S respectively win the competition. Then,

$$P(A) = \frac{1}{3}, P(B) = \frac{1}{4}, P(C) = \frac{1}{5} \text{ and } P(D) = \frac{1}{6}$$

Since only one competitor can win the competition. Therefore, A, B, C, D are mutually exclusive events.

$\therefore$ Required probability = $P(A \cup B \cup C \cup D)$

$$= P(A) + P(B) + P(C) + P(D) \quad [\text{By addition Theorem}]$$

$$= \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \frac{1}{6} = \frac{114}{120}$$

EXAMPLE 20 (i) Two dice are thrown together. What is the probability that the sum of the numbers on the two faces is neither divisible by 3 nor by 4? (ii) What is the probability that the sum of the numbers on the two faces is divisible by 3 or 4?

SOLUTION (i) Let S be the sample space associated with the experiment of throwing a pair of dice. Then, $n(S) = 36$.

$\therefore$ Total number of elementary events = 36

Consider the following events.

A = The sum of the numbers on two faces is divisible by 3

B = The sum of the numbers on two faces is divisible by 4.

Then, $A = \{(1, 2), (2, 1), (1, 5), (5, 1), (3, 3), (2, 4), (4, 2), (3, 6), (6, 3), (4, 5), (5, 4), (6, 6)\}$

$B = \{(2, 2), (1, 3), (3, 1), (2, 6), (6, 2), (4, 4), (3, 5), (5, 3), (6, 6)\}$ and, $A \cap B = \{(6, 6)\}$

$$\therefore P(A) = \frac{12}{36} = \frac{1}{3}, P(B) = \frac{9}{36} = \frac{1}{4} \text{ and } P(A \cap B) = \frac{1}{36}$$

(i) Required probability = $P(\overline{A \cap B}) = P(\overline{A \cup B}) = 1 - P(A \cup B)$

$$= 1 - \{P(A) + P(B) - P(A \cap B)\} = 1 - \left\{ \frac{1}{3} + \frac{1}{4} - \frac{1}{36} \right\} = \frac{4}{9}$$

(ii) Required probability = $P(A \cup B) = P(A) + P(B) - P(A \cap B) = \frac{1}{3} + \frac{1}{4} - \frac{1}{36} = \frac{5}{9}$

EXAMPLE 21 An urn contains twenty white slips of paper numbered from 1 through 20, ten red slips of paper numbered from 1 through 10, forty yellow slips of paper numbered from 1 through 40 and ten blue slips of paper numbered from 1 through 10. If these 80 slips of paper are thoroughly shuffled so that each slip has the same probability of being drawn. Find the probabilities of drawing a slip of paper that is (i) blue or white (ii) number 1, 2, 3, 4 or 5 (iii) red or yellow and numbered 1, 2, 3, or 4. (iv) numbered 5, 15, 25 or 35. (v) white and numbered higher than 12 or yellow and numbered higher than 26.

[NCERT EXEMPLAR]

SOLUTION There are 80 slips of paper out of which one slip can be chosen in ${}^{80}C_1 = 80$ ways.

So, total number of elementary events = 80

(i) There are 10 blue and 20 white slips out of which one slip can be chosen in ${}^{30}C_1 = 30$ ways.

$\therefore$ Favourable number of ways = 30

$$\text{Hence, } P(\text{Drawing a blue or white slip}) = \frac{30}{80} = \frac{3}{8}$$

(ii) Consider the following events:

W = Drawing a white slip numbered 1, 2, 3, 4 or 5,

R = Drawing a red slip numbered 1, 2, 3, 4 or 5,

Y = Drawing a yellow slip numbered 1, 2, 3, 4 or 5,

B = Drawing a blue slip numbered 1, 2, 3, 4 or 5

Clearly, these events are mutually exclusive.

Required probability = $P(W \cup R \cup Y \cup B)$

$$= P(W) + P(R) + P(Y) + P(B) = \frac{5}{80} + \frac{5}{80} + \frac{5}{80} + \frac{5}{80} = \frac{20}{80} = \frac{1}{4}$$

(iii) Consider the following events:

R = Drawing a red slip numbered 1, 2, 3 or 4,

Y = Drawing a yellow slip numbered 1, 2, 3, or 4

Clearly, R and Y are mutually exclusive events such that $P(R) = \frac{4}{80}$ and $P(Y) = \frac{4}{80}$.

$$\text{Required probability} = P(R \cup Y) = P(R) + P(Y) = \frac{4}{80} + \frac{4}{80} = \frac{1}{10}$$

(iv) Consider the following events:

A = Drawing a slip numbered 5, B = Drawing a slip numbered 15

C = Drawing a slip numbered 25, D = Drawing a slip numbered 35

We observe that A, B, C and D are mutually exclusive events such that

$$P(A) = \frac{4}{80} \quad [\because \text{There are 4 tickets, one of each colour numbered 5}]$$

$$P(B) = \frac{2}{80} \quad [\because \text{There is one white and one yellow ticket each numbered 15}]$$

$$P(C) = \frac{1}{80} \quad [\because \text{There is just one yellow ticket numbered 25}]$$

$$\text{and, } P(D) = \frac{1}{80} \quad [\because \text{There is just one yellow ticket numbered 35}]$$

$$\text{Required probability} = P(A \cup B \cup C \cup D) = P(A) + P(B) + P(C) + P(D) = \frac{4}{80} + \frac{2}{80} + \frac{1}{80} + \frac{1}{80} = \frac{1}{10}$$

(v) Consider the following events:

A = Drawing a white slip numbered higher than 12

B = Drawing a yellow slip numbered higher than 26

We observe that A and B are mutually exclusive events such that $P(A) = \frac{8}{80}$ and $P(B) = \frac{14}{80}$.

$$\text{Required probability} = P(A \cup B) = P(A) + P(B) = \frac{8}{80} + \frac{14}{80} = \frac{11}{40}$$

Type III PROBLEMS WHICH CAN BE SOLVED BY USING ADDITION THEOREMS ONLY

EXAMPLE 22 Two cards are drawn from a pack of 52 cards. What is the probability that either both are red or both are kings?

SOLUTION Out of 52 cards, two cards can be drawn in ${}^{52}C_2$ ways.

So, total number of elementary events = ${}^{52}C_2$.

Consider the following events:

A = Two cards drawn are red cards, B = Two cards drawn are kings.

Required probability = $P(A \cup B) = P(A) + P(B) - P(A \cap B)$ [By addition Theorem] ... (i)

Let us now find $P(A)$, $P(B)$ and $P(A \cap B)$.

There are 26 red cards, out of which 2 red cards can be drawn in ${}^{26}C_2$ ways.

$$\therefore P(A) = \frac{{}^{26}C_2}{{}^{52}C_2}$$

Since there are 4 kings, out of which 2 kings can be drawn in 4C_2 ways.

$$\therefore P(B) = \frac{{}^4C_2}{{}^{52}C_2}$$

There are 2 cards which are both red and kings.

$$\begin{aligned}\therefore P(A \cap B) &= \text{Probability of getting 2 cards which are both red and kings.} \\ &= \text{Probability of getting 2 red kings} = \frac{{}^2C_2}{{}^{52}C_2}\end{aligned}$$

$$\begin{aligned}\text{Required probability} &= P(A) + P(B) - P(A \cap B) \\ &= \frac{{}^{26}C_2}{{}^{52}C_2} + \frac{{}^4C_2}{{}^{52}C_2} - \frac{{}^2C_2}{{}^{52}C_2} = \frac{325}{1326} + \frac{1}{221} - \frac{1}{1326} = \frac{55}{221}\end{aligned}$$

EXAMPLE 23 A basket contains 20 apples and 10 oranges out of which 5 apples and 3 oranges are defective. If a person takes out 2 at random what is the probability that either both are apples or both are good?

SOLUTION Out of 30 items, two can be selected in ${}^{30}C_2$ ways.

So, total number of elementary events = ${}^{30}C_2$.

Consider the following events:

A = Getting two apples, B = Getting two good items

There are 20 apples, out of which 2 can be drawn in ${}^{20}C_2$ ways.

$$\therefore P(A) = \frac{{}^{20}C_2}{{}^{30}C_2}$$

There are 8 defective pieces and the remaining 22 are good. Out of 22 good pieces, two can be selected in ${}^{22}C_2$ ways.

$$\therefore P(B) = \frac{{}^{22}C_2}{{}^{30}C_2}$$

Since there are 15 pieces which are good apples out of which 2 can be selected in ${}^{15}C_2$ ways.

$$\therefore P(A \cap B) = \text{Probability of getting 2 pieces which are good apples} = \frac{{}^{15}C_2}{{}^{30}C_2}$$

$$\text{Required probability} = P(A) + P(B) - P(A \cap B) = \frac{{}^{20}C_2}{{}^{30}C_2} + \frac{{}^{22}C_2}{{}^{30}C_2} - \frac{{}^{15}C_2}{{}^{30}C_2} = \frac{316}{435}$$

EXAMPLE 24 A die has two faces each with number '1', three faces each with number '2' and one face with number '3'. If the die is rolled once, determine

- (i) $P(1)$ (ii) $P(1 \text{ or } 3)$ (iii) $P(\text{not } 3)$

SOLUTION Let A, B, C be three events defined by

A = Getting a face with number '1', B = Getting a face with number '2',

C = Getting a face with number '3'

$$\text{Then, } P(A) = \frac{2}{6} = \frac{1}{3}, P(B) = \frac{3}{6} = \frac{1}{2} \text{ and } P(C) = \frac{1}{6}$$

$$(i) \quad P(2) = P(A) = \frac{1}{3}$$

$$(ii) \quad P(1 \text{ or } 3) = P(A \cup C) = P(A) + P(C)$$

[$\because A$ and C are mutually exclusive]

$$\Rightarrow P(1 \text{ or } 3) = \frac{1}{3} + \frac{1}{6} = \frac{1}{2}$$

$$(iii) \quad P(\text{not } 3) = P(\bar{C}) = 1 - P(C) = 1 - \frac{1}{6} = \frac{5}{6}$$

EXAMPLE 25 The probability that a student will receive A, B, C or D grade are 0.40, 0.35, 0.15 and 0.10 respectively. Find the probability that a student will receive

(i) B or C grade

(ii) at most C grade.

SOLUTION Let E_1, E_2, E_3 and E_4 denote respectively the events that a student will receive A, B, C and D grades. Then,

$$P(E_1) = 0.40, P(E_2) = 0.35, P(E_3) = 0.15 \text{ and } P(E_4) = 0.10$$

(i) Required probability = $P(E_2 \cup E_3)$

$$= P(E_2) + P(E_3)$$

[$\because E_2$ and E_3 are mutually exclusive events]

$$= 0.35 + 0.15 = 0.50$$

(ii) Required probability = Probability that the student receives C or D grade

$$= P(E_3 \cup E_4)$$

$$= P(E_3) + P(E_4)$$

[$\because E_3$ and E_4 are mutually exclusive]

$$= 0.15 + 0.10 = 0.25$$

EXAMPLE 26 The probability that a person will get an electric contract is $\frac{2}{5}$ and the probability that he will not get plumbing contract is $\frac{4}{7}$. If the probability of getting at least one contract is $\frac{2}{3}$, what is the probability that he will get both?

SOLUTION Consider the following events:

A = Person gets an electric contract, B = Person gets plumbing contract

Clearly,

$$P(A) = \frac{2}{5}, P(\bar{B}) = \frac{4}{7} \text{ and } P(A \cup B) = \frac{2}{3}$$

We have to find $P(A \cap B)$. By addition theorem, we have

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$\Rightarrow \frac{2}{3} = \frac{2}{5} + \left(1 - \frac{4}{7}\right) - P(A \cap B) \Rightarrow P(A \cap B) = \frac{2}{5} + \frac{3}{7} - \frac{2}{3} = \frac{17}{105}$$

EXAMPLE 27 If a person visits his dentist, suppose the probability that he will have his teeth cleaned is 0.48, the probability that he will have cavity filled is 0.25, probability that he will have a tooth extracted is 0.20, the probability that he will have a teeth cleaned and cavity filled is 0.09, the probability that he will have his teeth cleaned and a tooth extracted is 0.12, the probability that he will have a cavity filled and tooth extracted is 0.07, and the probability that he will have his teeth cleaned, cavity filled, and tooth extracted is 0.03. What is the probability that a person visiting his dentist will have at least one of these things done to him?

[NCERT EXEMPLAR]

SOLUTION Consider the following events:

C = The person will have his teeth cleaned, F = The person will have cavity filled

E = The person will have a tooth extracted

It is given that $P(C) = 0.48, P(F) = 0.25, P(E) = 0.20,$

$$P(C \cap F) = 0.09, P(C \cap E) = 0.12, P(E \cap F) = 0.07 \text{ and } P(C \cap F \cap E) = 0.03.$$

$\therefore$ Required probability = $P(C \cup F \cup E)$

$$= P(C) + P(F) + P(E) - P(C \cap F) - P(F \cap E) - P(C \cap E) + P(C \cap F \cap E)$$

$$= 0.48 + 0.25 + 0.20 - 0.09 - 0.12 - 0.07 + 0.03 = 0.68$$

EXAMPLE 28 The probability that a patient visiting a dentist will have a tooth extracted is 0.06, the probability that he will have a cavity filled is 0.2 and the probability that he will have a tooth extracted as well as cavity filled is 0.03. What is the probability of that a patient has either a tooth extracted or a cavity filled?

SOLUTION Let A be the event that the patient will have his tooth extracted, B the event that he will have a cavity filled.

We have, $P(A) = 0.06$, $P(B) = 0.2$ and $P(A \cap B) = 0.03$

$\therefore$ Required probability $= P(A \cup B) = P(A) + P(B) - P(A \cap B) = 0.06 + 0.2 - 0.03 = 0.23$

EXAMPLE 29 The probability that a person visiting a dentist will have his teeth cleaned is 0.44, the probability that he will have a cavity filled is 0.24. The probability that he will have his teeth cleaned or a cavity filled is 0.60. What is the probability that a person visiting a dentist will have his teeth cleaned and cavity filled?

SOLUTION Let A be the event that the patient will have his teeth cleaned and B be the event that he will have cavity filled.

We have, $P(A) = 0.44$, $P(B) = 0.24$ and $P(A \cup B) = 0.60$

$\therefore$ Required probability $= P(A \cap B) = P(A) + P(B) - P(A \cup B) = 0.44 + 0.24 - 0.60 = 0.08$

EXAMPLE 30 Probability that Hameed passes in Mathematics is $\frac{2}{3}$ and the probability that he passes in English is $\frac{4}{9}$. If the probability of passing both courses is $\frac{1}{4}$, what is the probability that Hameed will pass in at least one of these subjects?

SOLUTION Let A be the event that Hameed passes in Mathematics and B be the event that he passes in English.

We have, $P(A) = \frac{2}{3}$, $P(B) = \frac{4}{9}$ and $P(A \cap B) = \frac{1}{4}$

$\therefore$ Required probability $= P(A \cup B) = P(A) + P(B) - P(A \cap B) = \frac{2}{3} + \frac{4}{9} - \frac{1}{4} = \frac{31}{36}$

EXAMPLE 31 Find the probability of at most two tails or at least two heads in a toss of three coins.

SOLUTION Consider the following events:

A = Getting at most two tails in a toss of three coins.

B = Getting at least two heads in a toss of three coins.

We have, $S = \{HHH, HHT, HTH, THH, HTT, THT, TTH, TTT\}$,

$A = \{HHH, HHT, HTH, THH, TTH, HTT, THT\}$, $B = \{HHT, HTH, THH, HHH\}$

$\therefore P(A) = \frac{7}{8}$, $P(B) = \frac{4}{8} = \frac{1}{2}$ and $P(A \cap B) = \frac{4}{8} = \frac{1}{2}$

Required probability $= P(A \cup B) = P(A) + P(B) - P(A \cap B) = \frac{7}{8} + \frac{1}{2} - \frac{1}{2} = \frac{7}{8}$

EXAMPLE 32 In a town of 6000 people 1200 are over 50 years old and 2000 are female. It is known that 30% of the females are over 50 years. What is the probability that a random chosen individual from the town either female or over 50 years?

SOLUTION Consider the following events :

A = A randomly chosen individual is a female

B = A randomly chosen individual is over 50 years old.

Clearly,

$$P(A) = \frac{2000}{6000} = \frac{1}{3}, P(B) = \frac{1200}{6000} = \frac{1}{5}$$

$$\text{and, } P(A \cap B) = P(\text{An individual is a female over 50 years old}) = \frac{30\% \text{ of } 2000}{6000} = \frac{600}{6000} = \frac{1}{10}$$

$$\text{Required probability} = P(A \cup B) = P(A) + P(B) - P(A \cap B) = \frac{1}{3} + \frac{1}{5} - \frac{1}{10} = \frac{13}{30}$$

EXAMPLE 33 From the employees of a company, 5 persons are elected to represent them in the managing committee of the company. Particulars of the five persons are as follows: [NCERT]

S. No.	Person	Age (in years)
1	Male	30
2	Male	33
3	Female	46
4	Female	28
5	Male	41

A person is selected at random from this group to act as a spokespersons. What is the probability that a spokespersons will be either male or over 35 years?

SOLUTION Consider the following events:

A = The spokespersons is a male, B = The spokespersons is over 35 years.

$$\text{Clearly, } P(A) = \frac{3}{5}, P(B) = \frac{2}{5} \text{ and } P(A \cap B) = \frac{1}{5}$$

$$\therefore \text{ Required probability} = P(A \cup B) = P(A) + P(B) - P(A \cap B) = \frac{3}{5} + \frac{2}{5} - \frac{1}{5} = \frac{4}{5}$$

EXAMPLE 34 In class XI of a school, 40% of the students study Mathematics and 30% study Biology. 10% of the class study both Mathematics and Biology. If a student is selected at random from the class, find the probability that he will be studying Mathematics or Biology or both. [NCERT]

SOLUTION Consider the following events:

M = A student studies Mathematics, B = A student studies Biology.

We have,

$$P(M) = \frac{40}{100}, P(B) = \frac{30}{100} \text{ and } P(M \cap B) = \frac{10}{100}$$

$$\text{Required probability} = P(M \cup B)$$

$$= P(M) + P(B) - P(M \cap B) = \frac{40}{100} + \frac{30}{100} - \frac{10}{100} = \frac{60}{100} = \frac{3}{5}$$

EXAMPLE 35 Two students Anil and Ashima appeared in an examination. The probability that Anil will qualify the examination is 0.05 and that Ashima will qualify the examination is 0.10. The probability that both will qualify the examination is 0.02. Find the probability that:

- both Anil and Ashima will not qualify the exam. [NCERT]
- at least one of them will not qualify the exam.
- only one of them will qualify the exam.

SOLUTION Let E and F denote the events that Anil and Ashima will qualify the examination. Then, $P(E) = 0.05$, $P(F) = 0.10$ and $P(E \cap F) = 0.02$

- Required probability = $P(\bar{E} \cap \bar{F})$
 $= P(\overline{E \cup F}) = 1 - P(E \cup F)$
 $= 1 - [P(E) + P(F) - P(E \cap F)] = 1 - (0.05 + 0.10 - 0.02) = 0.87$
- Required probability = $P(\text{At least one of them will not qualify the exam})$

$$= 1 - P(\text{Both of them will qualify the exam})$$

$$= 1 - P(E \cap F) = 1 - 0.02 = 0.98$$

$$\begin{aligned} \text{(iii) } P(\text{Only one of them will qualify the exam}) &= P(E) + P(F) - 2P(E \cap F) \\ &= 0.05 + 0.10 - 2 \times 0.02 = 0.15 - 0.04 = 0.11 \end{aligned}$$

EXAMPLE 36 In a class of 60 students 30 opted for NCC, 32 opted for NSS and 24 opted for both NCC and NSS. If one of these students is selected at random, find the probability that:

(i) the student opted for NCC or NSS (ii) the student has opted neither NCC nor NSS

(iii) the student has opted NSS but not NCC.

[NCERT]

SOLUTION Consider the following events:

A = A student opted NCC, B = A student opted NSS

We have,

$$P(A) = \frac{30}{60}, P(B) = \frac{32}{60} \text{ and } P(A \cap B) = \frac{24}{60}$$

$$\text{(i) Required probability} = P(A \cup B) = P(A) + P(B) - P(A \cap B) = \frac{30}{60} + \frac{32}{60} - \frac{24}{60} = \frac{38}{60} = \frac{19}{30}$$

$$\text{(ii) Required probability} = P(\bar{A} \cap \bar{B}) = P(\overline{A \cup B}) = 1 - P(A \cup B) = 1 - \frac{19}{30} = \frac{11}{30}$$

$$\text{(iii) Required probability} = P(\bar{A} \cap B) = P(B) - P(A \cap B) = \frac{32}{60} - \frac{24}{60} = \frac{8}{60} = \frac{2}{15}$$

EXAMPLE 37 One of the four persons John, Rita, Aslam or Gurpreet will be promoted next month. Consequently the sample space S consists of four elementary outcomes as given below.

$S = \{\text{John promoted, Rita promoted, Aslam promoted, Gurpreet promoted}\}$

You are told that the chances of John's promotion is same as that of Gurpreet. Rita's chances of promotion are twice as likely as Johns. Aslam's chances are four times that of John.

(i) Determine $P(\text{John promoted})$, $P(\text{Rita promoted})$, $P(\text{Aslam promoted})$, $P(\text{Gurpreet promoted})$

(ii) If $A = \{\text{John promoted or Gurpreet promoted}\}$, find $P(A)$

[NCERT EXEMPLAR]

SOLUTION (i) Let $P(\text{John promoted}) = p$. Then, by hypothesis

$$P(\text{Gurpreet promoted}) = p, P(\text{Rita promoted}) = 2p \text{ and } P(\text{Aslam promoted}) = 4p.$$

We have,

$S = \{\text{John promoted, Rita promoted, Aslam promoted, Gurpreet promoted}\}$

$$\therefore P(\text{John promoted}) + P(\text{Rita promoted}) + P(\text{Aslam promoted}) + P(\text{Gurpreet promoted}) = 1$$

$$\Rightarrow p + 2p + 4p + p = 1$$

$$\Rightarrow p = \frac{1}{8}$$

$$\text{Hence, } P(\text{John promoted}) = \frac{1}{8}, P(\text{Rita promoted}) = 2p = \frac{2}{8} = \frac{1}{4},$$

$$P(\text{Aslam promoted}) = 4p = \frac{4}{8} = \frac{1}{2} \text{ and } P(\text{Gurpreet promoted}) = p = \frac{1}{8}.$$

(ii) $A = \{\text{John promoted or Gurpreet promoted}\}$

$$\therefore P(A) = P(\text{John promoted}) + P(\text{Gurpreet promoted}) = p + p = 2p = \frac{2}{8} = \frac{1}{4}$$

LEVEL-2

EXAMPLE 38 A, B, C are events such that $P(A) = 0.3, P(B) = 0.4, P(C) = 0.8, P(A \cap B) = 0.08, P(A \cap C) = 0.28, P(A \cap B \cap C) = 0.09$. If $P(A \cup B \cup C) \geq 0.75$, then show that $P(B \cap C)$ lies in the interval $(0.23, 0.48)$.

SOLUTION We know that the probability of occurrence of an event is always less than or equal to 1 and it is given that $P(A \cup B \cup C) \geq 0.75$

$$\therefore 0.75 \leq P(A \cup B \cup C) \leq 1$$

$$\Rightarrow 0.75 \leq P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C) - P(A \cap C) + P(A \cap B \cap C) \leq 1$$

$$\Rightarrow 0.75 \leq 0.3 + 0.4 + 0.8 - 0.08 - P(B \cap C) - 0.28 + 0.09 \leq 1$$

$$\Rightarrow 0.75 \leq 1.59 - 0.36 - P(B \cap C) \leq 1$$

$$\Rightarrow 0.75 \leq 1.23 - P(B \cap C) \leq 1$$

$$\Rightarrow -0.48 \leq -P(B \cap C) \leq -0.23 \Rightarrow 0.23 \leq P(B \cap C) \leq 0.48$$

EXAMPLE 39 If A and B are any two events such that $P(A \cup B) = \frac{1}{2}$ and $P(\bar{A}) = \frac{2}{3}$, find $P(\bar{A} \cap B)$.

SOLUTION Clearly, $\bar{A} \cap B$ and A are mutually exclusive events such that

$$A \cup B = A \cup (\bar{A} \cap B)$$

$$\Rightarrow P(A \cup B) = P(A) + P(\bar{A} \cap B)$$

$$\Rightarrow \frac{1}{2} = 1 - P(\bar{A}) + P(\bar{A} \cap B)$$

$$\Rightarrow \frac{1}{2} = 1 - \frac{2}{3} + P(\bar{A} \cap B)$$

$$\Rightarrow P(\bar{A} \cap B) = \frac{1}{6}$$

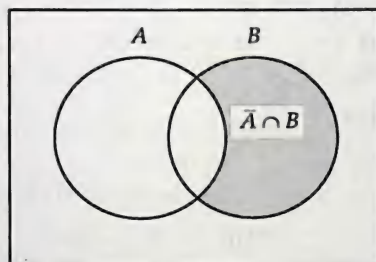


Fig. 33.5

EXAMPLE 40 Figure 33.6 shows three events A , B and C and also the probabilities of the various intersections (for instance $P(A \cap B) = 0.07$) determine

- | | | |
|--------------------------|--------------------------|--|
| (i) $P(A)$ | (ii) $P(B \cap \bar{C})$ | (iii) $P(A \cup B)$ |
| (iv) $P(A \cap \bar{B})$ | (v) $P(B \cap C)$ | (vi) Probability of exactly one of the three events. |

[NCERT EXEMPLAR]

SOLUTION (i) We have, $P(A \cap \bar{B}) = 0.13$ and $P(A \cap B) = 0.07$

Since $A \cap \bar{B}$ and $A \cap B$ are mutually exclusive events such that

$$A = (A \cap \bar{B}) \cup (A \cap B).$$

$$\therefore P(A) = P(A \cap \bar{B}) + P(A \cap B)$$

$$\Rightarrow P(A) = 0.13 + 0.07 = 0.20$$

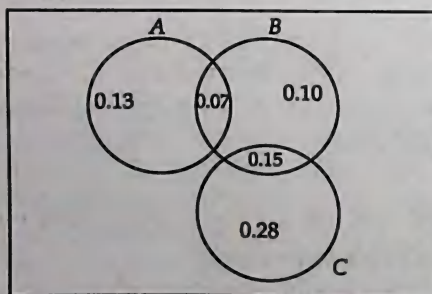


Fig. 33.6

(ii) It is evident from the Figure 33.6 that

$$P(A \cap B) = 0.07, P(B \cap C) = 0.15 \text{ and } P(\bar{A} \cap B \cap \bar{C}) = 0.10$$

Also, $A \cap B$, $B \cap C$ and $\bar{A} \cap B \cap \bar{C}$ are mutually exclusive events such that

$$B = (A \cap B) \cup (B \cap C) \cup (\bar{A} \cap B \cap \bar{C})$$

$$\therefore P(B) = P(A \cap B) + P(B \cap C) + P(\bar{A} \cap B \cap \bar{C})$$

$$\Rightarrow P(B) = 0.07 + 0.15 + 0.10 = 0.32$$

$$\text{Now, } P(B \cap \bar{C}) = P(B) - P(B \cap C)$$

$$\Rightarrow P(B \cap \bar{C}) = 0.32 - 0.15 = 0.17$$

$$\text{(iii) We have, } P(A) = 0.20, P(B) = 0.32 \text{ and } P(A \cap B) = 0.07$$

$$\therefore P(A \cup B) = P(A) + P(B) - P(A \cap B) = 0.20 + 0.32 - 0.07 = 0.45$$

$$\text{(iv) We have, } P(A) = 0.20 \text{ and } P(A \cap B) = 0.07$$

$$\therefore P(A \cap \bar{B}) = P(A) - P(A \cap B) = 0.20 - 0.07 = 0.13$$

$$\text{(v) It is evident from the Figure 33.6 that } P(B \cap C) = 0.15.$$

$$\text{(vi) Probability that exactly one of three events } A, B \text{ and } C \text{ occurs}$$

$$= P(A \cap \bar{B} \cap \bar{C}) + P(\bar{A} \cap B \cap \bar{C}) + P(\bar{A} \cap \bar{B} \cap C) = 0.13 + 0.10 + 0.28 = 0.51$$

EXAMPLE 41 Let A, B, C be three events. If the probability of occurring exactly one event out of A and B is $1 - x$, out of B and C is $1 - 2x$, out of C and A is $1 - x$, and that of occurring three events simultaneously is x^2 , then prove that the probability that atleast one out of A, B, C will occur is greater than $1/2$.

SOLUTION We have,

$$P(A) + P(B) - 2P(A \cap B) = 1 - x \quad \dots(i)$$

$$P(B) + P(C) - 2P(B \cap C) = 1 - 2x \quad \dots(ii)$$

$$P(C) + P(A) - 2P(C \cap A) = 1 - x \quad \dots(iii)$$

$$\text{and, } P(A \cap B \cap C) = x^2 \quad \dots(iv)$$

Adding (i), (ii) and (iii), we get

$$P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C) - P(C \cap A) = \frac{3 - 4x}{2} \quad \dots(v)$$

$$\therefore \text{Probability that at least one out of } A, B, C \text{ will occur}$$

$$= P(A \cup B \cup C)$$

$$= P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C) - P(A \cap C) + P(A \cap B \cap C)$$

$$= \frac{3 - 4x}{2} + x^2 \quad [\text{Using (iv) and (v)}]$$

$$= x^2 - 2x + \frac{3}{2} = (x - 1)^2 + \frac{1}{2} > \frac{1}{2}$$

EXAMPLE 42 For the three events A, B and C , $P(\text{exactly one of the events } A \text{ or } B \text{ occurs}) = P(\text{exactly one of the events } B \text{ or } C \text{ occurs}) = P(\text{exactly one of the events } C \text{ and } A \text{ occurs}) = p$ and $P(\text{all the three events occur simultaneously}) = p^2$, where $0 < p < 1/2$. Then, find the probability of occurrence of at least one of the three events A, B , and C .

SOLUTION It is given that

$$P(\text{Exactly one of the events } A \text{ or } B \text{ occurs}) = p, P(\text{Exactly one of the events } B \text{ or } C \text{ occurs}) = p$$

$$P(\text{Exactly one of the events } C \text{ or } A \text{ occurs}) = p$$

$$\text{and, } P(\text{All the three events occur simultaneously}) = p^2$$

$$\text{i.e. } P(A) + P(B) - 2P(A \cap B) = p \quad \dots(i)$$

$$P(B) + P(C) - 2P(B \cap C) = p \quad \dots(ii)$$

$$P(C) + P(A) - 2P(C \cap A) = p \quad \dots(iii)$$

$$\text{and, } P(A \cap B \cap C) = p^2 \quad \dots(iv)$$

Adding (i), (ii) and (iii), we get

$$P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C) - P(A \cap C) = \frac{3p}{2} \quad \dots(v)$$

Required probability = $P(A \cup B \cup C)$

$$\begin{aligned} &= P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C) - P(A \cap C) + P(A \cap B \cap C) \\ &= \frac{3p}{2} + p^2 = \frac{3p + 2p^2}{2} \end{aligned}$$

EXAMPLE 43 For a post three persons A, B and C appear in the interview. The probability of A being selected is twice that of B and the probability of B being selected is thrice that of C. What are the individual probabilities of A, B, C being selected?

SOLUTION Let A_1, A_2 and A_3 be three events as defined below:

A_1 = Person A is selected, A_2 = Person B is selected, A_3 = Person C is selected.

We have,

$$P(A_1) = 2P(A_2) \text{ and } P(A_2) = 3P(A_3) \Rightarrow P(A_1) = 6P(A_3) \text{ and } P(A_2) = 3P(A_3).$$

Since A_1, A_2, A_3 are mutually exclusive and exhaustive events.

$$\therefore A_1 \cup A_2 \cup A_3 = S$$

$$\Rightarrow P(A_1 \cup A_2 \cup A_3) = P(S)$$

$$\Rightarrow P(A_1) + P(A_2) + P(A_3) = 1$$

[$\because A_1, A_2, A_3$ are mutually exclusive]

$$\Rightarrow 6P(A_3) + 3P(A_3) + P(A_3) = 1$$

$$\Rightarrow 10P(A_3) = 1$$

$$\Rightarrow P(A_3) = \frac{1}{10}$$

$$\therefore P(A_1) = \frac{6}{10} \text{ and } P(A_2) = \frac{3}{10}$$

EXAMPLE 44 P and Q are two candidates seeking admission in I.I.T. The probability that P is selected is 0.5 and the probability that both P and Q are selected is at most 0.3. Prove that the probability of Q being selected is at most 0.8.

SOLUTION Let A_1 and A_2 be two events defined as follows:

A_1 = P is selected, A_2 = Q is selected.

We have, $P(A_1) = 0.5$ and $P(A_1 \cap A_2) \leq 0.3$

Now, $P(A_1 \cup A_2) \leq 1$

$$\Rightarrow P(A_1) + P(A_2) - P(A_1 \cap A_2) \leq 1$$

$$\Rightarrow 0.5 + P(A_2) - P(A_1 \cap A_2) \leq 1$$

$$\Rightarrow P(A_2) \leq 0.5 + P(A_1 \cap A_2) \Rightarrow P(A_2) \leq 0.5 + 0.3 \Rightarrow P(A_2) \leq 0.8$$

EXAMPLE 45 A box contains 6 red, 4 white and 5 black balls. A person draws 4 balls from the box at random. Find the probability that among the balls drawn there is atleast one ball of each colour.

SOLUTION We observe that at least one ball of each colour can be drawn in one of the following mutually exclusive ways :

(i) 1 red, 1 white and 2 black balls. (ii) 2 red, 1 white and 1 black balls.

(iii) 1 red, 2 white and 1 black balls.

Thus, if we define three events A, B and C as follows:

A = Drawing 1 red, 1 white and 2 black balls, B = Drawing 2 red, 1 white and 1 black balls

C = Drawing 1 red, 2 white and 1 black balls

We observe that A, B, C are mutually exclusive events.

$$\begin{aligned}
 \therefore \text{ Required probability} &= P(A \cup B \cup C) \\
 &= P(A) + P(B) + P(C) \quad [\text{By addition Theorem}] \\
 &= \frac{{}^6C_1 \times {}^4C_1 \times {}^5C_2}{{}^{15}C_4} + \frac{{}^6C_2 \times {}^4C_1 \times {}^5C_1}{{}^{15}C_4} + \frac{{}^6C_1 \times {}^4C_2 \times {}^5C_1}{{}^{15}C_4} \\
 &= \frac{6 \times 4 \times 10 + 15 \times 4 \times 5 + 6 \times 6 \times 5}{{}^{15}C_4} = \frac{24 \times 720}{15 \times 14 \times 13 \times 12} = \frac{48}{91}
 \end{aligned}$$

EXAMPLE 46 A die is loaded in such a way that each odd number is twice as likely to occur as each even number. Find $P(G)$, where G is the event that a number greater than 3 occurs on a single roll of the die.

SOLUTION Let A_i denote the event "Getting number i on the upper face of the die", $i = 1, 2, 3, 4, 5, 6$. Clearly, $A_i; i = 1, 2, \dots, 6$ are mutually exclusive and exhaustive events.

It is given that $P(A_2) = P(A_4) = P(A_6) = p$ (say) and, $P(A_1) = P(A_3) = P(A_5) = 2p$.

Now, $A_1 \cup A_2 \cup A_3 \cup A_4 \cup A_5 \cup A_6 = S$

$$\Rightarrow P(A_1 \cup A_2 \cup A_3 \cup A_4 \cup A_5 \cup A_6) = P(S)$$

$$\Rightarrow P(A_1) + P(A_2) + P(A_3) + P(A_4) + P(A_5) + P(A_6) = 1$$

$$\Rightarrow 2p + p + 2p + p + 2p + p = 1 \Rightarrow 9p = 1 \Rightarrow p = \frac{1}{9}$$

Now, $G = A_4 \cup A_5 \cup A_6$

$$\Rightarrow P(G) = P(A_4 \cup A_5 \cup A_6)$$

$$\Rightarrow P(G) = P(A_4) + P(A_5) + P(A_6)$$

[$\because A_4, A_5, A_6$ are mutually exclusive]

$$\Rightarrow P(G) = p + 2p + p = 4p = \frac{4}{9}$$

EXAMPLE 47 A team of medical students doing their internship have to assist during surgeries at a city hospital. The probabilities of surgeries rated as very complex, complex, routine, simple and very simple are respectively 0.15, 0.20, 0.31, 0.26, 0.08. Find the probabilities that a particular surgery will be rated (i) Complex or very complex (ii) neither very complex nor very simple (iii) routine or complex (iv) routine or simple. [NCERT EXEMPLAR]

SOLUTION Consider the following events:

A = Surgery is very complex, B = Surgery is complex, C = Surgery is routine,

D = Surgery is simple, E = Surgery is very simple.

It is given that $P(A) = 0.15$, $P(B) = 0.20$, $P(C) = 0.31$, $P(D) = 0.26$ and $P(E) = 0.08$.

Clearly, A, B, C, D and E are mutually exclusive events.

$$(i) \text{ Required probability} = P(A \cup B) = P(A) + P(B) = 0.15 + 0.20 = 0.35$$

$$\begin{aligned}
 (ii) \text{ Required probability} &= P(\overline{A} \cap \overline{E}) \\
 &= P(\overline{A \cup E}) \\
 &= 1 - P(A \cup E) = 1 - \{P(A) + P(E)\} = 1 - (0.15 + 0.08) = 0.77
 \end{aligned}$$

$$(iii) \text{ Required probability} = P(C \cup B) = P(C) + P(B) = 0.31 + 0.20 = 0.51$$

$$(iv) \text{ Required probability} = P(C \cup D) = P(C) + P(D) = 0.31 + 0.26 = 0.57$$

EXERCISE 33.4

LEVEL-1

- (a) If A and B be mutually exclusive events associated with a random experiment such that $P(A) = 0.4$ and $P(B) = 0.5$, then find:
 - $P(A \cup B)$
 - $P(\overline{A} \cap \overline{B})$
 - $P(\overline{A} \cap B)$
 - $P(A \cap \overline{B})$

- (b) A and B are two events such that $P(A) = 0.54$, $P(B) = 0.69$ and $P(A \cap B) = 0.35$. Find (i) $P(A \cup B)$ (ii) $P(\bar{A} \cap \bar{B})$ (iii) $P(A \cap \bar{B})$ (iv) $P(B \cap \bar{A})$
- (c) Fill in the blanks in the following table:

	$P(A)$	$P(B)$	$P(A \cap B)$	$P(A \cup B)$
(i)	$\frac{1}{3}$	$\frac{1}{5}$	$\frac{1}{15}$	
(ii)	0.35		0.25	0.6
(iii)	0.5	0.35		0.7

- If A and B are two events associated with a random experiment such that $P(A) = 0.3$, $P(B) = 0.4$ and $P(A \cup B) = 0.5$, find $P(A \cap B)$.
 - If A and B are two events associated with a random experiment such that $P(A) = 0.5$, $P(B) = 0.3$ and $P(A \cap B) = 0.2$, find $P(A \cup B)$.
 - If A and B are two events associated with a random experiment such that $P(A \cup B) = 0.8$, $P(A \cap B) = 0.3$ and $P(\bar{A}) = 0.5$, find $P(B)$.
 - Given two mutually exclusive events A and B such that $P(A) = 1/2$ and $P(B) = 1/3$, find $P(A \text{ or } B)$.
 - There are three events A, B, C one of which must and only one can happen, the odds are 8 to 3 against A , 5 to 2 against B , find the odds against C .
 - One of the two events must happen. Given that the chance of one is two-third of the other, find the odds in favour of the other.
- NOTE** Students are advised to do the following exercises by using addition theorems and also by using the definition only i.e. by calculating exhaustive number of cases and favourable number of cases.
- A card is drawn at random from a well-shuffled deck of 52 cards. Find the probability of its being a spade or a king.
 - In a single throw of two dice, find the probability that neither a doublet nor a total of 9 will appear.
 - A natural number is chosen at random from amongst first 500. What is the probability that the number so chosen is divisible by 3 or 5?
 - A die is thrown twice. What is the probability that at least one of the two throws come up with the number 3?
 - A card is drawn from a deck of 52 cards. Find the probability of getting an ace or a spade card.
 - The probability that a student will pass the final examination in both English and Hindi is 0.5 and the probability of passing neither is 0.1. If the probability of passing the English examination is 0.75. What is the probability of passing the Hindi examination? [NCERT]
 - One number is chosen from numbers 1 to 100. Find the probability that it is divisible by 4 or 6?
 - From a well shuffled deck of 52 cards, 4 cards are drawn at random. What is the probability that all the drawn cards are of the same colour.
 - 100 students appeared for two examinations. 60 passed the first, 50 passed the second and 30 passed both. Find the probability that a student selected at random has passed at least one examination.
 - A box contains 10 white, 6 red and 10 black balls. A ball is drawn at random from the box. What is the probability that the ball drawn is either white or red?
 - In a race, the odds in favour of horses A, B, C, D are 1 : 3, 1 : 4, 1 : 5 and 1 : 6 respectively. Find probability that one of them wins the race.
 - The probability that a person will travel by plane is $3/5$ and that he will travel by train is $1/4$. What is the probability that he (she) will travel by plane or train?
 - Two cards are drawn from a well shuffled pack of 52 cards. Find the probability that either both are black or both are kings.

21. In an entrance test that is graded on the basis of two examinations, the probability of a randomly chosen student passing the first examination is 0.8 and the probability of passing the second examination is 0.7. The probability of passing at least one of them is 0.95. What is the probability of passing both? [NCERT]
22. A box contains 30 bolts and 40 nuts. Half of the bolts and half of the nuts are rusted. If two items are drawn at random, what is the probability that either both are rusted or both are bolts?
23. An integer is chosen at random from first 200 positive integers. Find the probability that the integer is divisible by 6 or 8.
24. Find the probability of getting 2 or 3 tails when a coin is tossed four times.
25. Suppose an integer from 1 through 1000 is chosen at random, find the probability that the integer is a multiple of 2 or a multiple of 9. [NCERT EXEMPLAR]
26. In a large metropolitan area, the probabilities are 0.87, 0.36, 0.30 that a family (randomly chosen for a sample survey) owns a colour television set, a black and white television set, or both kinds of sets. What is the probability that a family owns either any one or both kinds of sets? [NCERT EXEMPLAR]
27. If A and B are mutually exclusive events such that $P(A) = 0.35$ and $P(B) = 0.45$, find
(i) $P(A \cup B)$ (ii) $P(A \cap B)$ (iii) $P(A \cap \bar{B})$ (iv) $P(\bar{A} \cap \bar{B})$ [NCERT EXEMPLAR]
28. A sample space consists of 9 elementary event $E_1, E_2, E_3, \dots, E_8, E_9$ whose probabilities are $P(E_1) = P(E_2) = 0.08$, $P(E_3) = P(E_4) = 0.1$, $P(E_6) = P(E_7) = 0.2$, $P(E_8) = P(E_9) = 0.07$
Suppose $A = \{E_1, E_5, E_8\}$, $B = \{E_2, E_5, E_8, E_9\}$ [NCERT EXEMPLAR]
(i) Compute $P(A)$, $P(B)$ and $P(A \cap B)$.
(ii) Using the addition law of probability, find $P(A \cup B)$.
(iii) List the composition of the event $A \cup B$, and calculate $P(A \cup B)$ by adding the probabilities of the elementary events.
(iv) Calculate $P(\bar{B})$ from $P(B)$, also calculate $P(\bar{B})$ directly from the elementary events of $\bar{B}$.

ANSWERS

1. (a) (i) 0.9 (ii) 0.1 (iii) 0.5 (iv) 0.4 1. (b) (i) 0.88 (ii) 0.12 (iii) 0.19 (iv) 0.34
1. (c) (i) $\frac{7}{15}$ (ii) 0.5 (iii) 0.15 2. 0.2 3. 0.6 4. 0.6 5. $\frac{5}{6}$
6. 43 : 34 7. 3 : 2 8. $\frac{4}{13}$ 9. $\frac{13}{18}$ 10. $\frac{233}{500}$ 11. $\frac{11}{36}$
12. $\frac{4}{13}$ 13. 0.65 14. $\frac{33}{100}$ 15. $\frac{92}{833}$ 16. $\frac{4}{5}$ 17. $\frac{8}{13}$
18. $\frac{171}{420}$ 19. $\frac{17}{20}$ 20. $\frac{55}{221}$ 21. 0.556 22. $\frac{185}{483}$ 23. $\frac{1}{4}$
24. $\frac{5}{8}$ 25. 0.556 26. 0.93 27. (i) 0.8 (ii) 0 (iii) 0.35 (iv) 0.2
28. (i) 0.25, 0.32, 0.17 (ii) 0.40 (iii) 0.40 (iv) 0.68

HINTS TO NCERT & SELECTED PROBLEMS

1. (ii) Use : $P(\bar{A} \cap \bar{B}) = P(\overline{A \cup B}) = 1 - P(A \cup B)$
(iii) Use : $P(\bar{A} \cap B) = P(B) - P(A \cap B)$
(iv) Use : $P(A \cap \bar{B}) = P(A) - P(A \cap B)$
6. It is given that A, B, C are mutually exclusive and exhaustive.
 $\therefore A \cup B \cup C = S \Rightarrow P(A \cup B \cup C) = P(S) \Rightarrow P(A) + P(B) + P(C) = 1$

7. Let A, B be two events. Then A, B are mutually exclusive and exhaustive.

$$\therefore A \cup B = S$$

$$\Rightarrow P(A \cup B) = 1$$

$$\Rightarrow P(A) + P(B) = 1$$

$$\Rightarrow \frac{2}{3} P(B) + P(B) = 1$$

$$\left[\because P(A) = \frac{2}{3} P(B) \right]$$

$$\Rightarrow P(B) = \frac{3}{5}$$

Odds in favour of B are $P(B) : P(\bar{B})$ i.e. $3/5 : 2/5$ or $3 : 2$

9. Let $A =$ Getting a doublet, $B =$ Getting a total of 9. Then,

$$P(\bar{A} \cap \bar{B}) = P(\overline{A \cup B}) = 1 - P(A \cup B) = 1 - [P(A) + P(B) - P(A \cap B)]$$

12. $A =$ Getting an ace, $B =$ Getting a spade card.

$$\text{Required probability} = P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

13. Let E and H denote the events that the student will pass in English and Hindi examination respectively. Then, we have

$$P(E \cap H) = 0.5, P(\bar{E} \cap \bar{H}) = 0.1 \text{ and } P(E) = 0.75$$

Now,

$$P(\bar{E} \cap \bar{H}) = 0.1$$

$$\Rightarrow P(\overline{E \cup H}) = 0.1$$

$$\Rightarrow 1 - P(E \cup H) = 0.1$$

$$\Rightarrow P(E \cup H) = 0.9$$

$$\Rightarrow P(E) + P(H) - P(E \cap H) = 0.9 \Rightarrow 0.75 + P(H) - 0.5 = 0.9 \Rightarrow P(H) = 0.65.$$

15. Let $A =$ 4 cards drawn are red, $B =$ 4 cards drawn are black. Then, A, B are mutually exclusive events.

$$\text{So, required probability} = P(A \cup B) = P(A) + P(B)$$

21. Let A and B denote the events that a randomly chosen student passes first and second examinations respectively. Then, $P(A) = 0.8$, $P(B) = 0.7$ and $P(A \cup B) = 0.95$

$$\text{Required probability} = P(A \cap B) = P(A) + P(B) - P(A \cup B) = 0.8 + 0.7 - 0.95 = 0.55$$

25. Consider the following events:

$A =$ Integer chosen is a multiple of 2, $B =$ Integer chosen is a multiple of 9.

We have,

$$P(A) = \frac{500}{1000} = \frac{1}{2}, P(B) = \frac{111}{1000} \text{ and } P(A \cap B) = \frac{55}{1000}$$

$$\text{Required probability} = P(A \cup B) = P(A) + P(B) - P(A \cap B) = \frac{1}{2} + \frac{111}{1000} - \frac{55}{1000} = \frac{555}{1000}$$

26. Consider the following events:

$A =$ Family owns colour television set, $B =$ Family owns black and white television set

It is given that $P(A) = 0.87$, $P(B) = 0.36$ and $P(A \cap B) = 0.30$

$$\text{Required probability} = P(A \cup B) = P(A) + P(B) - P(A \cap B) = 0.87 + 0.36 - 0.30 = 0.93$$

27. It is given that A and B are mutually exclusive events. Therefore, $P(A \cap B) = 0$.

$$(i) P(A \cup B) = P(A) + P(B) = 0.35 + 0.45 = 0.8$$

$$(ii) P(A \cap B) = 0$$

$$(iii) P(A \cap \bar{B}) = P(A) - P(A \cap B) = P(A) = 0.35$$

$$(iv) P(\bar{A} \cap \bar{B}) = P(\overline{A \cup B}) = 1 - P(A \cup B) = 1 - [P(A) + P(B)] = 1 - 0.8 = 0.2$$

28. We have, $P(E_1) + P(E_2) + P(E_3) + P(E_4) + P(E_5) + P(E_6) + P(E_7) + P(E_8) = 1 \Rightarrow P(E_5) = 0.1$

$$(i) P(A) = P(E_1) + P(E_5) + P(E_8) = 0.08 + 0.1 + 0.07 = 0.25$$

$$P(B) = P(E_2) + P(E_5) + P(E_8) + P(E_9) = 0.08 + 0.1 + 0.07 + 0.07 = 0.32$$

$$P(A \cap B) = P(E_5) + P(E_8) = 0.1 + 0.07 = 0.17$$

$$(ii) P(A \cup B) = P(A) + P(B) - P(A \cap B) = 0.25 + 0.32 - 0.17 = 0.40$$

VERY SHORT ANSWER QUESTIONS (VSAQs)

Answer each of the following questions in one word or one sentence or as per exact requirement of the question:

- Three numbers are chosen at random from numbers 1 to 30. Write the probability that the chosen numbers are consecutive.
- $n (> 3)$ persons are sitting in a row. Two of them are selected. Write the probability that they are together.
- A single letter is selected at random from the word 'PROBABILITY'. What is the probability that it is a vowel?
- What is the probability that a leap year will have 53 Fridays or 53 Saturdays?
- Three dice are thrown simultaneously. What is the probability of getting 15 as the sum?
- If the letters of the word 'MISSISSIPPI' are written down at random in a row, what is the probability that four S's come together.
- What is the probability that the 13th days of a randomly chosen month is Friday?
- Three of the six vertices of a regular hexagon are chosen at random. What is the probability that the triangle with these vertices is equilateral.
- If E and E_2 are independent events, write the value of $P(E_1 \cup E_2) \cap (\bar{E} \cap \bar{E}_2)$.
- If A and B are two independent events such that $P(A \cap B) = \frac{1}{6}$ and $P(\bar{A} \cap \bar{B}) = \frac{1}{3}$, then write the values of $P(A)$ and $P(B)$.

ANSWERS

- $\frac{144}{145}$
- $\frac{2}{n}$
- $\frac{4}{11}$
- $\frac{3}{7}$
- $\frac{13}{216}$
- $\frac{4}{165}$
- $\frac{1}{84}$
- $\frac{1}{10}$
- 0
- $P(A) = \frac{1}{2}, P(B) = \frac{1}{3}$

MULTIPLE CHOICE QUESTIONS (MCQs)

Mark the correct alternative in each of the following:

- One card is drawn from a pack of 52 cards. The probability that it is the card of a king or spade is
(a) $\frac{1}{26}$ (b) $\frac{3}{26}$ (c) $\frac{4}{13}$ (d) $\frac{3}{13}$
- Two dice are thrown together. The probability that at least one will show its digit greater than 3 is
(a) $\frac{1}{4}$ (b) $\frac{3}{4}$ (c) $\frac{1}{2}$ (d) $\frac{1}{8}$
- Two dice are thrown simultaneously. The probability of obtaining a total score of 5 is
(a) $\frac{1}{18}$ (b) $\frac{1}{12}$ (c) $\frac{1}{9}$ (d) none of these
- Two dice are thrown simultaneously. The probability of obtaining total score of seven is
(a) $\frac{5}{36}$ (b) $\frac{6}{36}$ (c) $\frac{7}{36}$ (d) $\frac{8}{36}$
- The probability of getting a total of 10 in a single throw of two dice is
(a) $\frac{1}{9}$ (b) $\frac{1}{12}$ (c) $\frac{1}{6}$ (d) $\frac{5}{36}$

6. A card is drawn at random from a pack of 100 cards numbered 1 to 100. The probability of drawing a number which is a square is
 (a) $1/5$ (b) $2/5$ (c) $1/10$ (d) none of these.
7. A bag contains 3 red, 4 white and 5 blue balls. All balls are different. Two balls are drawn at random. The probability that they are of different colour is
 (a) $47/66$ (b) $10/33$ (c) $1/3$ (d) 1
8. Two dice are thrown together. The probability that neither they show equal digits nor the sum of their digits is 9 will be
 (a) $13/15$ (b) $13/18$ (c) $1/9$ (d) $8/9$
9. Four persons are selected at random out of 3 men, 2 women and 4 children. The probability that there are exactly 2 children in the selection is
 (a) $11/21$ (b) $9/21$ (c) $10/21$ (d) none of these
10. The probabilities of happening of two events A and B are 0.25 and 0.50 respectively. If the probability of happening of A and B together is 0.14, then probability that neither A nor B happens is
 (a) 0.39 (b) 0.25 (c) 0.11 (d) none of these
11. A die is rolled, then the probability that a number 1 or 6 may appear is
 (a) $2/3$ (b) $5/6$ (c) $1/3$ (d) $1/2$
12. Six boys and six girls sit in a row randomly. The probability that all girls sit together is
 (a) $1/122$ (b) $1/112$ (c) $1/102$ (d) $1/132$
13. The probabilities of three mutually exclusive events A , B and C are given by $2/3$, $1/4$ and $1/6$ respectively. The statement
 (a) is true (b) is false
 (c) nothing can be said (d) could be either
14. If $\frac{(1-3p)}{2}$, $\frac{(1+4p)}{3}$, $\frac{(1+p)}{6}$ are the probabilities of three mutually exclusive and exhaustive events, then the set of all values of p is
 (a) $(0,1)$ (b) $(-1/4, 1/3)$ (c) $(0, 1/3)$ (d) $(0, \infty)$
15. A pack of cards contains 4 aces, 4 kings, 4 queens and 4 jacks. Two cards are drawn at random. The probability that at least one of them is an ace is
 (a) $1/5$ (b) $3/16$ (c) $9/20$ (d) $1/9$
16. If three dice are throw simultaneously, then the probability of getting a score of 5 is
 (a) $5/216$ (b) $1/6$ (c) $1/36$ (d) none of these
17. One of the two events must occur. If the chance of one is $2/3$ of the other, then odds in favour of the other are
 (a) 1 : 3 (b) 3 : 1 (c) 2 : 3 (d) 3 : 2
18. The probability that a leap year will have 53 Fridays or 53 Saturdays is
 (a) $2/7$ (b) $3/7$ (c) $4/7$ (d) $1/7$
19. A person write 4 letters and addresses 4 envelopes. If the letters are placed in the envelopes at random, then the probability that all letters are not placed in the right envelopes, is
 (a) $1/4$ (b) $11/24$ (c) $15/24$ (d) $23/24$
20. A and B are two events such that $P(A) = 0.25$ and $P(B) = 0.50$. The probability of both happening together is 0.14. The probability of both A and B not happening is
 (a) 0.39 (b) 0.25 (c) 0.11 (d) none of these

21. If the probability of A to fail in an examination is $\frac{1}{5}$ and that of B is $\frac{3}{10}$. Then, the probability that either A or B fails is
 (a) $1/2$ (b) $11/25$ (c) $19/50$ (d) none of these
22. A box contains 10 good articles and 6 defective articles. One item is drawn at random. The probability that it is either good or has a defect, is
 (a) $64/64$ (b) $49/64$ (c) $40/64$ (d) $24/64$
23. Three integers are chosen at random from the first 20 integers. The probability that their product is even is
 (a) $2/19$ (b) $3/29$ (c) $17/19$ (d) $4/19$
24. Out of 30 consecutive integers, 2 are chosen at random. The probability that their sum is odd, is
 (a) $14/29$ (b) $16/29$ (c) $15/29$ (d) $10/29$
25. A bag contains 5 black balls, 4 white balls and 3 red balls. If a ball is selected randomwise, the probability that it is black or red ball is
 (a) $1/3$ (b) $1/4$ (c) $5/12$ (d) $2/3$
26. Two dice are thrown simultaneously. The probability of getting a pair of aces is
 (a) $1/36$ (b) $1/3$ (c) $1/6$ (d) none of these
27. An urn contains 9 balls two of which are red, three blue and four black. Three balls are drawn at random. The probability that they are of the same colour is
 (a) $5/84$ (b) $3/9$ (c) $3/7$ (d) $7/17$
28. Five persons entered the lift cabin on the ground floor of an 8 floor house. Suppose that each of them independently and with equal probability can leave the cabin at any floor beginning with the first, then the probability of all 5 persons leaving at different floor is
 (a) $\frac{{}^7P_5}{7^5}$ (b) $\frac{7^5}{{}^7P_5}$ (c) $\frac{6}{{}^6P_5}$ (d) $\frac{{}^5P_5}{5^5}$
29. A box contains 10 good articles and 6 with defects. One item is drawn at random. The probability that it is either good or has a defect is
 (a) $64/64$ (b) $49/64$ (c) $40/64$ (d) $24/64$
30. A box contains 6 nails and 10 nuts. Half of the nails and half of the nuts are rusted. If one item is chosen at random, the probability that it is rusted or is a nail is
 (a) $3/16$ (b) $5/16$ (c) $11/16$ (d) $14/16$
31. If S is the sample space and $P(A) = \frac{1}{3} P(B)$ and $S = A \cup B$, where A and B are two mutually exclusive events, then $P(A) =$
 (a) $1/4$ (b) $1/2$ (c) $3/4$ (d) $3/8$
32. One mapping is selected at random from all the mappings of the set $A = \{1, 2, 3, \dots, n\}$ into itself. The probability that the mapping selected is one to one is
 (a) $\frac{1}{n^n}$ (b) $\frac{1}{n!}$ (c) $\frac{(n-1)!}{n^{n-1}}$ (d) None of these
33. If A, B, C are three mutually exclusive and exhaustive events of an experiment such that $3P(A) = 2P(B) = P(C)$, then $P(A)$ is equal to
 (a) $\frac{1}{11}$ (b) $\frac{2}{11}$ (c) $\frac{5}{11}$ (d) $\frac{6}{11}$

34. If A and B are mutually exclusive events then
 (a) $P(A) \leq P(\bar{B})$ (b) $P(A) \geq P(\bar{B})$ (c) $P(A) < P(\bar{B})$ (d) None of these
35. If $P(A \cup B) = P(A \cap B)$ for any two events A and B , then
 (a) $P(A) = P(B)$ (b) $P(A) > P(B)$ (c) $P(A) < P(B)$ (d) None of these
36. Three numbers are chosen from 1 to 20. The probability that they are not consecutive is
 (a) $\frac{186}{190}$ (b) $\frac{187}{190}$ (c) $\frac{188}{190}$ (d) $\frac{18}{{}^{20}C_3}$
37. 6 boys and 6 girls sit in a row at random. The probability that all the girls sit together is
 (a) $\frac{1}{432}$ (b) $\frac{12}{431}$ (c) $\frac{1}{132}$ (d) None of these
38. Without repetition of the numbers, four digit numbers are formed with the numbers 0, 2, 3, 5. The probability of such a number divisible by 5 is
 (a) $\frac{1}{5}$ (b) $\frac{4}{5}$ (c) $\frac{1}{30}$ (d) $\frac{5}{9}$
39. If the probability for A to fail in an examination is 0.2 and that for B is 0.3, then the probability that either A or B fails is
 (a) > 0.5 (b) 0.5 (c) ≤ 0.5 (d) 0
40. Three digit numbers are formed using the digits 0, 2, 4, 6, 8. A number is chosen at random out of these numbers what is the probability that this number has the same digits?
 (a) $\frac{1}{16}$ (b) $\frac{16}{25}$ (c) $\frac{1}{645}$ (d) $\frac{1}{25}$

ANSWERS

- | | | | | | | | |
|---------|---------|---------|---------|---------|---------|---------|---------|
| 1. (c) | 2. (b) | 3. (c) | 4. (b) | 5. (b) | 6. (c) | 7. (a) | 8. (b) |
| 9. (c) | 10. (a) | 11. (c) | 12. (d) | 13. (b) | 14. (b) | 15. (c) | 16. (c) |
| 17. (d) | 18. (b) | 19. (d) | 20. (a) | 21. (c) | 22. (a) | 23. (c) | 24. (c) |
| 25. (d) | 26. (a) | 27. (a) | 28. (a) | 29. (a) | 30. (c) | 31. (a) | 32. (c) |
| 33. (b) | 34. (a) | 35. (a) | 36. (b) | 37. (c) | 38. (d) | 39. (c) | 40. (d) |

SUMMARY

1. An experiment whose outcomes cannot be predicted or determined in advance is called a random experiment.
2. Each outcome of a random experiment is known as an elementary event.
3. The set of all possible outcomes (elementary events) of a random experiment is called the sample space associated with it.
4. A subset of the sample space associated with a random experiment is called an event.
5. An event is said to occur if any one of the elementary events belonging to it is an outcome.
6. An event associated with a random experiment is called a certain event if it always occurs whenever the experiment is performed.
The sample space associated with a random experiment defines a certain event.
7. The null set of the sample space defines an impossible event.
8. An event associated with a random experiment is a compound event, if it is the disjoint union of two or more elementary events.

9. Two or more events associated with a random experiment are said to be mutually exclusive or incompatible events if the occurrence of any one of them prevents the occurrence of all others i.e. no two or more of them can occur simultaneously in the same trial.
If A and B are mutually exclusive events, then $A \cap B = \phi$.
10. Events $A_1, A_2, A_3, \dots, A_n$ associated with a random experiment with sample space S are exhaustive if $A_1 \cup A_2 \cup \dots \cup A_n = S$.
11. Let S be the sample space associated with a random experiment. A set of events $A_1, A_2, \dots, A_n$ is said to form a set of mutually exclusive and exhaustive system of events if
(i) $A_1 \cup A_2 \cup \dots \cup A_n = S$ (ii) $A_i \cap A_j = \phi$ for $i \neq j$
12. **Probability function:** Let $S = \{w_1, w_2, \dots, w_n\}$ be the sample space associated with a random experiment. Then a function P which assigns every event $A \subset S$ to a unique non-negative real number $P(A)$ is called the probability function if the following axioms hold :
- A - 1 : $0 \leq P(w_i) \leq 1$ for all $w_i \in S$
 A - 2 : $P(S) = 1$ i.e. $P(w_1) + P(w_2) + \dots + P(w_n) = 1$
 A - 3 : For any event $A \subset S$, $P(A) = \sum_{w_k \in A} P(w_k)$, the number $P(w_k)$ is called

probability of elementary event w_k .

13. **Probability of an event:** If there are n elementary events associated with a random experiment and m of them are favourable to an event A , then the probability of occurrence of A is defined as:

$$P(A) = \frac{m}{n} = \frac{\text{Favourable number of elementary events}}{\text{Total number of elementary events}}$$

The odds in favour of occurrence of the event A are defined by $m : (n - m)$

The odds against the occurrence of A are defined by $(n - m) : m$.

The probability of non-occurrence of A is given by $P(\bar{A}) = 1 - P(A)$.

14. If A and B are two events associated with a random experiment, then
 $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.
 If A and B are mutually exclusive events, then $P(A \cup B) = P(A) + P(B)$.
15. If A, B, C are three events associated with a random experiment, then
 $P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C) - P(C \cap A) + P(A \cap B \cap C)$
16. If A and B are two events associated with a random experiment, then
 (i) $P(\bar{A} \cap B) = P(B) - P(A \cap B)$ i.e. probability of occurrence of B only $= P(B) - P(A \cap B)$
 (ii) $P(A \cap \bar{B}) = P(A) - P(A \cap B)$ i.e. probability of occurrence of A only $= P(A) - P(A \cap B)$
 (iii) Probability of occurrence of exactly one of A and B is $P(A) + P(B) - 2P(A \cap B)$
 $= P(A \cup B) - P(A \cap B)$